SA 4059 ENTERGY LOUISIANA-LACASSINE SOLAR II GIA

EFFECTIVE 05/29/2025

SECOND REVISED SERVICE AGREEMENT NO. 4059

J1795

AMENDED AND RESTATED

GENERATOR INTERCONNECTION

AGREEMENT entered into by the

Midcontinent Independent System Operator, Inc., And

Lacassine Solar II, LLC,

And

Entergy Louisiana, LLC

GENERATOR INTERCONNECTION AGREEMENT (GIA)

THIS AMENDED AND RESTATED GENERATOR INTERCONNECTION AGREEMENT ("GIA") is made and entered into this 29th day of May, 2025, by and between **Lacassine Solar II, LLC**, a limited liability company organized and existing under the laws of the State of Delaware ("Interconnection Customer" with a Generating Facility), and **Entergy Louisiana, LLC**, a limited liability company organized and existing under the laws of the State of Texas ("Transmission Owner"), and the **Midcontinent Independent System Operator, Inc**., a non-profit, non-stock corporation organized and existing under the laws of the State of Delaware ("Transmission Provider"). Interconnection Customer, Transmission Owner and Transmission Provider each may be referred to as a "Party," or collectively as the "Parties." This GIA replaces and supersedes the Generator Interconnection Agreement executed on October 21, 2024.

RECITALS

WHEREAS, Transmission Provider has functional control of the operations of the Transmission System, as defined herein, and is responsible for providing Transmission Service and Interconnection Service on the transmission facilities under its control; and

WHEREAS, Interconnection Customer intends to own, lease and/or control and operate the Generating Facility identified as a Generating Facility in Appendix A to this GIA; and

WHEREAS, Transmission Owner owns or operates the Transmission System, whose operations are subject to the functional control of Transmission Provider, to which Interconnection Customer desires to connect the Generating Facility, and may therefore be required to construct certain Interconnection Facilities and Network Upgrades, as set forth in this GIA; and

WHEREAS, Interconnection Customer, Transmission Owner and Transmission Provider have agreed to enter into this GIA, and where applicable subject to Appendix H for a Provisional Generator Interconnection Agreement, for the purpose of interconnecting the Generating Facility with the Transmission System; and

WHEREAS, this GIA was amended to update milestone dates consistent with Interconnection Customer's notice of expected delays under 5.7 of this agreement;

WHEREAS, this GIA is now being amended to update milestones consistent with Transmission Owner's notice of expected delays under 5.1.1 of this agreement;

NOW, THEREFORE, in consideration of and subject to the mutual covenants contained herein, it is agreed:

ARTICLE 1. DEFINITIONS

When used in this GIA, terms with initial capitalization that are not defined in Article 1 shall have the meanings specified in the Article in which they are used. Those capitalized terms used in this GIA that are not otherwise defined in this GIA have the meaning set forth in the Tariff.

Adverse System Impact shall mean the negative effects due to technical or operational limits on conductors or equipment being exceeded that may compromise the safety and reliability of the electric system.

Affected System shall mean an electric transmission or distribution system or the electric system associated with an Existing Generating Facility or of a higher queued Generating Facility, which is an electric system other than the Transmission Owner's Transmission System that is affected by the Interconnection Request. An Affected System may or may not be subject to FERC jurisdiction.

Affected System Operator shall mean the entity that operates an Affected System.

Affiliate shall mean, with respect to a corporation, partnership or other entity, each such other corporation, partnership or other entity that directly or indirectly, through one or more intermediaries, controls, is controlled by, or is under common control with, such corporation, partnership or other entity.

Ancillary Services shall mean those services that are necessary to support the transmission of capacity and energy from resources to loads while maintaining reliable operation of the Transmission System in accordance with Good Utility Practice.

Applicable Laws and Regulations shall mean all duly promulgated applicable federal, state and local laws, regulations, rules, ordinances, codes, decrees, judgments, directives, or judicial or administrative orders, permits and other duly authorized actions of any Governmental Authority having jurisdiction over the Parties, their respective facilities and/or the respective services they provide.

Applicable Reliability Council shall mean the Regional Entity of NERC applicable to the Local Balancing Authority of the Transmission System to which the Generating Facility is directly interconnected.

Applicable Reliability Standards shall mean Reliability Standards approved by the Federal Energy Regulatory Commission (FERC) under section 215 of the Federal Power Act, as applicable.

Base Case shall mean the base case power flow, short circuit, and stability databases used for the Interconnection Studies by Transmission Provider or Interconnection Customer.

Breach shall mean the failure of a Party to perform or observe any material term or condition of this GIA.

Breaching Party shall mean a Party that is in Breach of this GIA.

Business Day shall mean Monday through Friday, excluding Federal Holidays.

Calendar Day shall mean any day including Saturday, Sunday or a Federal Holiday.

Commercial Operation shall mean the status of a Generating Facility that has commenced generating electricity for sale, excluding electricity generated during Trial Operation.

Commercial Operation Date (COD) of a unit shall mean the date on which the Generating Facility commences Commercial Operation as agreed to by the Parties pursuant to Appendix E to this GIA.

Common Use Upgrade (CUU) shall mean an Interconnection Facility, Network Upgrade, System Protection Facility, or any other classified addition, alteration, or improvement on the Transmission System or the transmission system of an Affected System, not classified under Attachment FF as a Baseline Reliability Project, Market Efficiency Project, or Multi-Value Project, that is needed for the interconnection of multiple Interconnection Customers' Generating Facilities and which is the shared responsibility of such Interconnection Customers.

Confidential Information shall mean any proprietary or commercially or competitively sensitive information, trade secret or information regarding a plan, specification, pattern, procedure, design, device, list, concept, policy or compilation relating to the present or planned business of a Party, or any other information as specified in Article 22, which is designated as confidential by the Party supplying the information, whether conveyed orally, electronically, in writing, through inspection, or otherwise, that is received by another Party.

Default shall mean the failure of a Breaching Party to cure its Breach in accordance with Article 17 of this GIA.

Definitive Planning Phase Queue Position shall mean the order of a valid Interconnection Request, relative to all other pending valid Interconnection Requests, in the Definitive Planning Phase. The Definitive Planning Phase Queue Position is established based upon the date Interconnection Customer satisfies all of the requirements of Section 7.2 to enter the Definitive Planning Phase.

Demonstrated Capability shall mean the continuous net real power output that the Generating Facility is required to demonstrate in compliance with Applicable Reliability Standards.

Dispute Resolution shall mean the procedure for resolution of a dispute between or among the Parties in which they will first attempt to resolve the dispute on an informal basis.

Distribution System shall mean the Transmission Owner's facilities and equipment, or the Distribution System of another party that is interconnected with the Transmission Owner's Transmission System, if any, connected to the Transmission System, over which facilities Transmission Service or Wholesale Distribution Service under the Tariff is available at the time Interconnection Customer has requested interconnection of a Generating Facility for the purpose of either transmitting electric energy in interstate commerce or selling electric energy at wholesale in interstate commerce and which are used to transmit electricity to ultimate usage points such as homes and industries directly from nearby generators or from interchanges with higher voltage transmission networks which transport bulk power over longer distances. The voltage levels at which distribution systems operate differ among Local Balancing Authorities and other entities owning distribution facilities interconnected to the Transmission System.

Distribution Upgrades shall mean the additions, modifications, and upgrades to the Distribution System at or beyond the Point of Interconnection to facilitate interconnection of the Generating Facility and render the delivery service necessary to affect Interconnection Customer's wholesale sale of electricity in interstate commerce. Distribution Upgrades do not include Interconnection Facilities.

Effective Date shall mean the date on which this GIA becomes effective upon execution by the Parties subject to acceptance by the Commission, or if filed unexecuted, upon the date specified by the Commission.

Emergency Condition shall mean a condition or situation: (1) that in the reasonable judgment of the Party making the claim is imminently likely to endanger, or is contributing to the endangerment of, life, property, or public health and safety; or (2) that, in the case of either Transmission Provider or Transmission Owner, is imminently likely (as determined in a non-discriminatory manner) to cause a material adverse effect on the security of, or damage to the Transmission System, Transmission Owner's Interconnection Facilities or the electric systems of others to which the Transmission System is directly connected; or (3) that, in the case of Interconnection Customer, is imminently likely (as determined in a non-discriminatory manner) to cause a material adverse effect on the security of, or damage to, the Generating Facility or Interconnection Customer's Interconnection Facilities. System restoration and blackstart shall be considered Emergency Conditions; provided that Interconnection Customer is not obligated by this GIA to possess blackstart capability. Any condition or situation that results from lack of sufficient generating capacity to meet load requirements or that results solely from economic conditions shall not constitute an Emergency Condition, unless one of the enumerated conditions or situations identified in this definition also exists.

Energy Displacement Agreement shall mean an agreement between an Interconnection Customer with an Existing Generating Facility on the Transmission Provider's Transmission System and an Interconnection Customer with a proposed Generating Facility seeking to interconnect with Surplus Interconnection Service. The Energy Displacement Agreement specifies the term of operation, the Generating Facility Interconnection Service limit, and the mode of operation for energy production (common or singular operation).

Energy Resource Interconnection Service (ER Interconnection Service) shall mean an Interconnection Service that allows Interconnection Customer to connect its Generating Facility to the Transmission System or Distribution System, as applicable, to be eligible to deliver the Generating Facility's electric output using the existing firm or non-firm capacity of the Transmission System on an as available basis. Energy Resource Interconnection Service does not convey transmission service.

Engineering & Procurement (E&P) Agreement shall mean an agreement that authorizes Transmission Owner to begin engineering and procurement of long lead-time items necessary for the establishment of the interconnection in order to advance the implementation of the Interconnection Request.

Environmental Law shall mean Applicable Laws or Regulations relating to pollution or protection of the environment or natural resources.

Federal Holiday shall mean a Federal Reserve Bank holiday for a Party that has its principal place of business in the United States and a Canadian Federal or Provincial banking holiday for a Party that has its principal place of business located in Canada.

Federal Power Act shall mean the Federal Power Act, as amended, 16 U.S.C. §§ 791a *et seq*.

FERC shall mean the Federal Energy Regulatory Commission, also known as Commission, or its successor.

Force Majeure shall mean any act of God, labor disturbance, act of the public enemy, war, insurrection, riot, fire, storm or flood, explosion, breakage or accident to machinery or equipment, any order, regulation or restriction imposed by governmental, military or lawfully established civilian authorities, or any other cause beyond a Party's control. A Force Majeure event does not include an act of negligence or intentional wrongdoing by the Party claiming Force Majeure.

Generating Facility shall mean Interconnection Customer's device(s) for the production and/or storage for later injection of electricity identified in the Interconnection Request, but shall not include the Interconnection Customer's Interconnection Facilities. A Generating Facility consists of one or more generating unit(s) and/or storage device(s) which usually can operate independently and be brought online or taken offline individually.

Generating Facility Capacity shall mean the net capacity of the Generating Facility and the aggregate net capacity of the Generating Facility where it includes multiple energy production devices.

Generating Facility Modification shall mean modification to an Existing Generating Facility, including comparable replacement of only a portion of its equipment at the Existing Generating Facility.

Generating Facility Replacement shall mean replacement of one or more generating units and/or storage devices at the Existing Generating Facility with one or more new generating units or storage devices at the same electrical Point of Interconnection as the generating units and/or storage devices that is/are being decommissioned and electrically disconnected.

Generator Interconnection Agreement (GIA) shall mean the form of interconnection agreement, set forth herein.

Generator Interconnection Procedures (GIP) shall mean the interconnection procedures set forth in Attachment X of the Tariff.

Generator Upgrades shall mean the additions, modifications, and upgrades to the electric system of an Existing Generating Facility or of a higher queued Generating Facility at or beyond the Point of Interconnection to facilitate interconnection of the Generating Facility and render the Transmission Service necessary to affect Interconnection Customer's wholesale sale of electricity in interstate commerce.

Good Utility Practice shall mean any of the practices, methods and acts engaged in or approved by a significant portion of the electric industry during the relevant time period, or any of the practices, methods and acts which, in the exercise of reasonable judgment in light of the facts known at the time the decision was made, could have been expected to accomplish the desired result at a reasonable cost consistent with good business practices, reliability, safety and expedition. Good Utility Practice is not intended to be limited to the optimum practice, method, or act to the exclusion of all others, but rather to be acceptable practices, methods, or acts generally accepted in the region.

Governmental Authority shall mean any federal, state, local or other governmental regulatory or administrative agency, court, commission, department, board, or other governmental subdivision, legislature, rulemaking board, tribunal, or other governmental authority having jurisdiction over the Parties, their respective facilities, or the respective services they provide, and exercising or entitled to exercise any administrative, executive, police, or taxing authority or power; provided, however, that such term does not include Interconnection Customer, Transmission Provider, Transmission Owner, or any Affiliate thereof.

Group Study(ies) shall mean the process whereby more than one Interconnection Request is studied together, instead of serially, for the purpose of conducting one or more of the required Studies.

Hazardous Substances shall mean any chemicals, materials or substances defined as or included in the definition of "hazardous substances," "hazardous wastes," "hazardous materials," "hazardous constituents," "restricted hazardous materials," "extremely hazardous substances," "toxic substances," "radioactive substances," "contaminants," "pollutants," "toxic pollutants" or words of similar meaning and regulatory effect under any applicable Environmental Law, or any other chemical, material or substance, exposure to which is prohibited, limited or regulated by any applicable Environmental Law.

HVDC Facilities shall mean the high voltage direct current transmission facilities, including associated alternating current facilities, if any, that are subject to Section 27A of the Tariff and that are specifically identified in (i) any Agency Agreement pertaining to such facilities between Transmission Provider and Transmission Owner that owns or operates such facilities, or (ii) in any other arrangement that permits or will permit Transmission Provider to provide HVDC Service over such facilities as set forth in Section 27A of the Tariff.

HVDC Service shall mean Firm and Non-Firm Point-To-Point Transmission Service provided by Transmission Provider on HVDC Facilities pursuant to Section 27A of the Tariff.

Initial Synchronization Date shall mean the date upon which the Generating Facility is initially synchronized and upon which Trial Operation begins.

In-Service Date (ISD) shall mean the date upon which Interconnection Customer reasonably expects it will be ready to begin use of the Transmission Owner's Interconnection Facilities to obtain backfeed power.

Interconnection Customer shall mean any entity, including Transmission Provider, Transmission Owner or any of the Affiliates or subsidiaries of either, that proposes to interconnect its Generating Facility with the Transmission System.

Interconnection Customer's Interconnection Facilities (ICIF) shall mean all facilities and equipment, as identified in Appendix A of this GIA, that are located between the Generating Facility and the Point of Change of Ownership, including any modification, addition, or upgrades to such facilities and equipment necessary to physically and electrically interconnect the Generating Facility to the Transmission System or Distribution System, as applicable. Interconnection Customer's Interconnection Facilities are sole use facilities.

Interconnection Facilities shall mean the Transmission Owner's Interconnection Facilities and the Interconnection Customer's Interconnection Facilities. Collectively, Interconnection Facilities include all facilities and equipment between the Generating Facility and the Point of Interconnection, including any modification, additions or upgrades that are necessary to physically and electrically interconnect the Generating Facility to the Transmission System. Interconnection Facilities shall not include Distribution Upgrades, Generator Upgrades, Stand Alone Network Upgrades or Network Upgrades.

Interconnection Facilities Study shall mean a study conducted by Transmission Provider, or its agent, for Interconnection Customer to determine a list of facilities (including Transmission Owner's Interconnection Facilities, System Protection Facilities, and if such upgrades have been determined, Network Upgrades, Distribution Upgrades, Generator Upgrades, Common Use Upgrades, and upgrades on Affected Systems, as identified in the Interconnection System Impact Study), the cost of those facilities, and the time required to interconnect the Generating Facility with the Transmission System.

Interconnection Facilities Study Agreement shall mean the form of agreement contained in Appendix 4 of the Generator Interconnection Procedures for conducting the Interconnection Facilities Study.

Interconnection Request shall mean an Interconnection Customer's request, in the form of Appendix 1 to the Generator Interconnection Procedures, to interconnect a new Generating Facility, or to increase the capacity of, or make a Material Modification to the operating characteristics of, an Existing Generating Facility that is interconnected with the Transmission System.

Interconnection Service shall mean the service provided by Transmission Provider associated with interconnecting the Generating Facility to the Transmission System and enabling it to receive electric energy and capacity from the Generating Facility at the Point of Interconnection, pursuant to the terms of this GIA and, if applicable, the Tariff.

Interconnection Study (or Study) shall mean any of the studies described in the Generator Interconnection Procedures.

Interconnection Study Agreement shall mean the form of agreement contained in Attachment B to Appendix 1 of the Generator Interconnection procedures for conducting all studies required by the Generator Interconnection Procedures.

Interconnection System Impact Study shall mean an engineering study that evaluates the impact of the proposed interconnection on the safety and reliability of Transmission System and, if applicable, an Affected System. The study shall identify and detail the system impacts that would result if the Generating Facility were interconnected without project modifications or system modifications, or to study potential impacts, including but not limited to those identified in the Scoping Meeting as described in the Generator Interconnection Procedures.

IRS shall mean the Internal Revenue Service.

Local Balancing Authority shall mean an operational entity or a Joint Registration Organization which is (i) responsible for compliance with the subset of NERC Balancing Authority Reliability Standards defined in the Balancing Authority Agreement for their local area within the MISO Balancing Authority Area, (ii) a Party to Balancing Authority Agreement, excluding MISO, and (iii) provided in the Balancing Authority Agreement.

Loss shall mean any and all damages, losses, claims, including claims and actions relating to injury to or death of any person or damage to property, demand, suits, recoveries, costs and expenses, court costs, attorney fees, and all other obligations by or to third parties, arising out of or resulting from the other Party's performance, or non-performance of its obligations under this GIA on behalf of the indemnifying Party, except in cases of gross negligence or intentional wrongdoing, by the indemnified party.

Material Modification shall mean: (1) modification to an Interconnection Request in the queue, that has a material adverse impact on the cost or timing of any other Interconnection

Request with a later queue priority date; or (2) planned modification to an Existing Generating Facility, that is undergoing evaluation for a Generating Facility Modification or Generating Facility Replacement, and has a material adverse impact on the Transmission System with respect to: i) steady-state thermal or voltage limits, ii) dynamic system stability and response, or iii) short-circuit capability limit; compared to the impacts of the Existing Generating Facility prior to the modification or replacement.

Metering Equipment shall mean all metering equipment installed or to be installed at the Generating Facility pursuant to this GIA at the metering points, including but not limited to instrument transformers, MWh-meters, data acquisition equipment, transducers, remote terminal unit, communications equipment, phone lines, and fiber optics.

Monitoring and Consent Agreement shall mean an agreement that defines the terms and conditions applicable to a Generating Facility acquiring Surplus Interconnection Service. The Monitoring and Consent Agreement will list the roles and responsibilities of an Interconnection Customer seeking to interconnect with Surplus Interconnection Service and Transmission Owner to maintain the total output of the Generating Facility inside the parameters delineated in the GIA.

NERC shall mean the North American Electric Reliability Corporation or its successor organization.

Network Customer shall have that meaning as provided in the Tariff.

Network Resource shall mean any designated generating resource owned, purchased, or leased by a Network Customer under the Tariff. Network Resources do not include any resource, or any portion thereof, that is committed for sale to third parties or otherwise cannot be called upon to meet the Network Customer's Network Load on a non-interruptible basis.

Network Resource Interconnection Service (NR Interconnection Service) shall mean an Interconnection Service that allows Interconnection Customer to integrate its Generating Facility with the Transmission System in the same manner as for any Generating Facility being designated as a Network Resource. Network Resource Interconnection Service does not convey transmission service. Network Resource Interconnection Service shall include any network resource interconnection service established under an agreement with, or the tariff of, a Transmission Owner prior to integration into MISO, that is determined to be deliverable through the integration deliverability study process.

Network Upgrades shall mean the additions, modifications, and upgrades to the Transmission System required at or beyond the point at which the Interconnection Facilities connect to the Transmission System or Distribution System, as applicable, to accommodate the interconnection of the Generating Facility to the Transmission System. Network Upgrade shall not include any HVDC Facility Upgrades.

Notice of Dispute shall mean a written notice of a dispute or claim that arises out of or in connection with this GIA or its performance.

Operating Horizon Study shall mean an Interconnection System Impact Study that includes in service transmission and generation for an identified timeframe to determine either the available injection capacity of an Interconnection Request or Interconnection Facilities and/or Transmission System changes required for the requested Interconnection Service**.**

Optional Interconnection Study shall mean a sensitivity analysis based on assumptions specified by Interconnection Customer in the Optional Interconnection Study Agreement.

Optional Interconnection Study Agreement shall mean the form of agreement contained in Appendix 5 of the Generator Interconnection Procedures for conducting the Optional Interconnection Study.

Party or Parties shall mean Transmission Provider, Transmission Owner, Interconnection Customer, or any combination of the above.

Planning Horizon Study shall mean an Interconnection System Impact Study that includes a future year study to determine either the available injection capacity of an Interconnection Request or Interconnection Facilities and/or Transmission System changes required for the requested Interconnection Service.

Point of Change of Ownership (PCO) shall mean the point, as set forth in Appendix A to the Generator Interconnection Agreement, where the Interconnection Customer's Interconnection Facilities connect to the Transmission Owner's Interconnection Facilities.

Point of Interconnection (POI) shall mean the point, as set forth in Appendix A of the GIA, where the Interconnection Facilities connect to the Transmission System.

Provisional Generator Interconnection Agreement shall mean the interconnection agreement for Provisional Interconnection Service established between the Transmission Provider and/or the Transmission Owner and the Interconnection Customer as set forth in Section 7.9 of this Attachment X. This agreement shall take the form of the Generator Interconnection Agreement modified for provisional purposes. Unless otherwise provided in the GIP, all requirements applicable to the Generator Interconnection Agreement shall apply to the Provisional Generator Interconnection Agreement.

Provisional Interconnection Service shall mean interconnection service provided by the Transmission Provider associated with interconnecting the Interconnection Customer's Generating Facility to the Transmission Provider's Transmission System and enabling that Transmission System to receive electric energy and capacity from the Generating Facility at the Point of Interconnection, pursuant to the terms of the Provisional Generator Interconnection Agreement and the Tariff.

Provisional Interconnection Study shall mean an engineering study, performed at Interconnection Customer's request, as a condition to entering into a Provisional Generator Interconnection Agreement, that evaluates the impact of the proposed interconnection on the safety and reliability of the Transmission System and, if applicable, any Affected System. The

study shall identify and detail the impacts on the Transmission System and, if applicable, an Affected System, from stability, short circuit, and voltage issues that would result if the Generating Facility were interconnected without project modifications or system modifications.

Queue Position shall mean the order of a valid Interconnection Request, relative to all other pending valid Interconnection Requests. The Queue Position is established based upon the date and time of receipt of the valid Interconnection Request by Transmission Provider.

Reasonable Efforts shall have that meaning as provided in the Tariff.

Replacement Generating Facility shall mean a Generating Facility that replaces an Existing Generating Facility, or a portion thereof, at the same electrical Point of Interconnection pursuant to Section 3.7 of this Attachment X.

Scoping Meeting shall mean the meeting between representatives of Interconnection Customer, Transmission Owner, Affected System Operator(s) and Transmission Provider conducted for the purpose of discussing alternative interconnection options, to exchange information including any transmission data and earlier study evaluations that would be reasonably expected to impact such interconnection options, to analyze such information, and to determine the potential feasible Points of Interconnection.

Shared Network Upgrade shall mean a Network Upgrade or Common Use Upgrade that is funded by an Interconnection Customer(s), including when the Transmission Owner elects to fund the capital cost of such a Network Upgrade or Common Use Upgrade under Section 11.3 of the GIA, and also benefits other Interconnection Customer(s) that are later identified as beneficiaries.

Site Control shall mean a documented right for one or more parcels of land for the purpose of constructing a Generating Facility, Interconnection Customer's Interconnection Facilities, and, if applicable (*i.e.*, when the Interconnection Customer is providing the site for such facilities), the Transmission Owner's Interconnection Facilities and Network Upgrades at the POI that the Interconnection Customer will develop. Such documented right shall be one of the following: (1) ownership of a site; (2) a leasehold interest in a site; or (3) an option to purchase or acquire a leasehold interest in a site; or (4) any other contractual or legal right to possess or occupy a site.

Small Generating Facility shall mean a Generating Facility that has an aggregate net Generating Facility Capacity of no more than five MW and meets the requirements of Section 14 and Appendix 3 of the GIP.

Special Protection System (SPS) shall mean an automatic protection system or remedial action scheme designed to detect abnormal or predetermined system conditions, and take corrective actions other than and/or in addition to the isolation of faulted components, to maintain system reliability. Such action may include changes in demand (MW and MVar), energy (MWh and MVarh), or system configuration to maintain system stability, acceptable voltage, or power flows. An SPS does not include (a) underfrequency or undervoltage load

shedding, (b) fault conditions that must be isolated, (c) out-of-step relaying not designed as an integral part of an SPS, or (d) Transmission Control Devices.

Stand Alone Network Upgrades shall mean Network Upgrades, that are not part of an Affected System that an Interconnection Customer may construct without affecting day-to-day operations of the Transmission System during their construction. Transmission Provider, Transmission Owner and Interconnection Customer must agree as to what constitutes Stand Alone Network Upgrades and identify them in Appendix A to this GIA. If the Transmission Provider or Transmission Owner and Interconnection Customer disagree about whether a particular Network Upgrade is a Stand Alone Network Upgrade, the Transmission Provider or Transmission Owner that disagrees with the Interconnection Customer must provide the Interconnection Customer a written technical explanation outlining why the Transmission Provider or Transmission Owner does not consider the Network Upgrade to be a Stand Alone Network Upgrade within 15 days of its determination.

Surplus Interconnection Service shall mean any Interconnection Service that is derived from the unneeded portion of Interconnection Service established in a GIA or in agreement with, or under the tariff of, a Transmission Owner prior to integration into MISO, such that if Surplus Interconnection Service is utilized the total amount of Interconnection Service at the Point of Interconnection would remain the same.

System Protection Facilities shall mean the equipment, including necessary protection signal communications equipment, required to protect (1) the Transmission System or other delivery systems or other generating systems from faults or other electrical disturbances occurring at the Generating Facility and (2) the Generating Facility from faults or other electrical system disturbances occurring on the Transmission System or on other delivery systems or other generating systems to which the Transmission System is directly connected.

Tariff shall mean the Transmission Provider's Tariff through which open access transmission service and Interconnection Service are offered, as filed with the Commission, and as amended or supplemented from time to time, or any successor tariff.

Transmission Control Devices shall mean a generally accepted transmission device that is planned and designed to provide dynamic control of electric system quantities, and are usually employed as solutions to specific system performance issues. Examples of such devices include fast valving, high response exciters, high voltage DC links, active or real power flow control and reactive compensation devices using power electronics (*e.g.*, unified power flow controllers), static var compensators, thyristor controlled series capacitors, braking resistors, and in some cases mechanically-switched capacitors and reactors. In general, such systems are not considered to be Special Protection Systems.

Transmission Owner shall mean that Transmission Owner as defined in the Tariff, which includes an entity that owns, leases or otherwise possesses an interest in the portion of the Transmission System at which Interconnection Customer proposes to interconnect or otherwise integrate the operation of the Generating Facility. Transmission Owner should be read to include any Independent Transmission Company that manages the transmission facilities of

Transmission Owner and shall include, as applicable, the owner and/or operator of distribution facilities interconnected to the Transmission System, over which facilities transmission service or Wholesale Distribution Service under the Tariff is available at the time Interconnection Customer requests Interconnection Service and to which Interconnection Customer has requested interconnection of a Generating Facility for the purpose of either transmitting electric energy in interstate commerce or selling electric energy at wholesale in interstate commerce.

Transmission Provider shall mean the Midcontinent Independent System Operator, Inc. ("MISO"), the Regional Transmission Organization that controls or operates the transmission facilities of its transmission-owning members used for the transmission of electricity in interstate commerce and provides transmission service under the Tariff.

Transmission Owner's Interconnection Facilities (TOIF) shall mean all facilities and equipment owned by Transmission Owner from the Point of Change of Ownership to the Point of Interconnection as identified in Appendix A to this GIA, including any modifications, additions or upgrades to such facilities and equipment. Transmission Owner's Interconnection Facilities are sole use facilities and shall not include Distribution Upgrades, Generator Upgrades, Stand Alone Network Upgrades or Network Upgrades.

Transmission System shall mean the facilities owned by Transmission Owner and controlled or operated by Transmission Provider or Transmission Owner that are used to provide Transmission Service (including HVDC Service) or Wholesale Distribution Service under the Tariff.

Trial Operation shall mean the period during which Interconnection Customer is engaged in on-site test operations and commissioning of the Generating Facility prior to Commercial Operation.

Variable Energy Resource shall mean a device for the production of electricity that is characterized by an energy source that: (1) is renewable; (2) cannot be stored by the facility owner or operator; and (3) has variability that is beyond the control of the facility owner or operator.

Wholesale Distribution Service shall have that meaning as provided in the Tariff. Wherever the term "transmission delivery service" is used, Wholesale Distribution Service shall also be implied.

ARTICLE 2. EFFECTIVE DATE, TERM AND TERMINATION

2.1 Effective Date. This GIA shall become effective upon execution by the Parties subject to acceptance by FERC (if applicable), or if filed unexecuted, upon the date specified by FERC. Transmission Provider shall promptly file this GIA with FERC upon execution in accordance with Article 3.1, if required.

2.2 Term of Agreement. Subject to the provisions of Article 2.3, this GIA shall remain in effect for a period of thirty (30) years from the Effective Date and shall be automatically renewed for each successive one-year period thereafter on the anniversary of the Effective Date.

2.3 Termination Procedures. This GIA may be terminated as follows:

2.3.1 Written Notice. This GIA may be terminated by Interconnection Customer after giving Transmission Provider and Transmission Owner ninety (90) Calendar Days advance written notice. This GIA shall be terminated by Transmission Provider if the Generating Facility or a portion of the Generating Facility fails to achieve Commercial Operation by the Commercial Operation Date established in accordance with Section 4.4.4 of Attachment X, including any extension provided thereunder, or has ceased Commercial Operation for three (3) consecutive years, beginning with the last date of Commercial Operation for the Generating Facility, after giving Interconnection Customer ninety (90) Calendar Days advance written notice. Where only a portion of the Generating Facility fails to achieve Commercial Operation by the Commercial Operation Date established in accordance with Section 4.4.4 of Attachment X, including any extension provided thereunder, Transmission Provider shall only terminate that portion of the GIA. Notwithstanding the foregoing, in the limited circumstance that the Interconnection Request is served by a contingent Network Upgrade with an in-service date that is farther out than the Commercial Operation Date permitted under Section 4.4.4 of Attachment X, Transmission Provider shall only terminate this GIA for failure to achieve Commercial Operation by that later in-service date of the contingent Network Upgrade. The Generating Facility will not be deemed to have ceased Commercial Operation for purposes of this Article 2.3.1 if Interconnection Customer can document that it has taken other significant steps to maintain or restore operational readiness of the Generating Facility for the purpose of returning the Generating Facility to Commercial Operation as soon as possible.

2.3.1.1 Surplus Interconnection Service. Where this GIA provides for Surplus Interconnection Service and the Energy Displacement Agreement or the Monitoring and Consent Agreement required for Surplus Interconnection Service are no longer in effect, Interconnection Customer shall immediately cease Commercial Operation of the Generating Facility and this GIA shall be deemed terminated. In the event that the Existing Generating Facility retires and/or permanently ceases commercial

operation, the Surplus Interconnection Service provided under this GIA shall terminate except as provided in Section 3.3.1.3 of the GIP.

2.3.2 Default. Any Party may terminate this GIA in accordance with Article 17.

2.3.3 Notwithstanding Articles 2.3.1 and 2.3.2, no termination shall become effective until the Parties have complied with all Applicable Laws and Regulations applicable to such termination, including the filing with FERC of a notice of termination of this GIA, if required, which notice has been accepted for filing by FERC.

2.4 Termination Costs. If a Party elects to terminate this GIA pursuant to Article 2.3 above, each Party shall pay all costs incurred for which that Party is responsible (including any cancellation costs relating to orders or contracts for Interconnection Facilities, applicable upgrades, and related equipment) or charges assessed by the other Parties, as of the date of the other Parties' receipt of such notice of termination, under this GIA. In the event that this GIA is terminated by Interconnection Customer, Interconnection Customer is responsible for all financial impact that is caused as a result of this termination or the termination of Interconnection Customer's FCA(s) or MPFCA(s), such financial impact being determined through the financial impact analysis performed in accordance with Section 7.8 of Attachment X. In the event of termination by a Party, the Parties shall use commercially Reasonable Efforts to mitigate the costs, damages and charges arising as a consequence of termination. Upon termination of this GIA, unless otherwise ordered or approved by FERC:

2.4.1 With respect to any portion of the Transmission Owner's Interconnection Facilities, Network Upgrades, System Protection Facilities, Distribution Upgrades, Generator Upgrades, and if so determined and made a part of this GIA, upgrades on Affected Systems, that have not yet been constructed or installed, Transmission Owner shall to the extent possible and to the extent of Interconnection Customer's written notice under Article 2.3.1, cancel any pending orders of, or return, any materials or equipment for, or contracts for construction of, such facilities; provided that in the event Interconnection Customer elects not to authorize such cancellation, Interconnection Customer shall assume all payment obligations with respect to such materials, equipment, and contracts, and Transmission Owner shall deliver such material and equipment, and, if necessary, assign such contracts, to Interconnection Customer as soon as practicable, at Interconnection Customer's expense. To the extent that Interconnection Customer has already paid Transmission Owner for any or all such costs of materials or equipment not taken by Interconnection Customer or upgrades not yet constructed, Transmission Owner shall promptly transfer such amounts to Transmission Provider, less any costs, including penalties incurred by Transmission Owner to cancel any pending orders of or return such materials, equipment, or contracts. Transmission Provider will perform a financial impact analysis in accordance with Section 7.8 of Attachment X to determine the amount that should be refunded to Interconnection Customer. Transmission Provider shall

refund such remaining amounts to Interconnection Customer, less any financial impact caused by the termination of this GIA or Interconnection Customer's FCA(s) or MPFCA(s), as determined through the analysis performed in accordance with Section 7.8 of Attachment X. If Interconnection Customer made its payment(s) through a letter of credit, surety bond, or parental guarantee, Transmission Owner will draw against that letter of credit, surety bond, or parental guarantee in an amount determined through the analysis performed in accordance with Section 7.8 of Attachment X, and transfer that amount to Transmission Provider, unless Interconnection Customer funds the financial impact through another means.

If an Interconnection Customer terminates this GIA, it shall be responsible for all costs incurred in association with that Interconnection Customer's interconnection, including any cancellation costs relating to orders or contracts for Interconnection Facilities and equipment, and other expenses including any upgrades or related equipment for which Transmission Owner has incurred expenses and has not been reimbursed by Interconnection Customer.

2.4.2 Transmission Owner may, at its option, retain any portion of such materials, equipment, or facilities that Interconnection Customer chooses not to accept delivery of, in which case Transmission Owner shall be responsible for all costs associated with procuring such materials, equipment, or facilities. If Transmission Owner does not so elect, then Interconnection Customer shall be responsible for such costs.

2.4.3 With respect to any portion of the Interconnection Facilities, and any other facilities already installed or constructed pursuant to the terms of this GIA, Interconnection Customer shall be responsible for all costs associated with the removal, relocation, reconfiguration or other disposition or retirement of such materials, equipment, or facilities, and such other expenses actually incurred by Transmission Owner necessary to return the Transmission, Distribution or Generator System, as applicable, to safe and reliable operation.

2.5 **Disconnection**. Upon termination of this GIA, the Parties will take all appropriate steps to disconnect the Generating Facility from the Transmission or Distribution System, as applicable. All costs required to effectuate such disconnection shall be borne by the terminating Party, unless such termination resulted from the non-terminating Party's Default of this GIA or such non-terminating Party otherwise is responsible for these costs under this GIA.

2.6 **Survival.** This GIA shall continue in effect after termination to the extent necessary to provide for final billings and payments and for costs incurred hereunder, including billings and payments pursuant to this GIA; to permit the determination and enforcement of liability and indemnification obligations arising from acts or events that occurred while this GIA was in effect; and to permit each Party to have access to the lands of the other

Party pursuant to this GIA or other applicable agreements, to disconnect, remove or salvage its own facilities and equipment.

ARTICLE 3. REGULATORY FILINGS

3.1 Filing. Transmission Provider shall file this GIA (and any amendment hereto) with the appropriate Governmental Authority, if required. A Party may request that any information so provided be subject to the confidentiality provisions of Article 22. If that Party has executed this GIA, or any amendment thereto, the Party shall reasonably cooperate with Transmission Provider with respect to such filing and to provide any information reasonably requested by Transmission Provider needed to comply with applicable regulatory requirements.

ARTICLE 4. SCOPE OF SERVICE

4.1 Interconnection Product Options. Interconnection Customer has selected the following (checked) type of Interconnection Service:

Check: _____ NZ or _X_ ER and/or ____ NR (See Appendix A for details)

4.1.1 Energy Resource Interconnection Service (ER Interconnection Service).

4.1.1.1 The Product. ER Interconnection Service allows Interconnection Customer to connect the Generating Facility to the Transmission or Distribution System, as applicable, and be eligible to deliver the Generating Facility's output using the existing firm or non-firm capacity of the Transmission System on an "as available" basis. To the extent Interconnection Customer wants to receive ER Interconnection Service, Transmission Owner shall construct facilities consistent with the studies identified in Appendix A.

An Interconnection Customer seeking ER Interconnection Service for new or added capacity at a Generating Facility may be granted conditional ER Interconnection Service status to the extent there is such capacity available on the Transmission System to accommodate the Interconnection Customer's Generating Facility. At the request of Interconnection Customer, conditional ER Interconnection Service status may be granted subject to the system being able to accommodate the interconnection without upgrades, until such time as a higher queued project(s) with a later service date affecting the same common elements is placed into service. The conditional ER Interconnection Service shall be terminated in the event Interconnection Customer fails to fund the necessary studies and the Network Upgrades necessary to grant the Interconnection Customer's ER Interconnection Service upon the completion of higher queued projects involving the same common elements.

4.1.1.2 Transmission Delivery Service Implications. Under ER Interconnection Service, Interconnection Customer will be eligible to inject power from the Generating Facility into and deliver power across the Transmission System on an "as available" basis up to the amount of MW identified in the applicable stability and steady state studies to the extent the upgrades initially required to qualify for ER Interconnection Service have been constructed. After that date FERC makes effective MISO's Energy Market Tariff filed in Docket No. ER04-691-000, Interconnection Customer may place a bid to sell into the market up to the maximum identified Generating Facility output, subject to any conditions specified in the Interconnection Service approval, and the Generating Facility will be dispatched to the extent the Interconnection Customer's bid clears. In all other instances, no transmission or other delivery service from the Generating Facility is assured, but Interconnection Customer may obtain Point-To-Point Transmission Service, Network Integration Transmission Service or be used for secondary network transmission service, pursuant to the Tariff, up to the maximum output identified in the stability and steady state studies. In those instances, in order for Interconnection Customer to obtain the right to deliver or inject energy beyond the Point of Interconnection or to improve its ability to do so, transmission delivery service must be obtained pursuant to the provisions of the Tariff. The Interconnection Customer's ability to inject its Generating Facility output beyond the Point of Interconnection, therefore, will depend on the existing capacity of the Transmission or Distribution System as applicable, at such time as a Transmission Service request is made that would accommodate such delivery. The provision of Firm Point-To-Point Transmission Service or Network Integration Transmission Service may require the construction of additional Network or Distribution Upgrades.

4.1.2 Network Resource Interconnection Service (NR Interconnection Service).

4.1.2.1 The Product. Transmission Provider must conduct the necessary studies and Transmission Owner shall construct the facilities identified in Appendix A of this GIA, subject to the approval of Governmental Authorities, needed to integrate the Generating Facility in the same manner as for any Generating Facility being designated as a Network Resource.

4.1.2.2 Transmission Delivery Service Implications. NR Interconnection Service allows the Generating Facility to be designated by any Network Customer under the Tariff on the Transmission System as a Network Resource, up to the Generating Facility's full output, on the same basis as existing Network Resources that are interconnected to the Transmission or Distribution System, as applicable, and to be studied as a Network Resource on the assumption that such a designation will

occur. Although NR Interconnection Service does not convey a reservation of Transmission Service, any Network Customer can utilize Network Integration Transmission Service under the Tariff to obtain delivery of energy from the Generating Facility in the same manner as it accesses Network Resources. A Generating Facility receiving NR Interconnection Service may also be used to provide Ancillary Services after technical studies and/or periodic analyses are performed with respect to the Generating Facility's ability to provide any applicable Ancillary Services, provided that such studies and analyses have been or would be required in connection with the provision of such Ancillary Services by any existing Network Resource. However, if the Generating Facility has not been designated as a Network Resource by any Network Customer, it cannot be required to provide Ancillary Services except to the extent such requirements extend to all generating facilities that are similarly situated. The provision of Network Integration Transmission Service or Firm Point-To-Point Transmission Service may require additional studies and the construction of additional upgrades. Because such studies and upgrades would be associated with a request for delivery service under the Tariff, cost responsibility for the studies and upgrades would be in accordance with FERC's policy for pricing transmission delivery services.

NR Interconnection Service does not necessarily provide Interconnection Customer with the capability to physically deliver the output of its Generating Facility to any particular load on the Transmission System without incurring congestion costs. In the event of transmission or distribution constraints on the Transmission or Distribution System, as applicable, the Generating Facility shall be subject to the applicable congestion management procedures in the Transmission System in the same manner as Network Resources.

There is no requirement either at the time of study or interconnection, or at any point in the future, that the Generating Facility be designated as a Network Resource by a Network Customer or that Interconnection Customer identify a specific buyer (or sink). To the extent a Network Customer does designate the Generating Facility as a Network Resource, it must do so pursuant to the Tariff.

Once an Interconnection Customer satisfies the requirements for obtaining NR Interconnection Service, any future Transmission Service request for delivery from the Generating Facility within the Transmission System of any amount of capacity and/or energy, up to the amount initially studied, will not require that any additional studies be performed or that any further upgrades associated with such Generating Facility be undertaken, regardless of whether such Generating Facility is ever designated by a Network Customer as a Network Resource and

regardless of changes in ownership of the Generating Facility. To the extent Interconnection Customer enters into an arrangement for long term Transmission Service for deliveries from the Generating Facility to customers other than the studied Network Customers, or for any Point-To-Point Transmission Service, such request may require additional studies and upgrades in order for Transmission Provider to grant such request. However, the reduction or elimination of congestion or redispatch costs may require additional studies and the construction of additional upgrades.

To the extent Interconnection Customer enters into an arrangement for long term Transmission Service for deliveries from the Generating Facility outside the Transmission System, such request may require additional studies and upgrades in order for Transmission Provider to grant such request.

4.1.2.3 Conditional NR Interconnection Service. An Interconnection Customer seeking NR Interconnection Service for new or added capacity at a Generating Facility may be granted conditional NR Interconnection Service status to the extent there is such capacity available on the Transmission System to accommodate the Interconnection Customer's Generating Facility. At the request of Interconnection Customer, conditional NR Interconnection Service status may be granted subject to the system being able to accommodate the interconnection without upgrades, until such time as higher queued project(s) with a later service date affecting the same common elements is placed into service. The conditional NR Interconnection Service status may be converted to ER Interconnection Service if either of the following occurs:

1) Interconnection Customer fails to fund necessary studies and Network Upgrades required to allow the Interconnection Customer's Generating Facility to receive NR Interconnection Service upon the completion of higher queued projects involving the same common elements; or

2) The higher queued project(s) or planned and required Network Upgrades are placed in service and the Network Upgrades required to provide NR Interconnection Service status to the Interconnection Customer's Generating Facility are not in service.

In the event Interconnection Customer fails to fund the necessary studies and Network Upgrades for NR Interconnection Service, the Interconnection Customer's conditional NR Interconnection Service status shall be converted to ER Interconnection Service status unless Interconnection Customer makes a new Interconnection Request. Such

new Interconnection Request shall be evaluated in accordance with the GIP and its new queue position.

Some or all of the conditional NR Interconnection Service status may be temporarily revoked if the Network Upgrades are not in service when the higher queued project(s) are placed in service. The availability of conditional NR Interconnection Service status will be determined by Transmission Provider's studies. Upon funding and completion of the Network Upgrades required to establish the Generating Facility's NR Interconnection Service status, the Generating Facility will be granted NR Interconnection Service status.

The Parties agree that the portion of the Generating Facility classified as NR Interconnection Service is the first portion of the output of the combined output of all the units at the Generating Facility except in circumstances where Interconnection Customer otherwise elects this GIA, as amended, to allocate that portion to the output of specific unit(s) at the Generating Facility, the total of which will not exceed the output eligible for NR Interconnection Service as shown by the additional studies. To the extent Interconnection Customer desires to obtain NR Interconnection Service for any portion of the Generating Facility in addition to that supported by such additional studies, Interconnection Customer will be required to request such additional NR Interconnection Service through a separate Interconnection Request in accordance with the GIP.

4.1.3 Surplus Interconnection Service.

4.1.3.1 The Product. Surplus Interconnection Service is restricted Interconnection Service that allows an Interconnection Customer to increase the gross generating capability at the same Point of Interconnection of an Existing Generating Facility without increasing the total amount of Interconnection Service at the Point of Interconnection.

4.1.3.2 Transmission Delivery Service Implications. Surplus Interconnection Service does not convey any right to deliver electricity to any specific customer or Point of Delivery.

4.2 Provision of Service. Transmission Provider shall provide Interconnection Service for the Generating Facility at the Point of Interconnection.

4.3 Performance Standards. Each Party shall perform all of its obligations under this GIA in accordance with Applicable Laws and Regulations, Applicable Reliability Standards, and Good Utility Practice. To the extent a Party is required or prevented or limited in taking any action by such regulations and standards, or if the obligations of any Party may become limited by a change in Applicable Laws and Regulations, Applicable Reliability Standards, and Good Utility Practice after the execution of this GIA, that

Party shall not be deemed to be in Breach of this GIA for its compliance therewith. The Party so limited shall notify the other Parties whereupon Transmission Provider shall amend this GIA in concurrence with the other Parties and submit the amendment to the Commission for approval.

4.4 No Transmission Delivery Service. The execution of this GIA does not constitute a request for, or the provision of, any transmission delivery service under the Tariff, and does not convey any right to deliver electricity to any specific customer or Point of Delivery.

4.5 Interconnection Customer Provided Services. The services provided by Interconnection Customer under this GIA are set forth in Article 9.6 and Article 13.4.1. Interconnection Customer shall be paid for such services in accordance with Article 11.7.

ARTICLE 5. INTERCONNECTION FACILITIES ENGINEERING, PROCUREMENT, AND CONSTRUCTION

5.1 Options. Unless otherwise mutually agreed to between the Parties, Interconnection Customer shall select: 1) the In-Service Date, Initial Synchronization Date, and Commercial Operation Date based on a reasonable construction schedule that will allow sufficient time for design, construction, equipment procurement, and permit acquisition of Transmission System equipment or right-of-way; and 2) either the Standard Option or Alternate Option set forth below and such dates and selected option shall be set forth in Appendix B. If the dates designated by Interconnection Customer are not acceptable to Transmission Owner, Transmission Owner shall so notify Interconnection Customer within thirty (30) Calendar Days. Upon receipt of the notification that Interconnection Customer's designated dates are not acceptable to Transmission Owner, the Interconnection Customer shall notify Transmission Owner within thirty (30) Calendar Days whether it elects to exercise the Option to Build if it has not already elected to exercise the Option to Build.

5.1.1 Standard Option. Transmission Owner shall design, procure, and construct the Transmission Owner's Interconnection Facilities, Network Upgrades, System Protection Facilities, Distribution Upgrades, and Generator Upgrades using Reasonable Efforts to complete the Transmission Owner's Interconnection Facilities, Network Upgrades, System Protection Facilities, Distribution Upgrades and Generator Upgrades by the dates set forth in Appendix B, Milestones, subject to the receipt of all approvals required from Governmental Authorities and the receipt of all land rights necessary to commence construction of such facilities, and such other permits or authorizations as may be required. Transmission Provider or Transmission Owner shall not be required to undertake any action which is inconsistent with its standard safety practices, its material and equipment specifications, its design criteria and construction procedures, its labor agreements, Applicable Laws and Regulations and Good Utility Practice. In the event Transmission Owner reasonably expects that it will not be able to complete the Transmission Owner's Interconnection Facilities, Network Upgrades, System

Protection Facilities, Distribution Upgrades and Generator Upgrades by the specified dates, Transmission Owner shall promptly provide written notice to Interconnection Customer and Transmission Provider and shall undertake Reasonable Efforts to meet the earliest dates thereafter.

5.1.2 Alternate Option. If the dates designated by Interconnection Customer are acceptable to Transmission Provider and Transmission Owner, Transmission Provider shall so notify Interconnection Customer within thirty (30) Calendar Days, and Transmission Owner shall assume responsibility for the design, procurement and construction of the Transmission Owner's Interconnection Facilities by the designated dates.

If Transmission Owner subsequently fails to complete the Transmission Owner's Interconnection Facilities by the In-Service Date, to the extent necessary to provide back feed power; or fails to complete Network Upgrades by the Initial Synchronization Date to the extent necessary to allow for Trial Operation at full power output, unless other arrangements are made by the Parties for such Trial Operation; or fails to complete the Network Upgrades by the Commercial Operation Date, as such dates are reflected in Appendix B, Milestones; Transmission Owner shall pay Interconnection Customer liquidated damages in accordance with Article 5.3, Liquidated Damages, provided, however, the dates designated by Interconnection Customer shall be extended day for day for each Calendar Day that Transmission Provider refuses to grant clearances to install equipment.

Transmission Owner and Interconnection Customer may adopt an incentive payment schedule that is mutually agreeable to encourage Transmission Owner to meet specified accelerated dates. Such payment by Interconnection Customer is not subject to refund.

5.1.3 Option to Build. Pursuant to Section 7.3.2.5 & 7.3.3.5 of the GIP, Interconnection Customer shall have the option to assume responsibility for the design, procurement and construction of the Transmission Owner's Interconnection Facilities and Stand Alone Network Upgrades by the dates originally designated by Interconnection Customer under Article 5.1.2. The Parties must agree as to what constitutes Stand Alone Network Upgrades and identify such Stand Alone Network Upgrades in Appendix A. Except for Stand Alone Network Upgrades, Interconnection Customer shall have no right to construct Network Upgrades under this option.

5.1.4 Negotiated Option. If the dates designated by Interconnection Customer pursuant to Article 5.1 are not acceptable to Transmission Owner, the Parties shall in good faith attempt to negotiate terms and conditions (including revision of the specified dates and liquidated damages, the provision of incentives, or the procurement and construction of all facilities other than Transmission Owner's

Interconnection Facilities and Stand Alone Network Upgrades if the Interconnection Customer elects to exercise the Option to Build under Article 5.1.3). If the Parties are unable to reach agreement on such terms and conditions, then, pursuant to Article 5.1.1 (Standard Option), Transmission Owner shall assume responsibility for the design, procurement and construction of all facilities other than Transmission Owner's Interconnection Facilities and Stand Alone Network Upgrades if the Interconnection Customer has elected to exercise the Option to Build.

Transmission Owner and Interconnection Customer may adopt an incentive payment schedule that is mutually agreeable to encourage Transmission Owner to meet specified accelerated dates. Such payment by Interconnection Customer is not subject to refund.

5.2 General Conditions Applicable to Option to Build. If Interconnection Customer assumes responsibility for the design, procurement and construction of the Transmission Owner's Interconnection Facilities and Stand Alone Network Upgrades after receipt of all required approvals from Governmental Authorities necessary to commence construction,

(1) Interconnection Customer shall engineer, procure equipment, and construct the Transmission Owner's Interconnection Facilities and Stand Alone Network Upgrades (or portions thereof) using Good Utility Practice and using standards and specifications provided in advance by Transmission Owner, or as required by any Governmental Authority;

(2) Interconnection Customer's engineering, procurement and construction of the Transmission Owner's Interconnection Facilities and Stand Alone Network Upgrades shall comply with all requirements of law or Governmental Authority to which Transmission Owner would be subject in the engineering, procurement or construction of the Transmission Owner's Interconnection Facilities and Stand Alone Network Upgrades;

(3) Transmission Provider, at Transmission Provider's option, and Transmission Owner shall be entitled to review and approve the engineering design, equipment acceptance tests(including witnessing of acceptance tests), and the construction (including monitoring of construction) of the Transmission Owner's Interconnection Facilities and Stand Alone Network Upgrades, and shall have the right to reject any design, procurement, construction or acceptance test of any equipment that does not meet the standards and specifications of Transmission Provider, Transmission Owner and any Governmental Authority;

(4) prior to commencement of construction, Interconnection Customer shall provide to Transmission Provider and Transmission Owner a schedule for construction of the Transmission Owner's Interconnection Facilities and Stand

Alone Network Upgrades, and shall promptly respond to requests for information from Transmission Provider and Transmission Owner;

(5) at any time during construction, Transmission Provider and Transmission Owner shall have unrestricted access to the construction site for the Transmission Owner's Interconnection Facilities and Stand Alone Network Upgrades and to conduct inspections of the same;

(6) at any time during construction, should any phase of the engineering, equipment procurement, or construction of the Transmission Owner's Interconnection Facilities and Stand Alone Network Upgrades not meet the standards and specifications provided by Transmission Owner, Interconnection Customer shall be obligated to remedy deficiencies in that portion of the Transmission Owner's Interconnection Facilities and Stand Alone Network Upgrades to meet the standards and specifications provided by Transmission Provider and Transmission Owner;

(7) Interconnection Customer shall indemnify Transmission Provider and Transmission Owner for claims arising from the Interconnection Customer's construction of the Transmission Owner's Interconnection Facilities and Stand Alone Network Upgrades under the terms and procedures applicable to Article 18.1, Indemnity;

(8) Interconnection Customer shall transfer control of the Transmission Owner's Interconnection Facilities and Stand Alone Network Upgrades to Transmission Owner;

(9) Unless Parties otherwise agree, Interconnection Customer shall transfer ownership of the Transmission Owner's Interconnection Facilities and Stand Alone Network Upgrades to Transmission Owner in accordance with Appendix B;

(10) Transmission Provider, at Transmission Provider's option, and Transmission Owner shall approve and accept for operation and maintenance the Transmission Owner's Interconnection Facilities and Stand Alone Network Upgrades to the extent engineered, procured, and constructed in accordance with this Article 5.2 only if the Transmission Owner's Interconnection Facilities and Stand Alone Network Upgrades meet the standards and specifications of Transmission Provider, Transmission Owner and any Governmental Authority.

(11) Interconnection Customer shall deliver to Transmission Owner "as-built" drawings, information, and any other documents that are reasonably required by Transmission Owner to assure that the Interconnection Facilities and Stand-Alone Network Upgrades are built to the standards and specifications required by Transmission Owner.

(12) If Interconnection Customer exercises the Option to Build pursuant to Article 5.1.3, Interconnection Customer shall pay Transmission Owner the agreed upon amount of [$ PLACEHOLDER] for Transmission Owner to execute the responsibilities enumerated to Transmission Owner under Article 5.2. Transmission Owner shall invoice Interconnection Customer for this total amount to be divided on a monthly basis pursuant to Article 12.

(13) If Interconnection Customer exercises the Option to Build pursuant to Article 5.1.3, and the Transmission Owner has elected to fund the costs of Network Upgrades pursuant to Article 11.3, then prior to Interconnection Customer incurring any construction costs relating to the Option to Build and by the date specified in Appendix B, Interconnection Customer shall invoice the Transmission Owner for the estimated amount to be expended by the Interconnection Customer to construct any Stand Alone Network Upgrades for which the Interconnection Customer has exercised its Option to Build in accordance with Appendix B. The Transmission Owner shall be required to reimburse Interconnection Customer for the full amount of such invoiced costs by the date specified in Appendix B, which shall be prior to the date by which Interconnection Customer must make any construction payment for such Stand Alone Network Upgrades. After completion of the construction of Stand Alone Network Upgrades by the Interconnection Customer and by the date specified in Appendix B for the Interconnection Customer to transfer such Stand Alone Network Upgrades to the Transmission Owner, Interconnection Customer shall provide an invoice of the final cost of the construction of Stand Alone Upgrades and shall set forth such costs in sufficient detail to enable the Transmission Owner to compare the actual costs with the estimates and to ascertain deviations, if any, from the cost estimates. In the event that the actual costs exceed the estimated costs previously invoiced by Interconnection Customer and paid by Transmission Owner, Transmission Owner shall pay to Interconnection Customer the difference between the amount previously paid and the actual costs within thirty (30) Calendar Days after receipt of a final construction invoice from Interconnection Customer. In the event that the actual costs are less than the estimated costs previously invoiced by Interconnection Customer and paid by Transmission Owner, Interconnection Customer shall refund, with interest (calculated in accordance with 18 C.F.R. Section 35.19a(a)(2)(iii), to Transmission Owner any amount by which the actual payment by Transmission Owner for estimated costs exceeds the actual costs of construction within thirty (30) Calendar Days of the issuance of such final construction invoice. Following the transfer of the Stand Alone Network Upgrades from the Interconnection Customer to the Transmission Owner, the Interconnection Customer shall make payments for such facilities to the Transmission Owner pursuant to an agreement between and among the Parties.

5.3 Liquidated Damages. The actual damages to Interconnection Customer, in the event the Transmission Owner's Interconnection Facilities or Network Upgrades are not completed by the dates designated by Interconnection Customer and accepted by Transmission

Provider and Transmission Owner pursuant to subparagraphs 5.1.2 or 5.1.4, above, may include Interconnection Customer's fixed operation and maintenance costs and lost opportunity costs. Such actual damages are uncertain and impossible to determine at this time. Because of such uncertainty, any liquidated damages paid by Transmission Owner to Interconnection Customer in the event that Transmission Owner does not complete any portion of the Transmission Owner's Interconnection Facilities or Network Upgrades by the applicable dates, shall be an amount equal to ½ of 1 percent per day of the actual cost of the Transmission Owner's Interconnection Facilities and Network Upgrades, in the aggregate, for which Transmission Owner has assumed responsibility to design, procure and construct.

However, in no event shall the total liquidated damages exceed 20 percent of the actual cost of the Transmission Owner's Interconnection Facilities and Network Upgrades for which Transmission Owner has assumed responsibility to design, procure, and construct. The foregoing payments will be made by Transmission Owner to Interconnection Customer as just compensation for the damages caused to Interconnection Customer, which actual damages are uncertain and impossible to determine at this time, and as reasonable liquidated damages, but not as a penalty or a method to secure performance of this GIA. Liquidated damages, when the Parties agree to them, are the exclusive remedy for the Transmission Owner's failure to meet its schedule.

No liquidated damages shall be paid to Interconnection Customer if: (1) Interconnection Customer is not ready to commence use of the Transmission Owner's Interconnection Facilities or Network Upgrades to take the delivery of power for the Generating Facility's Trial Operation or to export power from the Generating Facility on the specified dates, unless Interconnection Customer would have been able to commence use of the Transmission Owner's Interconnection Facilities or Network Upgrades to take the delivery of power for Generating Facility's Trial Operation or to export power from the Generating Facility, but for Transmission Owner's delay; (2) the Transmission Owner's failure to meet the specified dates is the result of the action or inaction of Transmission Provider, Interconnection Customer or any other earlier queued Interconnection Customer who has entered into an earlier GIA with Transmission Provider and/or a Transmission Owner or with an Affected System Operator, or any cause beyond Transmission Owner's reasonable control or reasonable ability to cure;
(3) Interconnection Customer has assumed responsibility for the design, procurement and construction of the Transmission Owner's Interconnection Facilities and Stand Alone Network Upgrades; (4) the delay is due to the inability of Transmission Owner to obtain all required approvals from Governmental Authorities in a timely manner for the construction of any element of the Interconnection Facilities, Network Upgrades or Stand Alone Network Upgrades, or any other permit or authorization required, or any land rights or other private authorizations that may be required, and Transmission Owner has exercised Reasonable Efforts in procuring such approvals, permits, rights or authorizations; or (5) the Parties have otherwise agreed.

5.4 Power System Stabilizers. Interconnection Customer shall procure, install, maintain and operate power system stabilizers in accordance with the guidelines and procedures

established by the Applicable Reliability Council. Transmission Provider and Transmission Owner reserve the right to reasonably establish minimum acceptable settings for any installed power system stabilizers, subject to the design and operating limitations of the Generating Facility. If the Generating Facility's power system stabilizers are removed from service or are not capable of automatic operation, Interconnection Customer shall immediately notify the Transmission Provider's system operator, or its designated representative. The requirements of this paragraph shall not apply to induction generators.

5.5 Equipment Procurement. If responsibility for construction of the Transmission Owner's Interconnection Facilities, Network Upgrades and/or Distribution Upgrades is to be borne by Transmission Owner, then Transmission Owner shall commence design of the Transmission Owner's Interconnection Facilities, Network Upgrades and/or Distribution Upgrades, and procure necessary equipment as soon as practicable after all of the following conditions are satisfied, unless the Parties otherwise agree in writing:

5.5.1 Transmission Provider has completed the Interconnection Facilities Study pursuant to the Interconnection Facilities Study Agreement; and

5.5.2 Where applicable, Interconnection Customer has provided security to Transmission Owner in accordance with Article 11.6 by the dates specified in Appendix B, Milestones.

5.6 Construction Commencement. Transmission Owner shall commence construction of the Transmission Owner's Interconnection Facilities, Network Upgrades, Transmission Owner's System Protection Facilities, Distribution Upgrades, and Generator Upgrades for which it is responsible as soon as practicable after the following additional conditions are satisfied:

5.6.1 Approval of the appropriate Governmental Authority has been obtained for any facilities requiring regulatory approval; and

5.6.2 Where applicable, Interconnection Customer has provided security to Transmission Owner in accordance with Article 11.6 by the dates specified in Appendix B, Milestones.

5.7 Work Progress. Transmission Owner and Interconnection Customer will keep each other and Transmission Provider advised periodically as to the progress of their respective design, procurement and construction efforts. Either Transmission Owner or Interconnection Customer may, at any time, request a progress report from the other, with a copy to be provided to the other Parties. If, at any time, Interconnection Customer determines that the completion of the Transmission Owner's Interconnection Facilities, Network Upgrades, or Transmission Owner's System Protection Facilities will not be required until after the specified In-Service Date, Interconnection Customer will provide written notice to Transmission Provider and Transmission Owner of such later date upon which the completion of the Transmission Owner's Interconnection Facilities, Network

Upgrades or Transmission Owner's System Protection Facilities will be required. Transmission Owner may delay the In-Service Date of its facilities accordingly.

5.8 Information Exchange. As soon as reasonably practicable after the Effective Date, the Parties shall exchange information regarding the design and compatibility of the Interconnection Facilities and compatibility of the Interconnection Facilities with the Transmission System or Distribution System, as applicable, and shall work diligently and in good faith to make any necessary design changes.

5.9 Other Interconnection Options.

5.9.1 Limited Operation. If any of the Transmission Owner's Interconnection Facilities, Network Upgrades, or Transmission Owner's System Protection Facilities, Distribution Upgrades or Generator Upgrades are not reasonably expected to be completed prior to the Commercial Operation Date of the Generating Facility, Transmission Provider shall, upon the request and at the expense of Interconnection Customer, perform operating studies on a timely basis to determine the extent to which the Generating Facility and the Interconnection Customer's Interconnection Facilities may operate prior to the completion of the Transmission Owner's Interconnection Facilities, Network Upgrades, Transmission Owner's System Protection Facilities, Distribution Upgrades or Generator Upgrades consistent with Applicable Laws and Regulations, Applicable Reliability Standards, Good Utility Practice, and this GIA. Transmission Provider and Transmission Owner shall permit Interconnection Customer to operate the Generating Facility and the Interconnection Customer's Interconnection Facilities in accordance with the results of such studies; provided, however, such studies reveal that such operation may occur without detriment to the Transmission System as then configured and in accordance with the safety requirements of Transmission Owner and any Governmental Authority.

The maximum permissible output of the Generating Facility will be updated on a quarterly basis if the Network Upgrades necessary for the interconnection of the Generating Facility pursuant to this GIA are not in service within six (6) months following the Commercial Operation Date of the Generating Facility as specified in Appendix B of this GIA. These quarterly studies will be performed using the same methodology set forth in Section 11.5 of the GIP. These quarterly updates will end when all Network Upgrades necessary for the interconnection of the Generating Facility pursuant to this GIA are in service.

5.9.2 Provisional Interconnection Service.
Upon the request of Interconnection Customer, and prior to completion of requisite Interconnection Facilities, Network Upgrades, Distribution Upgrades, or System Protection Facilities Transmission Provider may execute a Provisional Generator Interconnection Agreement or Interconnection Customer may request the filing of an unexecuted Provisional Generator Interconnection Agreement with the Interconnection Customer for limited interconnection service at the

discretion of Transmission Provider based upon an evaluation that will consider the results of available studies. Transmission Provider shall determine, through available studies or additional studies as necessary, whether stability, short circuit, thermal, and/or voltage issues would arise if Interconnection Customer interconnects without modifications to the Generating Facility or Transmission Provider's system. Transmission Provider shall determine whether any Interconnection Facilities, Network Upgrades, Distribution Upgrades, or System Protection Facilities that are necessary to meet the requirements of NERC, or any applicable Regional Entity for the interconnection of a new, modified and/or expanded Generating Facility are in place prior to the commencement of interconnection service from the Generating Facility. Where available studies indicate that such Interconnection Facilities, Network Upgrades, Distribution Upgrades, and/or System Protection Facilities that are required for the interconnection of a new, modified and/or expanded Generating Facility are not currently in place, Transmission Provider will perform a study, at the Interconnection Customer's expense, to confirm the facilities that are required for Provisional Interconnection Service. The maximum permissible output of the Generating Facility in the Provisional Generator Interconnection Agreement shall be studied and updated on a quarterly basis. Interconnection Customer assumes all risk and liabilities with respect to changes between the Provisional Generator Interconnection Agreement and the Generator Interconnection Agreement, including changes in output limits and Interconnection Facilities, Network Upgrades, Distribution Upgrades, and/or System Protection Facilities cost responsibilities.

5.10 Interconnection Customer's Interconnection Facilities. Interconnection Customer shall, at its expense, design, procure, construct, own and install the ICIF, as set forth in Appendix A.

5.10.1 Interconnection Customer's Interconnection Facility Specifications. Interconnection Customer shall submit initial design and specifications for the ICIF, including Interconnection Customer's System Protection Facilities, to Transmission Provider and Transmission Owner at least one hundred eighty (180) Calendar Days prior to the Initial Synchronization Date; and final design and specifications for review and comment at least ninety (90) Calendar Days prior to the Initial Synchronization Date. Transmission Provider at Transmission Provider's option, and Transmission Owner shall review such specifications to ensure that the ICIF are compatible with their respective technical specifications, operational control, and safety requirements and comment on such design and specifications within thirty (30) Calendar Days of Interconnection Customer's submission. All specifications provided hereunder shall be deemed confidential.

5.10.2 Transmission Provider's and Transmission Owner's Review. Transmission Provider's and Transmission Owner's review of Interconnection Customer's final specifications shall not be construed as confirming, endorsing, or providing

a warranty as to the design, fitness, safety, durability or reliability of the Generating Facility, or the ICIF. Interconnection Customer shall make such changes to the ICIF as may reasonably be required by Transmission Provider and Transmission Owner, in accordance with Good Utility Practice, to ensure that the ICIF are compatible with the technical specifications, operational control and safety requirements of Transmission Provider and Transmission Owner.

5.10.3 ICIF Construction. The ICIF shall be designed and constructed in accordance with Good Utility Practice. Within one hundred twenty (120) Calendar Days after the Commercial Operation Date, unless the Parties agree on another mutually acceptable deadline, Interconnection Customer shall deliver to Transmission Provider and Transmission Owner "as-built" drawings, information and documents for the ICIF, such as: a one-line diagram, a site plan showing the Generating Facility and the ICIF, plan and elevation drawings showing the layout of the ICIF, a relay functional diagram, relaying AC and DC schematic wiring diagrams and relay settings for all facilities associated with the Interconnection Customer's step-up transformers, the facilities connecting the Generating Facility to the step-up transformers and the ICIF, and the impedances (determined by factory tests) for the associated step-up transformers and the Generating Facility. Interconnection Customer shall provide Transmission Provider and Transmission Owner with Interconnection Customer's specifications for the excitation system, automatic voltage regulator, Generating Facility control and protection settings, transformer tap settings, and communications, if applicable.

5.11 Transmission Owner's Interconnection Facilities Construction. The Transmission Owner's Interconnection Facilities shall be designed and constructed in accordance with Good Utility Practice. Upon request, within one hundred twenty (120) Calendar Days after the Commercial Operation Date, unless the Parties agree on another mutually acceptable deadline, Transmission Owner shall deliver to Transmission Provider (if requested) and Interconnection Customer the "as-built" drawings, information and documents for the Transmission Owner's Interconnection Facilities specified in Appendix C to this GIA.

Such drawings, information and documents shall be deemed Confidential Information.

Upon completion, the Transmission Owner's Interconnection Facilities and Stand Alone Network Upgrades shall be under the control of Transmission Provider or its designated representative.

5.12 Access Rights. Upon reasonable notice by a Party, and subject to any required or necessary regulatory approvals, a Party ("Granting Party") shall furnish *at no cost* to the other Party ("Access Party") any rights of use, licenses, rights of way and easements with respect to lands owned or controlled by the Granting Party, its agents (if allowed under the applicable agency agreement), or any Affiliate, that are necessary to enable the

Access Party to obtain ingress and egress to construct, operate, maintain, repair, test (or witness testing), inspect, replace or remove facilities and equipment to: (i) interconnect the Generating Facility with the Transmission System; (ii) operate and maintain the Generating Facility, the Interconnection Facilities and the Transmission System; and (iii) disconnect or remove the Access Party's facilities and equipment upon termination of this GIA. In exercising such licenses, rights of way and easements, the Access Party shall not unreasonably disrupt or interfere with normal operation of the Granting Party's business and shall adhere to the safety rules and procedures established in advance, as may be changed from time to time, by the Granting Party and provided to the Access Party.

5.13 Lands of Other Property Owners. If any part of the Transmission Owner's Interconnection Facilities, Network Upgrades, and/or Distribution Upgrades is to be installed on property owned by persons other than Interconnection Customer or Transmission Owner, Transmission Owner shall at Interconnection Customer's expense use efforts, similar in nature and extent to those that it typically undertakes on its own behalf or on behalf of its Affiliates, including use of its eminent domain authority to the extent permitted and consistent with Applicable Laws and Regulations and, to the extent consistent with such Applicable Laws and Regulations, to procure from such persons any rights of use, licenses, rights of way and easements that are necessary to construct, operate, maintain, test, inspect, replace or remove the Transmission Owner's Interconnection Facilities, Network Upgrades and/or Distribution Upgrades upon such property.

5.14 Permits. Transmission Provider or Transmission Owner and Interconnection Customer shall cooperate with each other in good faith in obtaining all permits, licenses and authorizations that are necessary to accomplish the interconnection in compliance with Applicable Laws and Regulations. With respect to this paragraph, Transmission Owner shall provide permitting assistance to Interconnection Customer comparable to that provided to the Transmission Owner's own, or an Affiliate's, generation to the extent that Transmission Owner or its Affiliate owns generation.

5.15 Early Construction of Base Case Facilities. (Includes facilities required for all queued projects with interconnection agreements).Interconnection Customer may request Transmission Owner to construct, and Transmission Owner shall construct, using Reasonable Efforts to accommodate Interconnection Customer's In-Service Date, all or any portion of any Network Upgrades, Transmission Owner's System Protection Facilities or Distribution Upgrades required for Interconnection Customer to be interconnected to the Transmission or Distribution System, as applicable, which are included in the Base Case of the Interconnection Facilities Study for Interconnection Customer, and which also are required to be constructed for another Interconnection Customer with a prior GIA, but where such construction is not scheduled to be completed in time to achieve Interconnection Customer's In-Service Date. Any such Network Upgrades, System Protection Facilities or Distribution Upgrades are included in the facilities to be constructed and as set forth in Appendix A to this GIA to the extent they are reasonably known.

5.16 Suspension.

5.16.1 Interconnection Customer's Right to Suspend for Force Majeure Event; Obligations. Provided that such suspension is permissible under the authorizations, permits or approvals granted for the construction of such Interconnection Facilities, Network Upgrades or Stand Alone Network Upgrades, Interconnection Customer will not suspend unless a Force Majeure event occurs.

Interconnection Customer must provide written notice of its request for suspension to Transmission Provider and Transmission Owner, and provide a description of the Force Majeure event that is acceptable to Transmission Provider. Suspension will only apply to Interconnection Customer milestones and Interconnection Facilities described in the Appendices of this GIA. Prior to suspension, Interconnection Customer must also provide security acceptable to Transmission Owner, equivalent to the higher of $5 million or the total cost of all Network Upgrades, Transmission Owner's System Protection Facilities, and Distribution Upgrades listed in Appendix A of this GIA. Network Upgrades and Transmission Owner's Interconnection Facilities will be constructed on the schedule described in the Appendices of this GIA unless: (1) construction is prevented by the order of a Governmental Authority; (2) the Network Upgrades are not needed by any other project; or (3) Transmission Owner or Transmission Provider determines that a Force Majeure event prevents construction. In the event of (1), (2), or (3) security shall be released upon the determination that the Network Upgrades will no longer be constructed.

If suspension occurs, the Transmission or Distribution System, as applicable, shall be left in a safe and reliable condition in accordance with Good Utility Practice and the Transmission Provider's and Transmission Owner's safety and reliability criteria. In such event, Interconnection Customer shall be responsible for all reasonable and necessary costs which Transmission Provider and Transmission Owner (i) have incurred pursuant to this GIA prior to the suspension and (ii) incur in suspending such work, including any costs incurred to perform such work as may be necessary to ensure the safety of persons and property and the integrity of the Transmission or Distribution System, as applicable, during such suspension and, if applicable, any costs incurred in connection with the cancellation or suspension of material, equipment and labor contracts which Transmission Provider and Transmission Owner cannot reasonably avoid; provided, however, that prior to canceling or suspending any such material, equipment or labor contract, Transmission Provider and Transmission Owner shall obtain Interconnection Customer's authorization to do so.

Transmission Provider and Transmission Owner shall each invoice Interconnection Customer for such costs pursuant to Article 12 and shall use Reasonable Efforts to minimize its costs. In the event Interconnection Customer suspends work by Transmission Owner required under this GIA pursuant to this Article 5.16, and has not requested Transmission Owner to recommence the work

required under this GIA on or before the expiration of three (3) years following commencement of such suspension, this GIA shall be deemed terminated. The three-year period shall begin on the date the suspension is requested, or the date of the written notice to Transmission Provider, if no effective date is specified.

5.16.2 Effect of Missed Interconnection Customer Milestones. If Interconnection Customer fails to provide notice of suspension pursuant to Article 5.16, and Interconnection Customer fails to fulfill or complete any Interconnection Customer Milestone provided in Appendix B ("Milestone"), this constitutes a Breach under this GIA. Depending upon the consequences of the Breach and effectiveness of the cure pursuant to Article 17, the Transmission Owners' Milestones may be revised, following consultation with Interconnection Customer, consistent with Reasonable Efforts, and in consideration of all relevant circumstances. Parties shall employ Reasonable Efforts to maintain their remaining respective Milestones.

5.16.3 Effect of Suspension; Parties Obligations. In the event that Interconnection Customer suspends work pursuant to this Article 5.16, no construction duration, timelines and schedules set forth in Appendix B shall be suspended during the period of suspension unless ordered by a Governmental Authority, with such order being the Force Majeure event causing the suspension. Should Interconnection Customer request that work be recommenced, Transmission Owner shall be obligated to proceed with Reasonable Efforts and in consideration of all relevant circumstances including regional outage schedules, construction availability and material procurement in performing the work as described in Appendix A and Appendix B. Transmission Owner will provide Interconnection Customer with a revised schedule for the design, procurement, construction, installation and testing of the Transmission Owner's Interconnection Facilities and Network Upgrades. Upon any suspension by Interconnection Customer pursuant to Article 5.16, Interconnection Customer shall be responsible for only those costs specified in this Article 5.16.

5.17 Taxes.

5.17.1 Interconnection Customer Payments Not Taxable. The Parties intend that all payments or property transfers made by Interconnection Customer to Transmission Owner for the installation of the Transmission Owner's Interconnection Facilities, Network Upgrades, Transmission Owner's System Protection Facilities, Distribution Upgrades and Generator Upgrades shall be non-taxable, either as contributions to capital, or as an advance, in accordance with the Internal Revenue Code and any applicable state income tax laws and shall not be taxable as contributions in aid of construction or otherwise under the Internal Revenue Code and any applicable state income tax laws. To the extent that Transmission Owner is a limited liability company and not a corporation, and has elected to be taxed as a partnership, then the following shall apply: Transmission Owner represents, and the Parties acknowledge, that Transmission Owner is a

limited liability company and is treated as a partnership for federal income tax purposes. Any payment made by Interconnection Customer to Transmission Owner for Network Upgrades is to be treated as an upfront payment in accordance with Rev Proc 2005-35. It is anticipated by the parties that any amounts paid by Interconnection Customer to Transmission Owner for Network Upgrades will be reimbursed to Interconnection Customer in accordance with the terms of this GIA, provided Interconnection Customer fulfills its obligations under this GIA.

5.17.2 Representations and Covenants. In accordance with IRS Notice 2016-36, Interconnection Customer represents and covenants that (i) ownership of the electricity generated at the Generating Facility will pass to another party prior to the transmission of the electricity on the Transmission System, (ii) for income tax purposes, the amount of any payments and the cost of any property transferred to Transmission Owner for the Transmission Owner's Interconnection Facilities will be capitalized by Interconnection Customer as an intangible asset and recovered using the straight-line method over a useful life of twenty (20) years, and (iii) any portion of the Transmission Owner's Interconnection Facilities that is a "dual-use intertie," within the meaning of IRS Notice 2016-36, is reasonably expected to carry only a de minimis amount of electricity in the direction of the Generating Facility. For this purpose, "de minimis amount" means no more than 5 percent of the total power flows in both directions, calculated in accordance with the "5 percent test" set forth in IRS Notice 2016-36. This is not intended to be an exclusive list of the relevant conditions that must be met to conform to IRS requirements for non-taxable treatment.

At Transmission Owner's request, Interconnection Customer shall provide Transmission Owner with a report from an independent engineer confirming its representation in clause (iii), above, with a copy to Transmission Provider. Transmission Owner represents and covenants that the cost of the Transmission Owner's Interconnection Facilities paid for by Interconnection Customer will have no net effect on the base upon which rates are determined.

5.17.3 Indemnification for the Cost Consequences of Current Tax Liability Upon Transmission Owner. Notwithstanding Article 5.17.1 and to the extent permitted by law, Interconnection Customer shall protect, indemnify and hold harmless Transmission Owner from the cost consequences of any tax liability imposed against Transmission Owner as the result of payments or property transfers made by Interconnection Customer to Transmission Owner under this GIA for Interconnection Facilities, as well as any interest and penalties, other than interest and penalties attributable to any delay caused by Transmission Owner.

Transmission Owner shall not include a gross-up for the cost consequences of any current tax liability in the amounts it charges Interconnection Customer

under this GIA unless (i) Transmission Owner has determined, in good faith, that the payments or property transfers made by Interconnection Customer to Transmission Owner should be reported as income subject to taxation or (ii) any Governmental Authority directs Transmission Owner to report payments or property as income subject to taxation; provided, however, that Transmission Owner may require Interconnection Customer to provide security for Interconnection Facilities, in a form reasonably acceptable to Transmission Owner (such as a parental guarantee or a letter of credit), in an amount equal to the cost consequences or any current tax liability under this Article 5.17. Interconnection Customer shall reimburse Transmission Owner for such costs on a fully grossed-up basis, in accordance with Article 5.17.4, within thirty (30) Calendar Days of receiving written notification from Transmission Owner of the amount due, including detail about how the amount was calculated.

The indemnification obligation shall terminate at the earlier of (1) the expiration of the ten-year testing period and the applicable statute of limitation, as it may be extended by Transmission Owner upon request of the IRS, to keep these years open for audit or adjustment, or (2) the occurrence of a subsequent taxable event and the payment of any related indemnification obligations as contemplated by this Article 5.17.

5.17.4 Tax Gross-Up Amount. Interconnection Customer's liability for the cost consequences of any current tax liability under this Article 5.17 shall be calculated on a fully grossed-up basis. Except as may otherwise be agreed to by the parties, this means that Interconnection Customer will pay Transmission Owner, in addition to the amount paid for the Interconnection Facilities, Network Upgrades, Transmission Owner's System Protection Facilities, and/or Distribution Upgrades, an amount equal to (1) the current taxes imposed on Transmission Owner ("Current Taxes") on the excess of (a) the gross income realized by Transmission Owner as a result of payments or property transfers made by Interconnection Customer to Transmission Owner under this GIA (without regard to any payments under this Article 5.17) (the "Gross Income Amount") over (b) the present value of future tax deductions for depreciation that will be available as a result of such payments or property transfers (the "Present Value Depreciation Amount"), plus (2) an additional amount sufficient to permit Transmission Owner to receive and retain, after the payment of all Current Taxes, an amount equal to the net amount described in clause (1).

For this purpose, (i) Current Taxes shall be computed based on Transmission Owner's composite federal and state tax rates at the time the payments or property transfers are received and Transmission Owner will be treated as being subject to tax at the highest marginal rates in effect at that time (the "Current Tax Rate"), and (ii) the Present Value Depreciation Amount shall be computed by discounting Transmission Owner's anticipated tax depreciation deductions as a result of such payments or property transfers by Transmission Owner's current weighted average cost of capital. Thus, the formula for calculating

Interconnection Customer's liability to Transmission Owner pursuant to this Article 5.17.4 can be expressed as follows: (Current Tax Rate x (Gross Income Amount – Present Value of Tax Depreciation))/(1-Current Tax Rate). Interconnection Customer's estimated tax liability in the event taxes are imposed shall be stated in Appendix A, Interconnection Facilities, Network Upgrades and Distribution Upgrades.

5.17.5 Private Letter Ruling or Change or Clarification of Law. At Interconnection Customer's request and expense, Transmission Owner shall file with the IRS a request for a private letter ruling as to whether any property transferred or sums paid, or to be paid, by Interconnection Customer to Transmission Owner under this GIA are subject to federal income taxation. Interconnection Customer will prepare the initial draft of the request for a private letter ruling, and will certify under penalties of perjury that all facts represented in such request are true and accurate to the best of Interconnection Customer's knowledge. Transmission Owner and Interconnection Customer shall cooperate in good faith with respect to the submission of such request.

Transmission Owner shall keep Interconnection Customer fully informed of the status of such request for a private letter ruling and shall execute either a privacy act waiver or a limited power of attorney, in a form acceptable to the IRS, that authorizes Interconnection Customer to participate in all discussions with the IRS regarding such request for a private letter ruling. Transmission Owner shall allow Interconnection Customer to attend all meetings with IRS officials about the request and shall permit Interconnection Customer to prepare the initial drafts of any follow-up letters in connection with the request.

5.17.6 Subsequent Taxable Events. If, within 10 years from the date on which the relevant Transmission Owner's Interconnection Facilities are placed in service, (i) Interconnection Customer breaches the covenant contained in Article 5.17.2, (ii) a "disqualification event" occurs within the meaning of IRS Notice 88-129, or (iii) this GIA terminates and Transmission Owner retains ownership of the Interconnection Facilities, Network Upgrades, Transmission Owner's System Protection Facilities, and/or Distribution Upgrades, Interconnection Customer shall pay a tax gross-up for the cost consequences of any current tax liability imposed on Transmission Owner, calculated using the methodology described in Article 5.17.4 and in accordance with IRS Notice 90-60.

5.17.7 Contests. In the event any Governmental Authority determines that Transmission Owner's receipt of payments or property constitutes income that is subject to taxation, Transmission Owner shall notify Interconnection Customer, in writing, within thirty (30) Calendar Days of receiving notification of such determination by a Governmental Authority. Upon the timely written request by Interconnection Customer and at Interconnection Customer's sole expense, Transmission Owner may appeal, protest, seek abatement of, or otherwise oppose such determination. Upon Interconnection Customer's written

request and sole expense, Transmission Owner shall file a claim for refund with respect to any taxes paid under this Article 5.17, whether or not it has received such a determination. Transmission Owner reserves the right to make all decisions with regard to the prosecution of such appeal, protest, abatement or other contest, including the selection of counsel and compromise or settlement of the claim, but Transmission Owner shall keep Interconnection Customer informed, shall consider in good faith suggestions from Interconnection Customer about the conduct of the contest, and shall reasonably permit Interconnection Customer or an Interconnection Customer representative to attend contest proceedings.

Interconnection Customer shall pay to Transmission Owner on a periodic basis, as invoiced by Transmission Owner, Transmission Owner's documented reasonable costs of prosecuting such appeal, protest, abatement or other contest. At any time during the contest, Transmission Owner may agree to a settlement either with Interconnection Customer's consent or after obtaining written advice from nationally-recognized tax counsel, selected by Transmission Owner, but reasonably acceptable to Interconnection Customer, that the proposed settlement represents a reasonable settlement given the hazards of litigation. Interconnection Customer's obligation shall be based on the amount of the settlement agreed to by Interconnection Customer, or if a higher amount, so much of the settlement that is supported by the written advice from nationally-recognized tax counsel selected under the terms of the preceding sentence. The settlement amount shall be calculated on a fully grossed-up basis to cover any related cost consequences of the current tax liability. Any settlement without Interconnection Customer's consent or such written advice will relieve Interconnection Customer from any obligation to indemnify Transmission Owner for the tax at issue in the contest.

5.17.8 Refund. In the event that (a) a private letter ruling is issued to Transmission Owner which holds that any amount paid or the value of any property transferred by Interconnection Customer to Transmission Owner under the terms of this GIA is not subject to federal income taxation, (b) any legislative change or administrative announcement, notice, ruling or other determination makes it reasonably clear to Transmission Owner in good faith that any amount paid or the value of any property transferred by Interconnection Customer to Transmission Owner under the terms of this GIA is not taxable to Transmission Owner, (c) any abatement, appeal, protest, or other contest results in a determination that any payments or transfers made by Interconnection Customer to Transmission Owner are not subject to federal income tax, or (d) if Transmission Owner receives a refund from any taxing authority for any overpayment of tax attributable to any payment or property transfer made by Interconnection Customer to Transmission Owner pursuant to this GIA, Transmission Owner shall promptly refund to Interconnection Customer the following:

any payment made by Interconnection Customer under this Article 5.17 for taxes that is attributable to the amount determined to be non-taxable, together with interest thereon,

(ii) interest on any amounts paid by Interconnection Customer to Transmission Owner for such taxes which Transmission Owner did not submit to the taxing authority, calculated in accordance with the methodology set forth in 18 C.F.R. Section 35.19a(a)(2)(iii) from the date payment was made by Interconnection Customer to the date Transmission Owner refunds such payment to Interconnection Customer, and

(iii) with respect to any such taxes paid by Transmission Owner, any refund or credit Transmission Owner receives or to which it may be entitled from any Governmental Authority, interest (or that portion thereof attributable to the payment described in clause (i), above) owed to Transmission Owner for such overpayment of taxes (including any reduction in interest otherwise payable by Transmission Owner to any Governmental Authority resulting from an offset or credit); provided, however, that Transmission Owner will remit such amount promptly to Interconnection Customer only after and to the extent that Transmission Owner has received a tax refund, credit or offset from any Governmental Authority for any applicable overpayment of income tax related to the Transmission Owner's Interconnection Facilities.

The intent of this provision is to leave both parties, to the extent practicable, in the event that no taxes are due with respect to any payment for Interconnection Facilities and Network Upgrades hereunder, in the same position they would have been in had no such tax payments been made.

5.17.9 Taxes Other Than Income Taxes. Upon the timely request by Interconnection Customer, and at Interconnection Customer's sole expense, Transmission Owner shall appeal, protest, seek abatement of, or otherwise contest any tax (other than federal or state income tax) asserted or assessed against Transmission Owner for which Interconnection Customer may be required to reimburse Transmission Owner under the terms of this GIA. Interconnection Customer shall pay to Transmission Owner on a periodic basis, as invoiced by Transmission Owner, Transmission Owner's documented reasonable costs of prosecuting such appeal, protest, abatement, or other contest. Interconnection Customer and Transmission Owner shall cooperate in good faith with respect to any such contest. Unless the payment of such taxes is a prerequisite to an appeal or abatement or cannot be deferred, no amount shall be payable by Interconnection Customer to Transmission Owner for such taxes until they are assessed by a final, non-appealable order by any court or agency of competent jurisdiction. In the event that a tax payment is withheld and ultimately due and payable after appeal, Interconnection Customer will be responsible for all taxes,

interest and penalties, other than penalties attributable to any delay caused by Transmission Owner.

5.18 Tax Status. Each Party shall cooperate with the other Parties to maintain each Party's tax status. Nothing in this GIA is intended to adversely affect any Party's tax-exempt status with respect to the issuance of bonds including, but not limited to, Local Furnishing Bonds.

5.19 Modification.

5.19.1 General. Either Party may undertake modifications to its facilities. If a Party plans to undertake a modification that reasonably may be expected to affect another Party's facilities, that Party shall provide to the other Parties sufficient information regarding such modification so that the other Parties may evaluate the potential impact of such modification prior to commencement of the work. Such information shall be deemed to be Confidential Information hereunder and shall include information concerning the timing of such modifications and whether such modifications are expected to interrupt the flow of electricity from the Generating Facility. The Party desiring to perform such work shall provide the relevant drawings, plans, and specifications to the other Parties at least ninety (90) Calendar Days in advance of the commencement of the work or such shorter period upon which the Parties may agree, which agreement shall not unreasonably be withheld, conditioned or delayed.

In the case of Generating Facility modifications that do not require Interconnection Customer to submit an Interconnection Request, Transmission Provider shall provide, within thirty (30) Calendar Days (or such other time as the Parties may agree), an estimate of any additional modifications to the Transmission or Distribution System as applicable, Transmission Owner's Interconnection Facilities, Network Upgrades, Transmission Owner's System Protection Facilities, and/or Distribution Upgrades necessitated by such Interconnection Customer modification and a good faith estimate of the costs thereof which shall be the responsibility of Interconnection Customer.

5.19.2 Standards. Any additions, modifications, or replacements made to a Party's facilities shall be designed, constructed and operated in accordance with this GIA and Good Utility Practice.

5.19.3 Modification Costs. Interconnection Customer shall not be directly assigned the costs of any additions, modifications, or replacements that Transmission Owner makes to the Transmission Owner's Interconnection Facilities, Network Upgrades, Transmission Owner's System Protection Facilities, Distribution Upgrades, or the Transmission or Distribution System, as applicable, to facilitate the interconnection of a third party to the Transmission Owner's Interconnection Facilities or the Transmission or Distribution System, as applicable, or to provide transmission service to a third party under the Tariff.

Interconnection Customer shall be responsible for the costs of any additions, modifications, or replacements to the Interconnection Customer's Interconnection Facilities that may be necessary to maintain or upgrade such Interconnection Customer's Interconnection Facilities consistent with Applicable Laws and Regulations, Applicable Reliability Standards or Good Utility Practice.

ARTICLE 6. TESTING AND INSPECTION

6.1 Pre-Commercial Operation Date Testing and Modifications. Prior to the Commercial Operation Date, Transmission Owner shall test the Transmission Owner's Interconnection Facilities, Network Upgrades, Transmission Owner's System Protection Facilities and Distribution Upgrades, and Interconnection Customer shall test each electric production device at the Generating Facility, Interconnection Customer's System Protection Facilities, including control equipment to limit injection at the POI to the level of Interconnection Service set forth in Appendix A and the Interconnection Customer's Interconnection Facilities to ensure their safe and reliable operation. Similar testing may be required after initial operation. Transmission Owner and Interconnection Customer shall make any modifications to their respective facilities that are found to be necessary as a result of such testing. Interconnection Customer shall bear the cost of all such testing and modifications. Interconnection Customer shall generate test energy at the Generating Facility only if it has arranged for the delivery of such test energy.

6.2 Post-Commercial Operation Date Testing and Modifications. Each Party shall at its own expense perform routine inspection and testing of its facilities and equipment in accordance with Good Utility Practice as may be necessary to ensure the continued interconnection of the Generating Facility with the Transmission or Distribution System, as applicable, in a safe and reliable manner. Each Party shall have the right, upon advance written notice, to require reasonable additional testing of the Interconnection Facilities, at the requesting Party's expense, as may be in accordance with Good Utility Practice.

6.3 Right to Observe Testing. Each Party shall notify the other Parties in advance of its performance of tests of its Interconnection Facilities. The other Parties shall each have the right, at its own expense, to observe such testing.

6.4 Right to Inspect. Each Party shall have the right, but shall have no obligation to: (i) observe Transmission Owner's and Interconnection Customer's tests and/or inspection of any of their respective System Protection Facilities and other protective equipment, including power system stabilizers and control equipment; (ii) review the settings of the System Protection Facilities and other protective equipment; and (iii) review the maintenance records relative to the Interconnection Facilities, the System Protection Facilities and other protective equipment. A Party may exercise these rights from time to time as it deems necessary upon reasonable notice to the other Parties. The exercise or non-exercise by a Party of any such rights shall not be construed as an endorsement or confirmation of any element or condition of the Interconnection Facilities or the System

Protection Facilities or other protective equipment or the operation thereof, or as a warranty as to the fitness, safety, desirability, or reliability of same. Any information that a Party obtains through the exercise of any of its rights under this Article 6.4 shall be deemed to be Confidential Information and treated pursuant to Article 22 of this GIA.

ARTICLE 7. METERING

7.1 General. Each Party shall comply with the Applicable Reliability Council requirements. Unless the Interconnection Customer is utilizing shared Interconnection Customer Interconnection Facilities, or unless otherwise agreed by the Parties, Transmission Owner, at its election, or otherwise Interconnection Customer, shall install Metering Equipment (the "Metering Party") at the Point of Interconnection prior to any operation of the Generating Facility. Transmission Owner, at its election, or otherwise Interconnection Customer shall own, operate, test and maintain such Metering Equipment. Power flows to and from the Generating Facility shall be measured at or, at the Metering Party's option, compensated to, the Point of Interconnection. If Interconnection Customer will share Interconnection Facilities with any other projects, Interconnection Customer shall install Metering Equipment either on its own Generating Facility or its own non-shared facilities sufficient to measure the output of such Interconnection Customer's Generating Facility separately from any other Generating Facilities with which it will share Interconnection Facilities. The Metering Party shall provide metering quantities, in analog and/or digital form, to the other Parties upon request. Interconnection Customer shall bear all reasonable documented costs associated with the purchase, installation, operation, testing and maintenance of the Metering Equipment.

7.2 Check Meters. Interconnection Customer, at its option and expense, may install and operate, on its premises and on its side of the Point of Interconnection, one or more check meters to check the Metering Equipment owned by the Metering Party. Such check meters shall be for check purposes only and shall not be used for the measurement of power flows for purposes of this GIA, except as provided in Article 7.4 below. The check meters shall be subject at all reasonable times to inspection and examination by Transmission Provider, Transmission Owner or their designees. The installation, operation and maintenance thereof shall be performed entirely by Interconnection Customer in accordance with Good Utility Practice.

7.3 Standards. The Metering Party shall install, calibrate, and test revenue quality Metering Equipment in accordance with applicable ANSI standards.

7.4 Testing of Metering Equipment. The Metering Party shall inspect and test Metering Equipment upon installation and at least once every two (2) years thereafter. If requested to do so by a Party, the Metering Party shall, at the requesting Party's expense, inspect or test Metering Equipment more frequently than every two (2) years. The Metering Party shall give reasonable notice to the other Parties of the time when any inspection or test shall take place, and the other Parties may have representatives present at the test or inspection. If at any time Metering Equipment is found to be inaccurate or defective, it

shall be adjusted, repaired or replaced at Interconnection Customer's expense, in order to provide accurate metering, unless the inaccuracy or defect is due to the Metering Party's failure to maintain, then the Metering Party shall pay. If Metering Equipment fails to register, or if the measurement made by Metering Equipment during a test varies by more than two percent (2%) from the measurement made by the standard meter used in the test, the Metering Party shall adjust the measurements by correcting all measurements for the period during which Metering Equipment was in error by using Interconnection Customer's check meters, if installed. If no such check meters are installed or if the period cannot be reasonably ascertained, the adjustment shall be for the period immediately preceding the test of the Metering Equipment equal to one-half the time from the date of the previous test of the Metering Equipment.

7.5 Metering Data. At Interconnection Customer's expense, the metered data shall be telemetered to one or more locations designated by Transmission Provider and Transmission Owner and one or more locations designated by Interconnection Customer. Such telemetered data shall be used, under normal operating conditions, as the official measurement of the amount of energy delivered from the Generating Facility to the Point of Interconnection.

ARTICLE 8. COMMUNICATIONS

8.1 Interconnection Customer Obligations. Interconnection Customer shall maintain satisfactory operating communications with Transmission Provider's Transmission System dispatcher or representative designated by Transmission Provider. Interconnection Customer shall provide standard voice line, dedicated voice line and facsimile communications at its Generating Facility control room or central dispatch facility through use of either the public telephone system, or a voice communications system that does not rely on the public telephone system. Interconnection Customer shall also provide the dedicated data circuit(s) necessary to provide Interconnection Customer data to Transmission Provider as set forth in Appendix D, Security Arrangements Details. The data circuit(s) shall extend from the Generating Facility to the location(s) specified by Transmission Provider. Any required maintenance of such communications equipment shall be performed by and at the cost of Interconnection Customer. Operational communications shall be activated and maintained under, but not be limited to, the following events: system paralleling or separation, scheduled and unscheduled shutdowns, equipment clearances, and hourly and daily load data.

Unless the Generating Facility is an Intermittent Resource not relying on wind or solar as a fuel source, Interconnection Customer shall install communication and control equipment such that the Generating Facility can receive and respond to the appropriate dispatch signals while operating under the Tariff. Where applicable, the requirements of the communication and control equipment will be enumerated in Appendix C to this GIA.

8.2 Remote Terminal Unit (RTU). Prior to the Initial Synchronization Date of the Generating Facility, a remote terminal unit, or equivalent data collection and transfer equipment acceptable to both Parties, shall be installed by Interconnection Customer, or

by Transmission Owner at Interconnection Customer's expense, to gather accumulated and instantaneous data to be telemetered to the location(s) designated by Transmission Owner and Transmission Provider through use of a dedicated point-to-point data circuit(s) as indicated in Article 8.1. The communication protocol for the data circuit(s) shall be specified by Transmission Owner and Transmission Provider. Instantaneous bi-directional analog real power and reactive power flow information must be telemetered directly to the location(s) specified by Transmission Provider and Transmission Owner.

Each Party will promptly advise the other Parties if it detects or otherwise learns of any metering, telemetry or communications equipment errors or malfunctions that require the attention and/or correction. The Party owning such equipment shall correct such error or malfunction as soon as reasonably feasible.

8.3 No Annexation. Any and all equipment placed on the premises of a Party shall be and remain the property of the Party providing such equipment regardless of the mode and manner of annexation or attachment to real property, unless otherwise mutually agreed by the Parties.

8.4 Provision of Data from a Variable Energy Resource. The Interconnection Customer whose Generating Facility is a Variable Energy Resource shall provide meteorological and forced outage data to the Transmission Provider to the extent necessary for the Transmission Provider's development and deployment of power production forecasts for that class of Variable Energy Resources. The Interconnection Customer with a Variable Energy Resource having wind as the energy source will, upon request by the Transmission Provider, be required to provide the Transmission Provider with site-specific meteorological data including: temperature, wind speed, wind direction, and atmospheric pressure. The Interconnection Customer with a Variable Energy Resource having solar as the energy source will, upon request by the Transmission Provider, be required to provide the Transmission Provider with site-specific meteorological data including: temperature, atmospheric pressure, and irradiance. The Transmission Provider and Interconnection Customer whose Generating Facility is a Variable Energy Resource shall mutually agree to any additional meteorological data that are required for the development and deployment of a power production forecast. The Interconnection Customer whose Generating Facility is a Variable Energy Resource also shall submit data to the Transmission Provider regarding all forced outages to the extent necessary for the Transmission Provider's development and deployment of power production forecasts for that class of Variable Energy Resources. The exact specifications of the meteorological and forced outage data to be provided by the Interconnection Customer to the Transmission Provider, including the frequency and timing of data submittals, shall be made taking into account the size and configuration of the Variable Energy Resource, its characteristics, location, and its importance in maintaining generation resource adequacy and transmission system reliability in its area. All requirements for meteorological and forced outage data must be commensurate with the power production forecasting employed by the Transmission Provider. Data requirements for meteorological and forced outage data will be negotiated by the Transmission Provider and the Interconnection Customer, and will be set forth in Appendix C, Interconnection Details,

of this GIA.

ARTICLE 9. OPERATIONS

9.1 General. Each Party shall comply with the Applicable Reliability Council requirements. Each Party shall provide to any Party all information that may reasonably be required by that Party to comply with Applicable Laws and Regulations and Applicable Reliability Standards.

9.2 Local Balancing Authority Notification. At least three (3) months before Initial Synchronization Date, Interconnection Customer shall notify Transmission Provider and Transmission Owner in writing of the Local Balancing Authority in which the Generating Facility will be located. If Interconnection Customer elects to locate the Generating Facility through dynamic metering/scheduling in a Local Balancing Authority other than the Local Balancing Authority in which the Generating Facility is physically located, and if permitted to do so by the relevant transmission tariffs, all necessary arrangements, including but not limited to those set forth in Article 7 and Article 8 of this GIA, and remote Local Balancing Authority generator interchange agreements, if applicable, and the appropriate measures under such agreements, shall be executed and implemented prior to the placement of the Generating Facility in the other Local Balancing Authority.

9.3 Transmission Provider and Transmission Owner Obligations. Transmission Provider shall cause the Transmission System and the Transmission Owner's Interconnection Facilities to be operated, maintained and controlled in a safe and reliable manner in accordance with this GIA. Transmission Provider, or its designee, may provide operating instructions to Interconnection Customer consistent with this GIA and the Tariff and, if applicable, Transmission Owner's operating protocols and procedures as they may change from time to time. Transmission Provider will consider changes to its operating protocols and procedures proposed by Interconnection Customer.

9.4 Interconnection Customer Obligations. Interconnection Customer shall at its own expense operate, maintain and control the Generating Facility and the Interconnection Customer's Interconnection Facilities in a safe and reliable manner and in accordance with this GIA. The Generating Facility must be operated in accordance with the operating limits, if any, in the Interconnection Facilities Study and specified in Appendix C of this GIA. Interconnection Customer shall operate the Generating Facility and the Interconnection Customer's Interconnection Facilities in accordance with all applicable requirements of Transmission Provider or its designated Local Balancing Authority Operator of which the Generating Facility is part, as such requirements are set forth in Appendix C, Interconnection Details, of this GIA. Appendix C, Interconnection Details, will be modified to reflect changes to the requirements as they may change from time to time. Any Party may request that a Party provide copies of the requirements set forth in Appendix C, Interconnection Details, of this GIA.

9.5 Start-Up and Synchronization. Consistent with the Parties' mutually acceptable procedures, Interconnection Customer is responsible for the proper synchronization of the Generating Facility to the Transmission or Distribution System, as applicable.

9.6 Reactive Power and Primary Frequency Response.

9.6.1 Power Factor Design Criteria.

9.6.1.1 Synchronous Generation. Interconnection Customer shall design the Generating Facility to be capable of maintaining a composite power delivery at continuous rated power output at the Point of Interconnection at all power factors over 0.95 leading to 0.95 lagging, unless the Transmission Provider has established different requirements that apply to all synchronous generators in the Local Balancing Authority on a comparable basis. The applicable Local Balancing Authority power factor requirements are listed on the Transmission Provider's website at

https://cdn.misoenergy.org/Reactive_Generator_Requirements108137.pdf

and may be referenced in the Appendices to this GIA. The Generating Facility shall be capable of continuous dynamic operation throughout the power factor design range as measured at the Point of Interconnection. Such operation shall account for the net effect of all energy production devices on the Interconnection Customer's side of the Point of Interconnection.

9.6.1.2 Non-Synchronous Generation. Interconnection Customer shall design the Generating Facility to be capable of maintaining a composite power delivery at continuous rated power output at the high-side of the generator substation at all power factors over 0.95 leading to 0.95 lagging, unless the Transmission Provider has established different requirements that apply to all non-synchronous generators in the Local Balancing Authority on a comparable basis. The applicable Local Balancing Authority power factor requirements are listed on the Transmission Provider's website at

https://cdn.misoenergy.org/Reactive_Generator_Requirements108137.pdf

and may be referenced in the Appendices to this GIA. This power factor range standard shall be dynamic and can be met using, for example, power electronics designed to supply this level of reactive capability (taking into account any limitations due to voltage level, real power output, etc.) or fixed and switched capacitors, or a combination of the two. This requirement shall only apply to newly interconnecting non-synchronous generators that have not yet completed a System Impact Study as of the effective date of the Final Rule establishing this requirement (Order No. 827). These requirements apply to existing non-synchronous generators making upgrades that require a new Generator Interconnection Agreement only where the Transmission Provider's System

Impact Study shows the need for reactive power as a result of an upgrade. If applicable, these requirements will be memorialized in Appendix C to this GIA.

9.6.2 Voltage Schedules. Once Interconnection Customer has synchronized the Generating Facility with the Transmission System, Transmission Provider shall require Interconnection Customer to operate the Generating Facility to produce or absorb reactive power within the design limitations of the Generating Facility set forth in Article 9.6.1 (Power Factor Design Criteria), to maintain the output voltage or power factor at the Point of Interconnection as specified by Transmission Provider. Transmission Provider's voltage schedules shall treat all sources of reactive power in the Local Balancing Authority in an equitable and not unduly discriminatory manner. Transmission Provider shall exercise Reasonable Efforts to provide Interconnection Customer with such schedules at least one (1) Calendar Day in advance, and may make changes to such schedules as necessary to maintain the reliability of the Transmission or Distribution System as applicable. Interconnection Customer shall operate the Generating Facility to maintain the specified output voltage or power factor at the Point of Interconnection within the design limitations of the Generating Facility set forth in Article 9.6.1 (Power Factor Design Criteria). If Interconnection Customer is unable to maintain the specified voltage or power factor, it shall promptly notify Transmission Provider's system operator, or its designated representative.

9.6.2.1 Voltage Regulators. Whenever the Generating Facility is operated in parallel with the Transmission or Distribution System as applicable and voltage regulators are capable of operation, Interconnection Customer shall operate the Generating Facility with its speed governors and voltage regulators in automatic operation. If the Generating Facility's voltage regulators are not capable of such automatic operation, Interconnection Customer shall immediately notify Transmission Provider's system operator, or its designated representative, and ensure that such Generating Facility's reactive power production or absorption (measured in MVARs) are within the design capability of the Generating Facility's generating unit(s) and steady state stability limits. Interconnection Customer shall not cause its Generating Facility to disconnect automatically or instantaneously from the Transmission or Distribution System, as applicable, or trip any generating unit comprising the Generating Facility for an under or over frequency condition unless the abnormal frequency condition persists for a time period beyond the limits set forth in ANSI/IEEE Standard C37.106, or such other standard as applied to other generators in the Local Balancing Authority on a comparable basis.

9.6.3 Payment for Reactive Power. Payments for reactive power shall be pursuant to any tariff or rate schedule filed by Transmission Provider and approved by the FERC.

9.6.4 Primary Frequency Response. This Section 9.6.4 shall only apply in the event that the Interconnection Request for the Generating Facility completed Definitive Planning Phase Interconnection Customer Decision Point 2 after May 15, 2018.

Interconnection Customer shall ensure the primary frequency response capability of its Generating Facility by installing, maintaining, and operating a functioning governor or equivalent controls. The term "functioning governor or equivalent controls" as used herein shall mean the required hardware and/or software that provides frequency responsive real power control with the ability to sense changes in system frequency and autonomously adjust the Generating Facility's real power output in accordance with the droop and deadband parameters and in the direction needed to correct frequency deviations. Interconnection Customer is required to install a governor or equivalent controls with the capability of operating: (1) with a maximum 5 percent droop and ±0.036 Hz deadband; or (2) in accordance with the relevant droop, deadband, and timely and sustained response settings from an approved NERC Reliability Standard providing for equivalent or more stringent parameters. The droop characteristic shall be: (1) based on the nameplate capacity of the Generating Facility, and shall be linear in the range of frequencies between 59 to 61 Hz that are outside of the deadband parameter; or (2) based on an approved NERC Reliability Standard providing for an equivalent or more stringent parameter. The deadband parameter shall be: the range of frequencies above and below nominal (60 Hz) in which the governor or equivalent controls is not expected to adjust the Generating Facility's real power output in response to frequency deviations. The deadband shall be implemented: (1) without a step to the droop curve, that is, once the frequency deviation exceeds the deadband parameter, the expected change in the Generating Facility's real power output in response to frequency deviations shall start from zero and then increase (for under-frequency deviations) or decrease (for over-frequency deviations) linearly in proportion to the magnitude of the frequency deviation; or (2) in accordance with an approved NERC Reliability Standard providing for an equivalent or more stringent parameter. Interconnection Customer shall notify Transmission Provider that the primary frequency response capability of the Generating Facility has been tested and confirmed during commissioning. Once Interconnection Customer has synchronized the Generating Facility with the Transmission System, Interconnection Customer shall operate the Generating Facility consistent with the provisions specified in Sections 9.6.4.1 and 9.6.4.2 of this GIA. The primary frequency response requirements contained herein shall apply to both synchronous and non-synchronous Generating Facilities.

9.6.4.1 Governor or Equivalent Controls. Whenever the Generating Facility is operated in parallel with the Transmission System, Interconnection Customer shall operate the Generating Facility with its governor or equivalent controls in service and responsive to frequency. Interconnection Customer shall: (1) in coordination with Transmission Provider and/or the relevant balancing authority, set the deadband parameter to: (a) a maximum of ±0.036 Hz and set the droop parameter

to a maximum of 5 percent; or (b) implement the relevant droop and deadband settings from an approved NERC Reliability Standard that provides for equivalent or more stringent parameters. Interconnection Customer shall be required to provide the status and settings of the governor or equivalent controls to Transmission Provider and/or the relevant balancing authority upon request. If Interconnection Customer needs to operate the Generating Facility with its governor or equivalent controls not in service, Interconnection Customer shall immediately notify Transmission Provider and the relevant balancing authority, and provide both with the following information: (1) the operating status of the governor or equivalent controls (i.e., whether it is currently out of service or when it will be taken out of service); (2) the reasons for removing the governor or equivalent controls from service; and (3) a reasonable estimate of when the governor or equivalent controls will be returned to service. Interconnection Customer shall make Reasonable Efforts to return its governor or equivalent controls into service as soon as practicable. Interconnection Customer shall make Reasonable Efforts to keep outages of the Generating Facility's governor or equivalent controls to a minimum whenever the Generating Facility is operated in parallel with the Transmission System.

9.6.4.2 Timely and Sustained Response. Interconnection Customer shall ensure that the Generating Facility's real power response to sustained frequency deviations outside of the deadband setting is automatically provided and shall begin immediately after frequency deviates outside of the deadband, and to the extent the Generating Facility has operating capability in the direction needed to correct the frequency deviation. Interconnection Customer shall not block or otherwise inhibit the ability of the governor or equivalent controls to respond and shall ensure that the response is not inhibited, except under certain operational constraints including, but not limited to, ambient temperature limitations, physical energy limitations, outages of mechanical equipment, or regulatory requirements. The Generating Facility shall sustain the real power response at least until system frequency returns to a value within the deadband setting of the governor or equivalent controls. A Commission-approved Reliability Standard with equivalent or more stringent requirements shall supersede the above requirements.

9.6.4.3 Exemptions. Generating Facilities that are regulated by the United States Nuclear Regulatory Commission shall be exempt from Sections 9.6.4, 9.6.4.1, and 9.6.4.2 of this GIA. Generating Facilities that are behind the meter generation that is sized-to-load (i.e., the thermal load and the generation are near-balanced in real-time operation and the generation is primarily controlled to maintain the unique thermal, chemical, or mechanical output necessary for the operating requirements of its host facility) shall be required to install primary frequency

response capability in accordance with the droop and deadband capability requirements specified in Section 9.6.4, but shall be otherwise exempt from the operating requirements in Sections 9.6.4, 9.6.4.1, 9.6.4.2, and 9.6.4.4 of this GIA.

9.6.4.4 Electric Storage Resources. Interconnection Customer interconnecting an electric storage resource shall establish an operating range in Appendix C that specifies a minimum state of charge and a maximum state of charge between which the electric storage resource will be required to provide primary frequency response consistent with the conditions set forth in Sections 9.6.4, 9.6.4.1, 9.6.4.2 and 9.6.4.3 of this GIA. Appendix C shall specify whether the operating range is static or dynamic, and shall consider (1) the expected magnitude of frequency deviations in the interconnection; (2) the expected duration that system frequency will remain outside of the deadband parameter in the interconnection; (3) the expected incidence of frequency deviations outside of the deadband parameter in the interconnection; (4) the physical capabilities of the electric storage resource; (5) operational limitations of the electric storage resource due to manufacturer specifications; and (6) any other relevant factors agreed to by Transmission Provider and Interconnection Customer, and in consultation with the relevant transmission owner or balancing authority as appropriate. If the operating range is dynamic, then Appendix C must establish how frequently the operating range will be reevaluated and the factors that may be considered during its reevaluation.

Interconnection Customer's electric storage resource is required to provide timely and sustained primary frequency response consistent with Section 9.6.4.2 of this GIA when it is online and dispatched to inject electricity to the Transmission System and/or receive electricity from the Transmission System. This excludes circumstances when the electric storage resource is not dispatched to inject electricity to the Transmission System and/or dispatched to receive electricity from the Transmission System. If Interconnection Customer's electric storage resource is charging at the time of a frequency deviation outside of its deadband parameter, it is to increase (for over-frequency deviations) or decrease (for under-frequency deviations) the rate at which it is charging in accordance with its droop parameter. Interconnection Customer's electric storage resource is not required to change from charging to discharging, or vice versa, unless the response necessitated by the droop and deadband settings requires it to do so and it is technically capable of making such a transition.

9.7 Outages and Interruptions.

9.7.1 Outages.

9.7.1.1 Outage Authority and Coordination. Interconnection Customer and Transmission Owner may each in accordance with Good Utility Practice in coordination with the other Party and Transmission Provider remove from service any of its respective Interconnection Facilities, System Protection Facilities, Network Upgrades, System Protection Facilities or Distribution Upgrades that may impact the other Party's facilities as necessary to perform maintenance or testing or to install or replace equipment. Absent an Emergency Condition, the Party scheduling a removal of such facility(ies) from service will use Reasonable Efforts to notify one another and schedule such removal on a date and time mutually acceptable to the Parties. In all circumstances, any Party planning to remove such facility(ies) from service shall use Reasonable Efforts to minimize the effect on the other Parties of such removal.

9.7.1.2 Outage Schedules. Transmission Provider shall post scheduled outages of transmission facilities on the OASIS. Interconnection Customer shall submit its planned maintenance schedules for the Generating Facility to Transmission Provider and Transmission Owner for a minimum of a rolling twenty-four (24) month period in accordance with the Transmission Provider's procedures. Interconnection Customer shall update its planned maintenance schedules as necessary. Transmission Provider may request Interconnection Customer to reschedule its maintenance as necessary to maintain the reliability of the Transmission System; provided, however, adequacy of generation supply shall not be a criterion in determining Transmission System reliability.

Transmission Provider shall compensate, pursuant to applicable Transmission Provider tariff or rate schedule, Interconnection Customer for any additional direct costs that Interconnection Customer incurs as a result of having to reschedule maintenance, including any additional overtime, breaking of maintenance contracts or other costs above and beyond the cost Interconnection Customer would have incurred absent the Transmission Provider's request to reschedule maintenance. Interconnection Customer will not be eligible to receive compensation, if during the twelve (12) months prior to the date of the scheduled maintenance, Interconnection Customer had modified its schedule of maintenance activities.

Costs shall be determined by negotiation between Transmission Provider and Interconnection Customer prior to implementation of the voluntary change in outage schedules, or if such request is made by or on behalf of a Transmission Customer requesting firm service, costs and recovery of costs shall be determined through a bilateral agreement between the Transmission Customer and Interconnection Customer. Voluntary changes to outage schedules under this Article 9.7.1.2 are separate from actions and compensation required under Article 13 and for which costs

are recovered in accordance with Transmission Provider's applicable tariff or rate schedule.

9.7.1.3 Outage Restoration. If an outage on either the Interconnection Customer's or Transmission Owner's Interconnection Facilities, Network Upgrades, System Protection Facilities or Distribution Upgrades adversely affects a Party's operations or facilities, the Party that owns or controls the facility that is out of service shall use Reasonable Efforts to promptly restore such facility(ies) to a normal operating condition consistent with the nature of the outage. The Party that owns or controls the facility that is out of service shall provide the other Parties, to the extent such information is known, information on the nature of the Emergency Condition, an estimated time of restoration, and any corrective actions required. Initial verbal notice shall be followed up as soon as practicable with written notice to the other Parties explaining the nature of the outage.

9.7.2 Interruption of Service. If required by Good Utility Practice to do so, Transmission Provider may require Interconnection Customer to interrupt or reduce deliveries of electricity if such delivery of electricity could adversely affect Transmission Provider's ability to perform such activities as are necessary to safely and reliably operate and maintain the Transmission System. The following provisions shall apply to any interruption or reduction permitted under this Article 9.7.2:

9.7.2.1 The interruption or reduction shall continue only for so long as reasonably necessary under Good Utility Practice;

9.7.2.2 Any such interruption or reduction shall be made on an equitable, non-discriminatory basis with respect to all generating facilities directly connected to the Transmission or Distribution System, as applicable;

9.7.2.3 When the interruption or reduction must be made under circumstances which do not allow for advance notice, Transmission Provider shall notify Interconnection Customer by telephone as soon as practicable of the reasons for the curtailment, interruption, or reduction, and, if known, its expected duration. Telephone notification shall be followed by written notification as soon as practicable;

9.7.2.4 Except during the existence of an Emergency Condition, when the interruption or reduction can be scheduled without advance notice, Transmission Provider shall notify Interconnection Customer in advance regarding the timing of such scheduling and further notify Interconnection Customer of the expected duration. Transmission Provider shall coordinate with Interconnection Customer using Good Utility Practice to schedule the interruption or reduction during periods

of least impact to Interconnection Customer, Transmission Owner and Transmission Provider;

9.7.2.5 The Parties shall cooperate and coordinate with each other to the extent necessary in order to restore the Generating Facility, Interconnection Facilities, and the Transmission or Distribution System, as applicable to their normal operating state, consistent with system conditions and Good Utility Practice.

9.7.3 Under-Frequency, Over-Frequency, Under-Voltage, and Over-Voltage Conditions. The Transmission System is designed to automatically activate a load-shed program as required by the Applicable Reliability Council in the event of an under-frequency or under-voltage system disturbance. Interconnection Customer shall implement under-frequency, over-frequency, under-voltage, and over-voltage relay set points for the Generating Facility as required by the Applicable Reliability Council to ensure "ride through" capability of the Transmission System. Generating Facility response to frequency and/or voltage deviations of pre-determined magnitudes, including under-frequency, over-frequency, under-voltage, and over-voltage, shall be studied and coordinated with Transmission Provider in accordance with Good Utility Practice. The term "ride through" as used herein shall mean the ability of a Generating Facility to stay connected to and synchronized with the Transmission System during system disturbances within a range of under-frequency, over-frequency, under-voltage, and over-voltage conditions, in accordance with Good Utility Practice.

9.7.4 System Protection and Other Control Requirements.

9.7.4.1 System Protection Facilities. Interconnection Customer shall, at its expense, install, operate and maintain its System Protection Facilities as a part of the Generating Facility or the Interconnection Customer's Interconnection Facilities. Transmission Owner shall install at Interconnection Customer's expense any Transmission Owner's System Protection Facilities that may be required on the Transmission Owner's Interconnection Facilities or the Transmission Owner's transmission or distribution facilities as a result of the interconnection of the Generating Facility and the Interconnection Customer's Interconnection Facilities.

9.7.4.2 Interconnection Customer's and Transmission Owner's System Protection Facilities shall be designed and coordinated with Affected Systems in accordance with Good Utility Practice.

9.7.4.3 Each Party shall be responsible for protection of its facilities consistent with Good Utility Practice.

9.7.4.4 Each Party's protective relay design shall incorporate the necessary test switches to perform the tests required in Article 6. The required test

switches will be placed such that they allow operation of lockout relays while preventing breaker failure schemes from operating and causing unnecessary breaker operations and/or the tripping of the Generating Facility.

9.7.4.5 Each Party will test, operate and maintain their respective System Protection Facilities in accordance with Good Utility Practice.

9.7.4.6 Prior to the In-Service Date, and again prior to the Commercial Operation Date, Interconnection Customer or Transmission Owner, or their respective agents, shall perform a complete calibration test and functional trip test of the System Protection Facilities. At intervals suggested by Good Utility Practice and following any apparent malfunction of the System Protection Facilities, Interconnection Customer or Transmission Owner shall each perform both calibration and functional trip tests of their respective System Protection Facilities. These tests do not require the tripping of any in-service generating unit. These tests do, however, require that all protective relays and lockout contacts be activated.

9.7.5 Requirements for Protection. In compliance with Good Utility Practice, Interconnection Customer shall provide, install, own, and maintain relays, circuit breakers and all other devices necessary to remove any fault contribution of the Generating Facility to any short circuit occurring on the Transmission or Distribution System, as applicable, not otherwise isolated by Transmission Owner's equipment, such that the removal of the fault contribution shall be coordinated with the protective requirements of the Transmission or Distribution System, as applicable. Such protective equipment shall include, without limitation, a disconnecting device or switch with load-interrupting capability located between the Generating Facility and the Transmission or Distribution System, as applicable, at a site selected upon mutual agreement (not to be unreasonably withheld, conditioned or delayed) of the Parties. Interconnection Customer shall be responsible for protection of the Generating Facility and Interconnection Customer's other equipment from such conditions as negative sequence currents, over- or under-frequency, sudden load rejection, over- or under-voltage, and generator loss-of-field. Interconnection Customer shall be solely responsible to disconnect the Generating Facility and Interconnection Customer's other equipment if conditions on the Transmission or Distribution System, as applicable, could adversely affect the Generating Facility.

9.7.6 Power Quality. Neither Party's facilities shall cause excessive voltage flicker nor introduce excessive distortion to the sinusoidal voltage or current waves as defined by ANSI Standard C84.1-1989, in accordance with IEEE Standard 519, or any applicable superseding electric industry standard. In the event of a conflict between ANSI Standard C84.1-1989, and any applicable superseding electric

industry standard, the applicable superseding electric industry standard shall control.

9.8 Switching and Tagging Rules. Prior to the Initial Synchronization Date, each Party shall provide the other Parties a copy of its switching and tagging rules that are applicable to the other Parties' activities. Such switching and tagging rules shall be developed on a non-discriminatory basis. The Parties shall comply with applicable switching and tagging rules, as amended from time to time, in obtaining clearances for work or for switching operations on equipment.

9.9 Use of Interconnection Facilities by Other Parties.

9.9.1 Purpose of Interconnection Facilities. Except as may be required by Applicable Laws and Regulations, or as otherwise agreed to among the Parties, the Interconnection Facilities shall be constructed for the sole purpose of interconnecting the Generating Facility to the Transmission or Distribution System, as applicable, and shall be used for no other purpose.

9.9.2 Other Users. If required by Applicable Laws and Regulations or if the Parties mutually agree, such agreement not to be unreasonably withheld or delayed, to allow one or more Parties to use the Transmission Owner's Interconnection Facilities, or any part thereof, Interconnection Customer will be entitled to compensation for the capital expenses it incurred in connection with the Interconnection Facilities based upon the pro rata use of the Interconnection Facilities by Transmission Owner, all non-Party users, and Interconnection Customer, in accordance with Applicable Laws and Regulations or upon some other mutually-agreed upon methodology. In addition, cost responsibility for ongoing costs, including operation and maintenance costs associated with the Interconnection Facilities, will be allocated between Interconnection Customer and any non-Party users based upon the pro rata use of the Interconnection Facilities by Transmission Owner, all non-Party users, and Interconnection Customer, in accordance with Applicable Laws and Regulations or upon some other mutually agreed upon methodology. If the issue of such compensation or allocation cannot be resolved through such negotiations, it shall be submitted to Dispute Resolution pursuant to Section 12 of the Tariff.

9.10 Disturbance Analysis Data Exchange. The Parties will cooperate with one another in the analysis of disturbances to either the Generating Facility or the Transmission System by gathering and providing access to any information relating to any disturbance, including information from oscillography, protective relay targets, breaker operations and sequence of events records, and any disturbance information required by Good Utility Practice.

ARTICLE 10. MAINTENANCE

10.1 Transmission Owner Obligations. Transmission Owner shall maintain the Transmission Owner's Interconnection Facilities in a safe and reliable manner and in accordance with this GIA and all Applicable Laws and Regulations.

10.2 Interconnection Customer Obligations. Interconnection Customer shall maintain the Generating Facility and the Interconnection Customer's Interconnection Facilities in a safe and reliable manner and in accordance with this GIA and all Applicable Laws and Regulations.

10.3 Coordination. The Parties shall confer regularly to coordinate the planning, scheduling and performance of preventive and corrective maintenance on the Generating Facility and the Interconnection Facilities.

10.4 Secondary Systems. Each Party shall cooperate with the other in the inspection, maintenance, and testing of control or power circuits that operate below 600 volts, AC or DC, including, but not limited to, any hardware, control or protective devices, cables, conductors, electric raceways, secondary equipment panels, transducers, batteries, chargers, and voltage and current transformers that directly affect the operation of a Party's facilities and equipment which may reasonably be expected to impact another Party. Each Party shall provide advance notice to the other Parties before undertaking any work on such circuits, especially on electrical circuits involving circuit breaker trip and close contacts, current transformers, or potential transformers.

10.5 Operating and Maintenance Expenses. Subject to the provisions herein addressing the use of facilities by others, and except for operations and maintenance expenses associated with modifications made for providing Interconnection Service or Transmission Service to a non-Party and such non-Party pays for such expenses, Interconnection Customer shall be responsible for all reasonable expenses including overheads, associated with: (1) owning, operating, maintaining, repairing, and replacing Interconnection Customer's Interconnection Facilities; and (2) operation, maintenance, repair and replacement of Transmission Owner's Interconnection Facilities to the extent required by Transmission Owner on a comparable basis.

ARTICLE 11. PERFORMANCE OBLIGATION

11.1 Interconnection Customer's Interconnection Facilities. Interconnection Customer shall design, procure, construct, install, own and/or control the Interconnection Customer's Interconnection Facilities described in Appendix A at its sole expense.

11.2 Transmission Owner's Interconnection Facilities. Transmission Owner shall design, procure, construct, install, own and/or control the Transmission Owner's Interconnection Facilities described in Appendix A at the sole expense of Interconnection Customer.

11.3 Network Upgrades, System Protection Facilities and Distribution Upgrades. Transmission Owner shall design, procure, construct, install, and own the Network Upgrades, Transmission Owner's System Protection Facilities and Distribution Upgrades

described in Appendix A. Interconnection Customer shall be responsible for all costs related to Distribution Upgrades and/or Generator Upgrades. As required by Section 7.3.2.1 and 7.3.3.1 of Attachment X, Transmission Owner shall have provided Transmission Provider with written notice if Transmission Owner elects to fund the capital for the Network Upgrades and Transmission Owner's System Protection Facilities; otherwise, such facilities, if any, shall be solely funded by Interconnection Customer. In the event that the Transmission Owner elects to fund the capital costs of the Network Upgrades and/or Transmission Owner's System Protection Facilities, then the Parties shall enter into a Facilities Service Agreement. The Facilities Service Agreement shall take the form of the *pro forma* Facilities Service Agreement that is included as Appendix 14 of Attachment X to the MISO Tariff.

11.3.1 Contingencies Affecting Network Upgrades, System Protection Facilities and Distribution Upgrades**.** Network Upgrades, System Protection Facilities and Distribution Upgrades that are required to accommodate the Generating Facility may be modified because (1) a higher queued interconnection request withdrew or was deemed to have withdrawn, (2) the interconnection agreement associated with a higher queued interconnection request was terminated prior to the project's In-Service Date, (3) the Commercial Operation Date for a higher queued interconnection request is delayed, or the project itself is delayed (including due to suspension) such that facilities required to accommodate lower queued projects or the project itself may be altered, (4) the queue position is reinstated for a higher-queued interconnection request whose queue position was subject to dispute resolution, (5) changes occur in Transmission Provider or Transmission Owner equipment design standards or reliability criteria giving rise to the need for restudy, (6) the facilities required to accommodate a higher queued Interconnection Request were modified constituting a Material Modification pursuant to Section 4.4 of the GIP, (7) a GIA with an effective date prior to this GIA is terminated, (8) when ordered to restudy by FERC, or (9) the Network Upgrade Facilities Study for a Network Upgrade, which is to be included in this GIA, is completed after execution of this GIA. The higher queued Interconnection Requests that could impact the Network Upgrades, System Protection Facilities and Distribution Upgrades required to accommodate the Generating Facility, and possible Modifications that may result from the above listed events affecting the higher queued Interconnection Requests, to the extent such modifications are reasonably known and can be determined, and estimates of the costs associated with such required Network Upgrades, System Protection Facilities and Distribution Upgrades, are provided in Appendix A.

11.3.2 Agreement to Restudy and Cost Reallocation**.** In the event that one of the contingencies listed in Article 11.3.1 occurs, at any time before the Network Upgrades, Common Use Upgrades, Shared Network Upgrades, System Protection Facilities and/or Distribution Upgrades associated with higher queued Interconnection Requests with GIA in effect prior to this GIA are completed, Transmission Provider may determine, in its discretion, that a restudy is required. If a restudy is required, Transmission Provider will provide notice to

Interconnection Customer and Interconnection Customer agrees to enter into an Interconnection Study Agreement for such restudy. Transmission Provider will reevaluate the need for any Common Use Upgrade(s) and/or Shared Network Upgrade(s), and if still required, reallocate the cost and responsibility for any Common Use Upgrade and/or Shared Network Upgrade, without a restudy when possible, or with a restudy if the Transmission Provider deems it necessary in order to ensure reliability of the Transmission System. The Parties agree to amend Appendix A to this GIA in accordance with Article 30.10 to reflect the results of any cost reallocation required under this Article 11.3.2.

11.3.3 Agreement to Fund Shared Network Upgrades. Interconnection Customer agrees to fund Shared Network Upgrades, as determined by Transmission Provider. Where applicable, payments to fund Shared Network Upgrade(s) that are made to Transmission Provider by Interconnection Customer will be disbursed by Transmission Provider to the appropriate entities that funded the Shared Network Upgrades in accordance with Attachment X and Attachment FF of the Tariff. In the event that Interconnection Customer fails to meet its obligation to fund Shared Network Upgrades, Transmission Owner and Transmission Provider shall not be responsible for the Interconnection Customer's funding obligation.

11.4 Transmission Credits.

11.4.1 Repayment of Amounts Advanced for Network Upgrades. Interconnection Customer shall be entitled to a cash repayment by Transmission Owner(s) and the Affected System Owner(s) that own the Network Upgrades, of the amount paid respectively to Transmission Owner and Affected System Operator, if any, for the Network Upgrades, as provided under Attachment FF of this Tariff and including any tax gross-up or other tax-related payments associated with the repayable portion of the Network Upgrades, and not repaid to Interconnection Customer pursuant to Article 5.17.8 or otherwise, to be paid to Interconnection Customer on a dollar-for-dollar basis for the non-usage sensitive portion of transmission charges, as payments are made under the Tariff and Affected System's Tariff for Transmission Services with respect to the Generating Facility. Any repayment shall include interest calculated in accordance with the methodology set forth in FERC's regulations at 18 C.F.R. § 35.19 (a)(2)(iii) from the date of any payment for Network Upgrades through the date on which Interconnection Customer receives a repayment of such payment pursuant to this subparagraph. Interest shall not accrue during periods in which Interconnection Customer has suspended construction pursuant to Article 11 or the Network Upgrades have been determined not to be needed pursuant to this Article 11.4.1. Interconnection Customer may assign such repayment rights to any person.

If the Generating Facility is designated a Network Resource under the Tariff, or if there are otherwise no incremental payments for Transmission Service resulting from the use of the Generating Facility by Transmission Customer, and

in the absence of another mutually agreeable payment schedule any repayments provided under Attachment FF shall be established equal to the applicable rate for Firm Point-To-Point Transmission Service for the pricing zone where the Network Load is located multiplied by the portion of the demonstrated output of the Generating Facility designated as a Network Resource by the Network Customer(s) or in the absence of such designation, equal to the monthly firm single system-wide rate defined under Schedule 7 of the Tariff multiplied by the portion of the demonstrated output of the Generating Facility under contract to Network Customer(s) and consistent with studies pursuant to Section 3.2.2.2 of the GIP.

Notwithstanding the foregoing, as applicable and consistent with the provisions of Attachment FF of this Tariff, Interconnection Customer, Transmission Provider, Transmission Owner, and Affected System Operator may adopt any alternative payment schedule that is mutually agreeable so long as Transmission Owner and Affected System Operator take one of the following actions no later than five (5) years from the Commercial Operation Date: (1) return to Interconnection Customer any amounts advanced for Network Upgrades not previously repaid, or (2) declare in writing that Transmission Owner or Affected System Operator will continue to provide payments to Interconnection Customer on a dollar-for-dollar basis for the non-usage sensitive portion of transmission charges, or develop an alternative schedule that is mutually agreeable and provides for the return of all amounts advanced for Network Upgrades not previously repaid; however, full reimbursement shall not extend beyond twenty (20) years from the Commercial Operation Date.

If the Generating Facility is installed in phases, the amount eligible for refund as each phase achieves Commercial Operation will be reduced by the proportional amount of generation capacity not yet installed. However, all facilities in Appendix A other than the Generating Facility shall be built without consideration for the phasing of the Generating Facility as though the entire Generating Facility will be placed in Commercial Operation for the full output or increased output of the Generating Facility constructed by Interconnection Customer under this GIA.

If the Generating Facility fails to achieve Commercial Operation, but it or another generating facility is later constructed and makes use of the Network Upgrades, Transmission Owner and Affected System Operator shall at that time reimburse Interconnection Customer for the remaining applicable amounts that may be refundable pursuant to Attachment FF of this Tariff that were advanced for the Network Upgrades on their respective systems as described above. Before any such reimbursement can occur, Interconnection Customer, or the entity that ultimately constructs the Generating Facility, if different, is responsible for identifying the entity to which the reimbursement must be made.

11.4.2 Special Provisions for Transmission Provider as an Affected System to be covered under Separate Agreements. When the Transmission Owner's Transmission or Distribution System (including for this Article 11.4.2 independent distribution systems connected to the Transmission System) is an Affected System for an interconnection in another electric system, Transmission Provider will coordinate the performance of Interconnection Studies with the other system. Transmission Provider will determine if any Network Upgrades or Distribution Upgrades, which may be required on the Transmission System as a result of the interconnection, would not have been needed but for the interconnection. Unless Transmission Owner provides, under the interconnection agreement between Interconnection Customer and the other system, for the repayment of amounts advanced to Transmission Provider or an impacted Transmission Owner for Network Upgrades, Interconnection Customer, Transmission Provider, and the impacted Transmission Owner(s) shall enter into an agreement that provides for such repayment by Transmission Owner(s) as directed by Transmission Provider. The agreement shall specify the terms governing payments to be made by Interconnection Customer to the Affected System Operator as well as the payment of refunds by the Affected System Operator.

11.4.3 Notwithstanding any other provision of this GIA, nothing herein shall be construed as relinquishing or foreclosing any rights, including but not limited to firm transmission rights, capacity rights, transmission congestion rights, or transmission credits, that Interconnection Customer, shall be entitled to, now or in the future under any other agreement or tariff as a result of, or otherwise associated with, the transmission capacity, if any, created by the Network Upgrades, including the right to obtain cash reimbursement or transmission credits for transmission service that is not associated with the Generating Facility.

11.5 Initial Payment. Interconnection Customer shall elect (and provide its election to the Transmission Provider within five days of the commencement of negotiation of the GIA pursuant to Section 11.2 of the GIP) to make either 1) an initial payment equal to twenty (20) percent of the total cost of Network Upgrades, Transmission Owner Interconnection Facilities, Transmission Owner's System Protection Facilities, Distribution Upgrades and/or Generator Upgrades (if the In-Service Date is less than or equal to five (5) years of the initial payment date); or 2) an initial payment equal to ten (10) percent of the total cost of Network Upgrades, Transmission Owner Interconnection Facilities, Transmission Owner's System Protection Facilities, Distribution Upgrades and/or Generator Upgrades (if the In-Service Date exceeds the initial payment date by more than five (5) years); or 3) the total cost of Network Upgrades, Transmission Owner Interconnection Facilities, Transmission Owner's System Protection Facilities, Distribution Upgrades and/or Generator Upgrades in the form of security pursuant to Article 11.6. The initial payment shall be provided to Transmission Owner by Interconnection Customer pursuant to this Article 11.5 within the later of a) forty-five (45) Calendar Days of the execution of the GIA by all Parties, or b) forty-five (45) Calendar Days of acceptance by FERC if the GIA

is filed unexecuted and the payment is being protested by Interconnection Customer, or c) forty-five (45) Calendar Days of the filing if the GIA is filed unexecuted and the initial payment is not being protested by Interconnection Customer. If the Interconnection Customer made its milestone payments in the form of cash and the Interconnection Customer elects a cash initial payment, then the Transmission Provider shall transfer those funds to the Transmission Owner on the Interconnection Customer's behalf.

11.6 Provision of Security. Unless otherwise provided in Appendix B, at least forty-five (45) Calendar Days prior to the commencement of the design, procurement, installation, or construction of a discrete portion of an element, not otherwise funded under Article 11.5, of the Transmission Owner's Interconnection Facilities, Transmission Owner's System Protection Facilities, Network Upgrades, Distribution Upgrades or Stand-Alone Network Upgrades, or at the request of Transmission Owner if regulatory approvals are required for the construction of such facilities, Interconnection Customer shall provide Transmission Owner, at Interconnection Customer's selection, a guarantee, a surety bond, letter of credit or other form of security that is reasonably acceptable to Transmission Owner and is consistent with the Uniform Commercial Code of the jurisdiction identified in Article 14.2.1. Such security for payment shall be in an amount sufficient to cover the applicable costs and cost commitments, in addition to those funded under Article 11.5, required of the Party responsible for building the facilities pursuant to the construction schedule developed in Appendix B for designing, engineering, seeking regulatory approval from any Governmental Authority, constructing, procuring and installing the applicable portion of the Transmission Owner's Interconnection Facilities, Transmission Owner's System Protection Facilities, Network Upgrades, Distribution Upgrades or Stand-Alone Network Upgrades and shall be reduced on a dollar-for-dollar basis for payments made to Transmission Owner for these purposes.

In addition:

11.6.1 The guarantee must be made by an entity that meets the creditworthiness requirements of Transmission Owner, and contain terms and conditions that guarantee payment of any amount that may be due from Interconnection Customer, up to an agreed-to maximum amount.

11.6.2 The letter of credit must be issued by a financial institution reasonably acceptable to Transmission Owner and must specify a reasonable expiration date.

11.6.3 The surety bond must be issued by an insurer reasonably acceptable to Transmission Owner and must specify a reasonable expiration date.

11.6.4 If the Shared Network Upgrade is not in service, Interconnection Customer will not be required to make a payment under Schedule 26-B until the Shared Network Upgrade is in service, but Interconnection Customer will provide, as applicable, an Irrevocable Letter of Credit to fund any Shared Network Upgrade pursuant to Attachment FF of the Tariff. The Irrevocable Letter of Credit shall

be in an amount sufficient to cover the Interconnection Customer's share of the applicable costs and cost commitments associated with the Shared Network Upgrades. Transmission Provider may periodically adjust the Interconnection Customer's share of the applicable costs and cost commitment of Shared Network Upgrades and may require Interconnection Customer to adjust the amount of the Irrevocable Letter of Credit accordingly.

11.7 Interconnection Customer Compensation. If Transmission Provider requests or directs Interconnection Customer to provide a service pursuant to Article 13.4 of this GIA, Transmission Provider shall compensate Interconnection Customer in accordance with any tariff or rate schedule filed by Transmission Provider and approved by the FERC.

ARTICLE 12. INVOICE

12.1 General. Each Party shall submit to the other Party, on a monthly basis, invoices of amounts due, if any, for the preceding month. Each invoice shall state the month to which the invoice applies and fully describe the services and equipment provided. The Parties may discharge mutual debts and payment obligations due and owing to each other on the same date through netting, in which case all amounts a Party owes to the other Party under this GIA, including interest payments or credits, shall be netted so that only the net amount remaining due shall be paid by the owing Party.

12.2 Final Invoice. Within six (6) months after completion of the construction of the Transmission Owner's Interconnection Facilities, Transmission Owner's System Protection Facilities, Distribution Upgrades and the Network Upgrades, Transmission Owner shall provide an invoice of the final cost of the construction of the Transmission Owner's Interconnection Facilities, Transmission Owner's System Protection Facilities, Distribution Upgrades and the Network Upgrades and shall set forth such costs in sufficient detail to enable Interconnection Customer to compare the actual costs with the estimates and to ascertain deviations, if any, from the cost estimates. Transmission Owner shall refund, with interest (calculated in accordance with 18 C.F.R. Section 35.19a(a)(2)(iii), to Interconnection Customer any amount by which the actual payment by Interconnection Customer for estimated costs exceeds the actual costs of construction within thirty (30) Calendar Days of the issuance of such final construction invoice.

12.3 Payment. Invoices shall be rendered to the paying Party at the address specified in Appendix F. The Party receiving the invoice shall pay the invoice within thirty (30) Calendar Days of receipt. All payments shall be made in immediately available funds payable to the other Party, or by wire transfer to a bank named and account designated by the invoicing Party. Payment of invoices by a Party will not constitute a waiver of any rights or claims that Party may have under this GIA.

12.4 Disputes. In the event of a billing dispute among the Parties, Transmission Provider shall continue to provide Interconnection Service under this GIA as long as Interconnection Customer: (i) continues to make all payments not in dispute; and (ii) pays to Transmission Provider or Transmission Owner or into an independent escrow

account the portion of the invoice in dispute, pending resolution of such dispute. If Interconnection Customer fails to meet these two requirements for continuation of service, then Transmission Provider may or, at Transmission Owner's request upon Interconnection Customer's failure to pay, Transmission Owner, shall provide notice to Interconnection Customer of a Default pursuant to Article 17. Within thirty (30) Calendar Days after the resolution of the dispute, the Party that owes money to another Party shall pay the amount due with interest calculated in accord with the methodology set forth in 18 C.F.R. § 35.19a(a)(2)(iii).

ARTICLE 13. EMERGENCIES

13.1 Obligations. Each Party shall comply with the Emergency Condition procedures of Transmission Provider, NERC, the Applicable Reliability Council, and Applicable Laws and Regulations.

13.2 Notice. Transmission Provider or Transmission Owner shall notify the other Parties promptly when it becomes aware of an Emergency Condition that affects the Transmission Owner's Interconnection Facilities or the Transmission or Distribution System, as applicable, that may reasonably be expected to affect Interconnection Customer's operation of the Generating Facility or the Interconnection Customer's Interconnection Facilities.

Interconnection Customer shall notify Transmission Provider and Transmission Owner, which includes by definition if applicable, the operator of a Distribution System, promptly when it becomes aware of an Emergency Condition that affects the Generating Facility or the Interconnection Customer's Interconnection Facilities that may reasonably be expected to affect the Transmission or Distribution System, as applicable, or the Transmission Owner's Interconnection Facilities.

To the extent information is known, the notification shall describe the Emergency Condition, the extent of the damage or deficiency, the expected effect on the operation of Interconnection Customer's or Transmission Provider's or Transmission Owner's facilities and operations, its anticipated duration and the corrective action taken and/or to be taken. The initial notice shall be followed as soon as practicable with written notice.

13.3 Immediate Action. Unless, in a Party's reasonable judgment, immediate action is required, the Party exercising such judgment shall notify and obtain the consent of the other Parties, such consent to not be unreasonably withheld, prior to performing any manual switching operations at the Generating Facility or the Interconnection Customer's Interconnection Facilities in response to an Emergency Condition either declared by Transmission Provider or otherwise regarding the Transmission or Distribution System, as applicable.

13.4 Transmission Provider and Transmission Owner Authority.

13.4.1 **General**. Transmission Provider or Transmission Owner may take whatever actions or inactions with regard to the Transmission System or the Transmission Owner's Interconnection Facilities it deems necessary during an Emergency Condition in order to (i) preserve public health and safety, (ii) preserve the reliability of the Transmission System or the Transmission Owner's Interconnection Facilities, (iii) limit or prevent damage, and (iv) expedite restoration of service.

Transmission Provider or Transmission Owner shall use Reasonable Efforts to minimize the effect of such actions or inactions on the Generating Facility or the Interconnection Customer's Interconnection Facilities. Transmission Provider or Transmission Owner may, on the basis of technical considerations, require the Generating Facility to mitigate an Emergency Condition by taking actions necessary and limited in scope to remedy the Emergency Condition, including, but not limited to, directing Interconnection Customer to shut-down, start-up, increase or decrease the real or reactive power output of the Generating Facility; implementing a reduction or disconnection pursuant to Article 13.4.2; directing Interconnection Customer to assist with blackstart (if available) or restoration efforts; or altering the outage schedules of the Generating Facility and the Interconnection Customer's Interconnection Facilities. Interconnection Customer shall comply with all of Transmission Provider's or Transmission Owner's operating instructions concerning Generating Facility real power and reactive power output within the manufacturer's design limitations of the Generating Facility's equipment that is in service and physically available for operation at the time, in compliance with Applicable Laws and Regulations.

13.4.2 Reduction and Disconnection. Transmission Provider or Transmission Owner may reduce Interconnection Service or disconnect the Generating Facility or the Interconnection Customer's Interconnection Facilities, when such reduction or disconnection is necessary under Good Utility Practice due to Emergency Conditions. These rights are separate and distinct from any right of curtailment of Transmission Provider pursuant to the Tariff. When Transmission Provider can schedule the reduction or disconnection in advance, Transmission Provider shall notify Interconnection Customer of the reasons, timing and expected duration of the reduction or disconnection. Transmission Provider shall coordinate with Interconnection Customer and Transmission Owner using Good Utility Practice to schedule the reduction or disconnection during periods of least impact to Interconnection Customer, Transmission Owner and Transmission Provider. Any reduction or disconnection shall continue only for so long as reasonably necessary pursuant to Good Utility Practice. The Parties shall cooperate with each other to restore the Generating Facility, the Interconnection Facilities, and the Transmission System to their normal operating state as soon as practicable consistent with Good Utility Practice.

13.5 **Interconnection Customer Authority**. Consistent with Good Utility Practice and this GIA and the GIP, Interconnection Customer may take whatever actions or inactions with

regard to the Generating Facility or the Interconnection Customer's Interconnection Facilities during an Emergency Condition in order to (i) preserve public health and safety, (ii) preserve the reliability of the Generating Facility or the Interconnection Customer's Interconnection Facilities, (iii) limit or prevent damage, and (iv) expedite restoration of service. Interconnection Customer shall use Reasonable Efforts to minimize the effect of such actions or inactions on the Transmission System and the Transmission Owner's Interconnection Facilities. Transmission Provider and Transmission Owner shall use Reasonable Efforts to assist Interconnection Customer in such actions.

13.6 Limited Liability. Except as otherwise provided in Article 11.6 of this GIA, no Party shall be liable to any other for any action it takes in responding to an Emergency Condition so long as such action is made in good faith and is consistent with Good Utility Practice.

13.7 Audit. In accordance with Article 25.3, any Party may audit the performance of another Party when that Party declared an Emergency Condition.

ARTICLE 14. REGULATORY REQUIREMENTS AND GOVERNING LAW

14.1 Regulatory Requirements. Each Party's obligations under this GIA shall be subject to its receipt of any required approval or certificate from one or more Governmental Authorities in the form and substance satisfactory to the applying Party, or the Party making any required filings with, or providing notice to, such Governmental Authorities, and the expiration of any time period associated therewith. Each Party shall in good faith seek, and if necessary assist the other Party and use its Reasonable Efforts to obtain such other approvals. Nothing in this GIA shall require Interconnection Customer to take any action that could result in its inability to obtain, or its loss of, status or exemption under the Federal Power Act, the Public Utility Holding Company Act of 2005, as amended, or the Public Utility Regulatory Policies Act of 1978.

14.2 Governing Law.

14.2.1 The validity, interpretation and performance of this GIA and each of its provisions shall be governed by the laws of the state where the Point of Interconnection is located, without regard to its conflicts of law principles.

14.2.2 This GIA is subject to all Applicable Laws and Regulations.

14.2.3 Each Party expressly reserves the right to seek changes in, appeal, or otherwise contest any laws, orders, rules, or regulations of a Governmental Authority.

ARTICLE 15. NOTICES

15.1 General. Unless otherwise provided in this GIA, any notice, demand or request required or permitted to be given by any Party to the other Parties and any instrument required or

permitted to be tendered or delivered by a Party in writing to the other Parties shall be effective when delivered and may be so given, tendered or delivered, by recognized national courier, or by depositing the same with the United States Postal Service with postage prepaid, for delivery by certified or registered mail, addressed to the Party, or personally delivered to the Party, at the address set out in Appendix F, Addresses for Delivery of Notices and Billings.

Either Party may change the notice information in this GIA by giving five (5) Business Days written notice prior to the effective date of the change.

15.2 Billings and Payments. Billings and payments shall be sent to the addresses set out in Appendix F.

15.3 Alternative Forms of Notice. Any notice or request required or permitted to be given by any Party to the other and not required by this GIA to be given in writing may be so given by telephone, facsimile or email to the telephone numbers and email addresses set out in Appendix F.

15.4 Operations and Maintenance Notice. Each Party shall notify the other Parties in writing of the identity of the person(s) that it designates as the point(s) of contact with respect to the implementation of Articles 9 and 10.

ARTICLE 16. FORCE MAJEURE

16.1 Force Majeure.

16.1.1 Economic hardship is not considered a Force Majeure event.

16.1.2 A Party shall not be considered to be in Default with respect to any obligation hereunder, (including obligations under Article 4 and 5), other than the obligation to pay money when due, if prevented from fulfilling such obligation by Force Majeure. A Party unable to fulfill any obligation hereunder (other than an obligation to pay money when due) by reason of Force Majeure shall give notice and the full particulars of such Force Majeure to the other Parties in writing or by telephone as soon as reasonably possible after the occurrence of the cause relied upon. Telephone, facsimile or email notices given pursuant to this Article shall be confirmed in writing as soon as reasonably possible and shall specifically state full particulars of the Force Majeure, the time and date when the Force Majeure occurred and when the Force Majeure is reasonably expected to cease. The Party affected shall exercise Reasonable Efforts to remove such disability with reasonable dispatch, but shall not be required to accede or agree to any provision not satisfactory to it in order to settle and terminate a strike or other labor disturbance.

ARTICLE 17. DEFAULT

17.1 Default

17.1.1 General. No Default shall exist where such failure to discharge an obligation (other than the payment of money) is the result of Force Majeure as defined in this GIA or the result of an act or omission of another Party. Upon a Breach, the non-Breaching Party or Parties shall give written notice of such Breach to the Breaching Party with a copy to the other Party if one Party gives notice of such Breach. Except as provided in Article 17.1.2, the Breaching Party shall have thirty (30) Calendar Days from receipt of the Breach notice within which to cure such Breach; provided however, if such Breach is not capable of cure within thirty (30) Calendar Days, the Breaching Party shall commence such cure within thirty (30) Calendar Days after notice and continuously and diligently complete such cure within ninety (90) Calendar Days from receipt of the Breach notice; and, if cured within such time, the Breach specified in such notice shall cease to exist.

17.1.2 Termination. If a Breach is not cured as provided in this Article, or if a Breach is not capable of being cured within the period provided for herein, the non-Breaching Party or Parties shall terminate this GIA, subject to Article 2.3.2 of this GIA, by written notice to the Breaching Party, with a copy to the other Party if one Party gives notice of termination, and be relieved of any further obligation hereunder and, whether or not that Party(ies) terminates this GIA, to recover from the Breaching Party all amounts due hereunder, plus all other damages and remedies to which it is (they are) entitled at law or in equity. The provisions of this Article will survive termination of this GIA.

17.1.3 Cross-Default. Pursuant to Article 11.3 of this GIA, the Parties shall enter into a Facilities Service Agreement if the Transmission Owner has elected to fund the capital for the Network Upgrades and/or Transmission Owner's System Protection Facilities. Notwithstanding anything to the contrary contained in this GIA, a breach by Interconnection Customer of any provision, covenant, or other term or condition contained in the Facilities Service Agreement shall be considered a breach under this GIA. Such breach shall be subject to the terms of Article 17 of this GIA. If a default under this GIA results from the Interconnection Customer's breach of the Facilities Service Agreement and subsequent failure to cure, the Transmission Owner and Transmission Provider shall be entitled, but in no event required, to apply all rights and remedies available by reason of default under the Facilities Service Agreement and this GIA.

ARTICLE 18. LIMITATION OF LIABILITY, INDEMNITY, CONSEQUENTIAL DAMAGES AND INSURANCE

18.1 Limitation of Liability. A Party shall not be liable to another Party or to any third party or other person for any damages arising out of actions under this GIA, including, but not limited to, any act or omission that results in an interruption, deficiency or imperfection of Interconnection Service, except as provided in this Tariff. The provisions set forth in

the Tariff shall be additionally applicable to any Party acting in good faith to implement or comply with its obligations under this GIA, regardless of whether the obligation is preceded by a specific directive.

18.2 Indemnity. To the extent permitted by law, an Indemnifying Party shall at all times indemnify, defend and hold the other Parties harmless from Loss.

18.2.1 Indemnified Party. If an Indemnified Party is entitled to indemnification under this Article 18 as a result of a claim by a non-Party, and the Indemnifying Party fails, after notice and reasonable opportunity to proceed under Article 18.2, to assume the defense of such claim, such Indemnified Party may at the expense of the Indemnifying Party contest, settle or consent to the entry of any judgment with respect to, or pay in full, such claim.

18.2.2 Indemnifying Party. If an Indemnifying Party is obligated to indemnify and hold any Indemnified Party harmless under this Article 18, the amount owing to the Indemnified Party shall be the amount of such Indemnified Party's actual Loss, net of any insurance or other recovery.

18.2.3 Indemnity Procedures. Promptly after receipt by an Indemnified Party of any claim or notice of the commencement of any action or administrative or legal proceeding or investigation as to which the indemnity provided for in Article 18.2 may apply, the Indemnified Party shall notify the Indemnifying Party of such fact. Any failure of or delay in such notification shall not affect a Party's indemnification obligation unless such failure or delay is materially prejudicial to the Indemnifying Party.

The Indemnifying Party shall have the right to assume the defense thereof with counsel designated by such Indemnifying Party and reasonably satisfactory to the Indemnified Party. If the defendants in any such action include one or more Indemnified Parties and the Indemnifying Party and if the Indemnified Party reasonably concludes that there may be legal defenses available to it and/or other Indemnified Parties which are different from or additional to those available to the Indemnifying Party, the Indemnified Party shall have the right to select separate counsel to assert such legal defenses and to otherwise participate in the defense of such action on its own behalf. In such instances, the Indemnifying Party shall only be required to pay the fees and expenses of one additional attorney to represent an Indemnified Party or Indemnified Parties having such differing or additional legal defenses.

The Indemnified Party shall be entitled, at its expense, to participate in any such action, suit or proceeding, the defense of which has been assumed by the Indemnifying Party. Notwithstanding the foregoing, the Indemnifying Party (i) shall not be entitled to assume and control the defense of any such action, suit or proceedings if and to the extent that, in the opinion of the Indemnified Party and its counsel, such action, suit or proceeding involves the potential

imposition of criminal liability on the Indemnified Party, or there exists a conflict or adversity of interest between the Indemnified Party and the Indemnifying Party, in such event the Indemnifying Party shall pay the reasonable expenses of the Indemnified Party, and (ii) shall not settle or consent to the entry of any judgment in any action, suit or proceeding without the consent of the Indemnified Party, which shall not be reasonably withheld, conditioned or delayed.

18.3 Consequential Damages. Other than the Liquidated Damages heretofore described, in no event shall either Party be liable under any provision of this GIA for any losses, damages, costs or expenses for any special, indirect, incidental, consequential, or punitive damages, including but not limited to loss of profit or revenue, loss of the use of equipment, cost of capital, cost of temporary equipment or services, whether based in whole or in part in contract, in tort, including negligence, strict liability, or any other theory of liability; provided; however, that damages for which a Party may be liable to the other Party under another agreement will not be considered to be special, indirect, incidental, or consequential damages hereunder.

18.4 Insurance. Transmission Owner and Interconnection Customer shall, at their own expense, maintain in force throughout the period of this GIA pursuant to 18.4.9, and until released by the other Party, the following minimum insurance coverages, with insurers authorized to do business or an approved surplus lines carrier in the state where the Point of Interconnection is located:

18.4.1 Employers' Liability and Workers' Compensation Insurance providing statutory benefits in accordance with the laws and regulations of the state in which the Point of Interconnection is located.

18.4.2 Commercial General Liability Insurance including premises and operations, personal injury, broad form property damage, broad form blanket contractual liability coverage (including coverage for the contractual indemnification) products and completed operations coverage, coverage for explosion, collapse and underground hazards, independent contractors coverage, coverage for pollution to the extent normally available and punitive damages to the extent normally available and a cross liability endorsement, with minimum limits of One Million Dollars ($1,000,000) per occurrence/One Million Dollars ($1,000,000) aggregate combined single limit for personal injury, bodily injury, including death and property damage.

18.4.3 Comprehensive Automobile Liability Insurance, for coverage of owned and non-owned and hired vehicles, trailers or semi-trailers licensed for travel on public roads, with a minimum combined single limit of One Million Dollars ($1,000,000) each occurrence for bodily injury, including death, and property damage.

18.4.4 Excess Public Liability Insurance over and above the Employer's Liability, Commercial General Liability and Comprehensive Automobile Liability Insurance coverage, with a minimum combined single limit of Twenty Million Dollars ($20,000,000) per occurrence/Twenty Million Dollars ($20,000,000) aggregate.

18.4.5 The Commercial General Liability Insurance, Comprehensive Automobile Insurance and Excess Public Liability Insurance policies shall name the other Parties, their parents, associated and Affiliate companies and their respective directors, officers, agents, servants and employees ("Other Party Group") as additional insured. All policies shall contain provisions whereby the insurers waive all rights of subrogation in accordance with the provisions of this GIA against the Other Party Groups and provide thirty (30) Calendar Days' advance written notice to the Other Party Groups prior to anniversary date of cancellation or any material change in coverage or condition.

18.4.6 The Commercial General Liability Insurance, Comprehensive Automobile Liability Insurance and Excess Public Liability Insurance policies shall contain provisions that specify that the policies are primary and shall apply to such extent without consideration for other policies separately carried and shall state that each insured is provided coverage as though a separate policy had been issued to each, except the insurer's liability shall not be increased beyond the amount for which the insurer would have been liable had only one insured been covered. Each Party shall be responsible for its respective deductibles or retentions.

18.4.7 The Commercial General Liability Insurance, Comprehensive Automobile Liability Insurance and Excess Public Liability Insurance policies, if written on a Claims First Made Basis, shall be maintained in full force and effect for two (2) years after termination of this GIA, which coverage may be in the form of tail coverage or extended reporting period coverage if agreed by Transmission Owner and Interconnection Customer.

18.4.8 The requirements contained herein as to the types and limits of all insurance to be maintained by Transmission Owner and Interconnection Customer are not intended to and shall not in any manner, limit or qualify the liabilities and obligations assumed by Transmission Owner and Interconnection Customer under this GIA.

18.4.9 As of the date set forth in Appendix B, Milestones, and as soon as practicable after the end of each fiscal year or at the renewal of the insurance policy and in any event within ninety (90) Calendar Days thereafter, Interconnection Customer and Transmission Owner shall provide the other Party with certification of all insurance required in this GIA, executed by each insurer or by an authorized representative of each insurer.

18.4.10 Notwithstanding the foregoing, Transmission Owner or Interconnection Customer may self-insure to meet the minimum insurance requirements of Articles 18.4.1 through 18.4.8, to the extent it maintains a self-insurance program; provided that, Transmission Owner's or Interconnection Customer's senior secured debt is rated at investment grade, or better, by Standard & Poor's and that its self-insurance program meets minimum insurance requirements under Articles 18.4.1 through 18.4.8. For any period of time that a Transmission Owner's or Interconnection Customer's senior secured debt is unrated by Standard & Poor's or is rated at less than investment grade by Standard & Poor's, such Party shall comply with the insurance requirements applicable to it under Articles 18.4.1 through 18.4.9. In the event that Transmission Owner or Interconnection Customer is permitted to self-insure pursuant to this article, it shall notify the other Party that it meets the requirements to self-insure and that its self-insurance program meets the minimum insurance requirements in a manner consistent with that specified in Article 18.4.9.

18.4.11 Transmission Owner and Interconnection Customer agree to report to each other in writing as soon as practical all accidents or occurrences resulting in injuries to any person, including death, and any property damage arising out of this GIA.

ARTICLE 19. ASSIGNMENT

19.1 Assignment. This GIA may be assigned by any Party only with the written consent of the other Parties; provided that a Party may assign this GIA without the consent of the other Parties to any Affiliate of the assigning Party with an equal or greater credit rating and with the legal authority and operational ability to satisfy the obligations of the assigning Party under this GIA; and provided further that Interconnection Customer shall have the right to assign this GIA, without the consent of either Transmission Provider or Transmission Owner, for collateral security purposes to aid in providing financing for the Generating Facility, provided that Interconnection Customer will promptly notify Transmission Provider of any such assignment. Any financing arrangement entered into by Interconnection Customer pursuant to this Article will provide that prior to or upon the exercise of the secured party's, trustee's or mortgagee's assignment rights pursuant to said arrangement, the secured creditor, the trustee or mortgagee will notify Transmission Provider of the date and particulars of any such exercise of assignment right(s), including providing Transmission Provider and Transmission Owner with proof that it meets the requirements of Article 11.5 and 18.4. Any attempted assignment that violates this Article is void and ineffective. Any assignment under this GIA shall not relieve a Party of its obligations, nor shall a Party's obligations be enlarged, in whole or in part, by reason thereof. Where required, consent to assignment will not be unreasonably withheld, conditioned or delayed.

ARTICLE 20. SEVERABILITY

20.1 **Severability**. If any provision in this GIA is finally determined to be invalid, void or unenforceable by any court or other Governmental Authority having jurisdiction, such determination shall not invalidate, void or make unenforceable any other provision, agreement or covenant of this GIA; provided that if Interconnection Customer (or any non-Party, but only if such non-Party is not acting at the direction of either Transmission Provider or Transmission Owner) seeks and obtains such a final determination with respect to any provision of the Alternate Option (Article 5.1.2), or the Negotiated Option (Article 5.1.4), then none of these provisions shall thereafter have any force or effect and the Parties' rights and obligations shall be governed solely by the Standard Option (Article 5.1.1).

ARTICLE 21. COMPARABILITY

21.1 **Comparability**. The Parties will comply with all applicable comparability and code of conduct laws, rules and regulations including such laws, rules and regulations of Governmental Authorities establishing standards of conduct, as amended from time to time.

ARTICLE 22. CONFIDENTIALITY

22.1 **Confidentiality.** Confidential Information shall include, without limitation, all information relating to a Party's technology, research and development, business affairs, and pricing, and any information supplied by a Party to another Party prior to the execution of this GIA.

Information is Confidential Information only if it is clearly designated or marked in writing as confidential on the face of the document, or, if the information is conveyed orally or by inspection, if the Party providing the information orally informs the Party receiving the information that the information is confidential. The Parties shall maintain as confidential any information that is provided and identified by a Party as Critical Energy Infrastructure Information (CEII), as that term is defined in 18 C.F.R. Section 388.113(c). Such confidentiality will be maintained in accordance with this Article 22.

If requested by the receiving Party, the disclosing Party shall provide in writing, the basis for asserting that the information referred to in this Article warrants confidential treatment, and the requesting Party may disclose such writing to the appropriate Governmental Authority. Each Party shall be responsible for the costs associated with affording confidential treatment to its information.

22.1.1 Term**.** During the term of this GIA, and for a period of three (3) years after the expiration or termination of this GIA, except as otherwise provided in this Article 22 or with regard to CEII, each Party shall hold in confidence and shall not disclose to any person Confidential Information. CEII shall be treated in accordance with Commission policy and regulations.

22.1.2 Scope. Confidential Information shall not include information that the receiving Party can demonstrate: (1) is generally available to the public other than as a result of a disclosure by the receiving Party; (2) was in the lawful possession of the receiving Party on a non-confidential basis before receiving it from the disclosing Party; (3) was supplied to the receiving Party without restriction by a non-Party, who, to the knowledge of the receiving Party after due inquiry, was under no obligation to the disclosing Party to keep such information confidential; (4) was independently developed by the receiving Party without reference to Confidential Information of the disclosing Party; (5) is, or becomes, publicly known, through no wrongful act or omission of the receiving Party or Breach of this GIA; or (6) is required, in accordance with Article 22.1.7 of this GIA, Order of Disclosure, to be disclosed by any Governmental Authority or is otherwise required to be disclosed by law or subpoena, or is necessary in any legal proceeding establishing rights and obligations under this GIA. Information designated as Confidential Information will no longer be deemed confidential if the Party that designated the information as confidential notifies the receiving Party that it no longer is confidential.

22.1.3 Release of Confidential Information. No Party shall release or disclose Confidential Information to any other person, except to its Affiliates (limited by the Standards of Conduct requirements), subcontractors, employees, agents, consultants, or to non-parties who may be or are considering providing financing to or equity participation with Interconnection Customer, or to potential purchasers or assignees of Interconnection Customer, on a need-to-know basis in connection with this GIA, unless such person has first been advised of the confidentiality provisions of this Article 22 and has agreed to comply with such provisions. Notwithstanding the foregoing, a Party providing Confidential Information to any person shall remain primarily responsible for any release of Confidential Information in contravention of this Article 22.

22.1.4 Rights. Each Party retains all rights, title, and interest in the Confidential Information that it discloses to the receiving Party. The disclosure by a Party to the receiving Party of Confidential Information shall not be deemed a waiver by the disclosing Party or any other person or entity of the right to protect the Confidential Information from public disclosure.

22.1.5 No Warranties. By providing Confidential Information, no Party makes any warranties or representations as to its accuracy or completeness. In addition, by supplying Confidential Information, no Party obligates itself to provide any particular information or Confidential Information to another Party nor to enter into any further agreements or proceed with any other relationship or joint venture.

22.1.6 Standard of Care. Each Party shall use at least the same standard of care to protect Confidential Information it receives as it uses to protect its own Confidential Information from unauthorized disclosure, publication or

dissemination. Each Party may use Confidential Information solely to fulfill its obligations to another Party under this GIA or its regulatory requirements.

22.1.7 Order of Disclosure. If a court or a Government Authority or entity with the right, power, and apparent authority to do so requests or requires any Party, by subpoena, oral deposition, interrogatories, requests for production of documents, administrative order, or otherwise, to disclose Confidential Information, that Party shall provide the disclosing Party with prompt notice of such request(s) or requirement(s) so that the disclosing Party may seek an appropriate protective order or waive compliance with the terms of this GIA. Notwithstanding the absence of a protective order or waiver, the Party may disclose such Confidential Information which, in the opinion of its counsel, the Party is legally compelled to disclose. Each Party will use Reasonable Efforts to obtain reliable assurance that confidential treatment will be accorded any Confidential Information so furnished.

22.1.8 Termination of Agreement. Upon termination of this GIA for any reason, each Party shall, within ten (10) Calendar Days of receipt of a written request from another Party, use Reasonable Efforts to destroy, erase, or delete (with such destruction, erasure, and deletion certified in writing to the requesting Party) or return to the requesting Party, without retaining copies thereof, any and all written or electronic Confidential Information received from the requesting Party, except that each Party may keep one copy for archival purposes, provided that the obligation to treat it as Confidential Information in accordance with this Article 22 shall survive such termination.

22.1.9 Remedies**.** The Parties agree that monetary damages would be inadequate to compensate a Party for another Party's Breach of its obligations under this Article 22. Each Party accordingly agrees that the disclosing Party shall be entitled to equitable relief, by way of injunction or otherwise, if the receiving Party Breaches or threatens to Breach its obligations under this Article 22, which equitable relief shall be granted without bond or proof of damages, and the Breaching Party shall not plead in defense that there would be an adequate remedy at law. Such remedy shall not be deemed an exclusive remedy for the Breach of this Article 22, but shall be in addition to all other remedies available at law or in equity. The Parties further acknowledge and agree that the covenants contained herein are necessary for the protection of legitimate business interests and are reasonable in scope. No Party, however, shall be liable for indirect, incidental, or consequential or punitive damages of any nature or kind resulting from or arising in connection with this Article 22.

22.1.10 Disclosure to FERC, its Staff or a State**.** Notwithstanding anything in this Article 22 to the contrary, and pursuant to 18 CFR § 1b.20, if FERC or its staff, during the course of an investigation or otherwise, requests information from a Party that is otherwise required to be maintained in confidence pursuant to this GIA, the Party shall provide the requested information to FERC or its staff,

within the time provided for in the request for information. In providing the information to FERC or its staff, the Party must, consistent with 18 CFR § 388.112, request that the information be treated as confidential and non-public by FERC and its staff and that the information be withheld from public disclosure. Parties are prohibited from notifying the other Parties to this GIA prior to the release of the Confidential Information to FERC or its staff. The Party shall notify the other Parties to this GIA when it is notified by FERC or its staff that a request to release Confidential Information has been received by FERC, at which time any of the Parties may respond before such information would be made public, pursuant to 18 CFR § 388.112. Requests from a state regulatory body conducting a confidential investigation shall be treated in a similar manner if consistent with the applicable state rules and regulations.

22.1.11 Subject to the exception in Article 22.1.10, any information that a disclosing Party claims is competitively sensitive, commercial or financial information under this GIA shall not be disclosed by the receiving Party to any person not employed or retained by the receiving Party, except to the extent disclosure is (i) required by law; (ii) reasonably deemed by the receiving Party to be required to be disclosed in connection with a dispute between or among the Parties, or the defense of litigation or dispute; (iii) otherwise permitted by consent of the disclosing Party, such consent not to be unreasonably withheld; or (iv) necessary to fulfill its obligations under this GIA or as the Regional Transmission Organization or a Local Balancing Authority operator including disclosing the Confidential Information to a regional or national reliability organization. The Party asserting confidentiality shall notify the receiving Party in writing of the information that Party claims is confidential. Prior to any disclosures of that Party's Confidential Information under this subparagraph, or if any non-Party or Governmental Authority makes any request or demand for any of the information described in this subparagraph, the Party who received the Confidential Information from the disclosing Party agrees to promptly notify the disclosing Party in writing and agrees to assert confidentiality and cooperate with the disclosing Party in seeking to protect the Confidential Information from public disclosure by confidentiality agreement, protective order or other reasonable measures.

ARTICLE 23. ENVIRONMENTAL RELEASES

23.1 Each Party shall notify the other Parties, first orally and then in writing, of the release of any Hazardous Substances, any asbestos or lead abatement activities, or any type of remediation activities related to the Generating Facility or the Interconnection Facilities, each of which may reasonably be expected to affect another Party. The notifying Party shall: (i) provide the notice as soon as practicable, provided such Party makes a good faith effort to provide the notice no later than twenty-four hours after such Party becomes aware of the occurrence; and (ii) promptly furnish to the other Parties copies of any publicly available reports filed with any Governmental Authorities addressing such events.

ARTICLE 24. INFORMATION REQUIREMENTS

24.1 Information Acquisition. Transmission Provider, Transmission Owner and Interconnection Customer shall submit specific information regarding the electrical characteristics of their respective facilities to each other as described below and in accordance with Applicable Reliability Standards.

24.2 Information Submission by Transmission Provider and Transmission Owner The initial information submission by Transmission Provider to Interconnection Customer, with copy provided to Transmission Owner, shall occur no later than one hundred eighty (180) Calendar Days prior to Trial Operation and shall include Transmission or Distribution System information, as applicable and available, necessary to allow Interconnection Customer to select equipment and meet any system protection and stability requirements, unless otherwise mutually agreed to by the Parties. On a monthly basis, Transmission Owner shall provide Interconnection Customer a status report on the construction and installation of Transmission Owner's Interconnection Facilities, Transmission Owner's System Protection Facilities, Distribution Upgrades and Network Upgrades, including, but not limited to, the following information: (1) progress to date; (2) a description of the activities since the last report (3) a description of the action items for the next period; and (4) the delivery status of equipment ordered.

24.3 Updated Information Submission by Interconnection Customer. The updated information submission by Interconnection Customer to Transmission Provider, with copy to Transmission Owner, including manufacturer information, shall occur no later than one hundred eighty (180) Calendar Days prior to the Trial Operation. Interconnection Customer shall submit to Transmission Provider and Transmission Owner a completed copy of the Generating Facility data requirements contained in Appendix 1 to the GIP. It shall also include any additional information provided to Transmission Provider for the Interconnection Facilities Study. Information in this submission shall be the most current Generating Facility design or expected performance data. Information submitted for stability models shall be compatible with Transmission Provider standard models. If there is no compatible model, Interconnection Customer will work with a consultant mutually agreed to by Transmission Provider and Interconnection Customer to develop and supply a standard model and associated information.

If the Interconnection Customer's data is materially different from what was originally provided to Transmission Provider pursuant to the Interconnection Study Agreement between Transmission Provider and Interconnection Customer, then Transmission Provider will conduct appropriate studies to determine the impact on the Transmission System based on the actual data submitted pursuant to this Article 24.3. Interconnection Customer shall not begin Trial Operation until such studies are completed.

24.4 Information Supplementation. Prior to the Commercial Operation Date, the Parties shall supplement their information submissions described above in this Article 24 with any and all "as-built" Generating Facility information or "as-tested" performance

information that differs from the initial submissions or, alternatively, written confirmation that no such differences exist. Interconnection Customer shall conduct tests on the Generating Facility as required by Good Utility Practice, such as an open circuit "step voltage" test on the Generating Facility to verify proper operation of the Generating Facility's automatic voltage regulator.

Unless otherwise agreed, the test conditions shall include: (1) Generating Facility at synchronous speed; (2) automatic voltage regulator on and in voltage control mode; and (3) a five percent (5 %) change in Generating Facility terminal voltage initiated by a change in the voltage regulators reference voltage. Interconnection Customer shall provide validated test recordings showing the responses in Generating Facility terminal and field voltages. In the event that direct recordings of these voltages is impractical, recordings of other voltages or currents that mirror the response of the Generating Facility's terminal or field voltage are acceptable if information necessary to translate these alternate quantities to actual Generating Facility terminal or field voltages is provided. Generating Facility testing shall be conducted and results provided to Transmission Provider and Transmission Owner for each individual generating unit in a station.

Subsequent to the Commercial Operation Date, Interconnection Customer shall provide Transmission Provider and Transmission Owner any information changes due to equipment replacement, repair, or adjustment. Transmission Owner shall provide Interconnection Customer, with copy to Transmission Provider, any information changes due to equipment replacement, repair or adjustment in the directly connected substation or any adjacent Transmission Owner substation that may affect the Interconnection Customer's Interconnection Facilities equipment ratings, protection or operating requirements. The Parties shall provide such information no later than thirty (30) Calendar Days after the date of the equipment replacement, repair or adjustment.

ARTICLE 25. INFORMATION ACCESS AND AUDIT RIGHTS

25.1 Information Access. Each Party (the "disclosing Party") shall make available to the other Parties information that is in the possession of the disclosing Party and is necessary in order for the other Parties to: (i) verify the costs incurred by the disclosing Party for which another Party is responsible under this GIA; and (ii) carry out its obligations and responsibilities under this GIA. The Parties shall not use such information for purposes other than those set forth in this Article 25.1 and to enforce their rights under this GIA.

25.2 Reporting of Non-Force Majeure Events. A Party (the "notifying Party") shall notify the other Parties when the notifying Party becomes aware of its inability to comply with the provisions of this GIA for a reason other than a Force Majeure event. The Parties agree to cooperate with each other and provide necessary information regarding such inability to comply, including the date, duration, reason for the inability to comply, and corrective actions taken or planned to be taken with respect to such inability to comply. Notwithstanding the foregoing, notification, cooperation or information provided under

this Article shall not entitle any Party receiving such notification to allege a cause for anticipatory breach of this GIA.

25.3 Audit Rights. Subject to the requirements of confidentiality under Article 22 of this GIA, each Party shall have the right, during normal business hours, and upon prior reasonable notice to the other Parties, to audit at its own expense the other Parties' accounts and records pertaining to the Parties' performance or the Parties' satisfaction of obligations under this GIA. Such audit rights shall include audits of the other Parties' costs, calculation of invoiced amounts, the Transmission Provider's efforts to allocate responsibility for the provision of reactive support to the Transmission or Distribution System, as applicable, the Transmission Provider's efforts to allocate responsibility for interruption or reduction of generation, and each Party's actions in an Emergency Condition. Any audit authorized by this Article shall be performed at the offices where such accounts and records are maintained and shall be limited to those portions of such accounts and records that relate to each Party's performance and satisfaction of obligations under this GIA. Each Party shall keep such accounts and records for a period equivalent to the audit rights periods described in Article 25.4.

25.4 Audit Rights Periods.

25.4.1 Audit Rights Period for Construction-Related Accounts and Records. Accounts and records related to the design, engineering, procurement, and construction of the Transmission Owner's Interconnection Facilities, Transmission Owner's System Protection Facilities, Distribution Upgrades and Network Upgrades shall be subject to audit for a period of twenty-four months following Transmission Owner's issuance of a final invoice in accordance with Article 12.2.

25.4.2 Audit Rights Period for All Other Accounts and Records. Accounts and records related to a Party's performance or satisfaction of all obligations under this GIA other than those described in Article 25.4.1 shall be subject to audit as follows: (i) for an audit relating to cost obligations, the applicable audit rights period shall be twenty-four (24) months after the auditing Party's receipt of an invoice giving rise to such cost obligations; and (ii) for an audit relating to all other obligations, the applicable audit rights period shall be twenty-four (24) months after the event for which the audit is sought.

25.5 Audit Results. If an audit by a Party determines that an overpayment or an underpayment has occurred, a notice of such overpayment or underpayment shall be given to the Party or from whom the overpayment or underpayment is owed together with those records from the audit which support such determination.

ARTICLE 26. SUBCONTRACTORS

26.1 General. Nothing in this GIA shall prevent a Party from utilizing the services of any subcontractor as it deems appropriate to perform its obligations under this GIA; provided,

however, that each Party shall require its subcontractors to comply with all applicable terms and conditions of this GIA in providing such services and each Party shall remain primarily liable to the other Parties for the performance of such subcontractor.

26.2 Responsibility of Principal. The creation of any subcontract relationship shall not relieve the hiring Party of any of its obligations under this GIA. The hiring Party shall be fully responsible to the other Parties for the acts or omissions of any subcontractor the hiring Party hires as if no subcontract had been made; provided, however, that in no event shall Transmission Provider or Transmission Owner be liable for the actions or inactions of Interconnection Customer or its subcontractors with respect to obligations of Interconnection Customer under Article 5 of this GIA. Any applicable obligation imposed by this GIA upon the hiring Party shall be equally binding upon, and shall be construed as having application to, any subcontractor of such Party.

26.3 No Limitation by Insurance. The obligations under this Article 26 will not be limited in any way by any limitation of subcontractor's insurance.

ARTICLE 27. DISPUTES

27.1 Submission. In the event any Party has a dispute, or asserts a claim, that arises out of or in connection with this GIA or its performance, such Party (the "disputing Party") shall provide the other Parties with written notice of the dispute or claim ("Notice of Dispute"). Such dispute or claim shall be referred to a designated senior representative of each Party for resolution on an informal basis as promptly as practicable after receipt of the Notice of Dispute by the non-disputing Parties. In the event the designated representatives are unable to resolve the claim or dispute through unassisted or assisted negotiations within thirty (30) Calendar Days of the non-disputing Parties' receipt of the Notice of Dispute, such claim or dispute shall be submitted for resolution in accordance with the dispute resolution procedures of the Tariff.

ARTICLE 28. REPRESENTATIONS, WARRANTIES AND COVENANTS

28.1 General. Each Party makes the following representations, warranties and covenants:

28.1.1 Good Standing. Such Party is duly organized, validly existing and in good standing under the laws of the state in which it is organized, formed, or incorporated, as applicable; that it is qualified to do business in the state or states in which the Generating Facility, Interconnection Facilities and Network Upgrades owned by such Party, as applicable, are located; and that it has the corporate power and authority to own its properties, to carry on its business as now being conducted and to enter into this GIA and carry out the transactions contemplated hereby and perform and carry out all covenants and obligations on its part to be performed under and pursuant to this GIA.

28.1.2 Authority**.** Such Party has the right, power and authority to enter into this GIA, to become a Party hereto and to perform its obligations hereunder. This GIA is

a legal, valid and binding obligation of such Party, enforceable against such Party in accordance with its terms, except as the enforceability thereof may be limited by applicable bankruptcy, insolvency, reorganization or other similar laws affecting creditors' rights generally and by general equitable principles (regardless of whether enforceability is sought in a proceeding in equity or at law).

28.1.3 No Conflict. The execution, delivery and performance of this GIA does not violate or conflict with the organizational or formation documents, or bylaws or operating agreement, of such Party, or any judgment, license, permit, order, material agreement or instrument applicable to or binding upon such Party or any of its assets.

28.1.4 Consent and Approval. Such Party has sought or obtained, or, in accordance with this GIA will seek or obtain, each consent, approval, authorization, order, or acceptance by any Governmental Authority in connection with the execution, delivery and performance of this GIA, and it will provide to any Governmental Authority notice of any actions under this GIA that are required by Applicable Laws and Regulations.

ARTICLE 29. {RESERVED}

ARTICLE 30. MISCELLANEOUS

30.1 Binding Effect. This GIA and the rights and obligations hereof, shall be binding upon and shall inure to the benefit of the successors and assigns of the Parties hereto.

30.1.1 Reversion. If offered pursuant to an Agency Agreement under which this GIA is executed by Transmission Provider as agent for the relevant Transmission Owner, in the event that the relevant Agency Agreement terminates, any HVDC Service offered by Transmission Provider under this GIA shall revert to the relevant Transmission Owner and Transmission Provider shall be released from all obligations and responsibilities under this GIA.

30.2 Conflicts. In the event of a conflict between the body of this GIA and any attachment, appendices or exhibits hereto, the terms and provisions of the body of this GIA shall prevail and be deemed the final intent of the Parties.

30.3 Rules of Interpretation. This GIA, unless a clear contrary intention appears, shall be construed and interpreted as follows: (1) the singular number includes the plural number and vice versa; (2) reference to any person includes such person's successors and assigns but, in the case of a Party, only if such successors and assigns are permitted by this GIA, and reference to a person in a particular capacity excludes such person in any other capacity or individually; (3) reference to any agreement (including this GIA), document, instrument or tariff means such agreement, document, instrument, or tariff as amended or modified and in effect from time to time in accordance with the terms thereof and, if

applicable, the terms hereof; (4) reference to any Applicable Laws and Regulations means such Applicable Laws and Regulations as amended, modified, codified, or reenacted, in whole or in part, and in effect from time to time, including, if applicable, rules and regulations promulgated thereunder; (5) unless expressly stated otherwise, reference to any Article, Section or Appendix means such Article of this GIA or such Appendix to this GIA, or such Section to the GIP or such Appendix to the GIP, as the case may be; (6) "hereunder", "hereof", "herein", "hereto" and words of similar import shall be deemed references to this GIA as a whole and not to any particular Article or other provision hereof or thereof; (7) "including" (and with correlative meaning "include") means including without limiting the generality of any description preceding such term; and (8) relative to the determination of any period of time, "from" means "from and including", "to" means "to but excluding" and "through" means "through and including."

30.4 Entire Agreement. This GIA, including all Appendices and attachments hereto, constitutes the entire agreement between the Parties with reference to the subject matter hereof, and supersedes all prior and contemporaneous understandings or agreements, oral or written, between the Parties with respect to the subject matter of this GIA. There are no other agreements, representations, warranties, or covenants, which constitute any part of the consideration for, or any condition to, any Party's compliance with its obligations under this GIA.

30.5 No Third Party Beneficiaries. This GIA is not intended to and does not create rights, remedies, or benefits of any character whatsoever in favor of any persons, corporations, associations, or entities other than the Parties, and the obligations herein assumed are solely for the use and benefit of the Parties, their successors in interest and, where permitted, their assigns.

30.6 Waiver. The failure of a Party to this GIA to insist, on any occasion, upon strict performance of any provision of this GIA will not be considered a waiver of any obligation, right, or duty of, or imposed upon, such Party.

Any waiver at any time by any Party of its rights with respect to this GIA shall not be deemed a continuing waiver or a waiver with respect to any other failure to comply with any other obligation, right, duty of this GIA. Termination or Default of this GIA for any reason by Interconnection Customer shall not constitute a waiver of the Interconnection Customer's legal rights to obtain Interconnection Service from Transmission Provider. Any waiver of this GIA shall, if requested, be provided in writing.

30.7 Headings. The descriptive headings of the various Articles of this GIA have been inserted for convenience of reference only and are of no significance in the interpretation or construction of this GIA.

30.8 Multiple Counterparts. This GIA may be executed in two or more counterparts, each of which is deemed an original but all constitute one and the same instrument.

30.9 Amendment. The Parties may by mutual agreement amend this GIA by a written instrument duly executed by all of the Parties.

30.10 Modification by the Parties. The Parties may by mutual agreement amend the Appendices to this GIA by a written instrument duly executed by all of the Parties. Such amendment shall become effective and a part of this GIA upon satisfaction of all Applicable Laws and Regulations.

30.11 Reservation of Rights. Transmission Provider shall have the right to make a unilateral filing with FERC to modify this GIA with respect to any rates, terms and conditions, charges, classifications of service, rule or regulation under Section 205 or any other applicable provision of the Federal Power Act and FERC's rules and regulations thereunder, and Transmission Owner and Interconnection Customer shall have the right to make a unilateral filing with FERC to modify this GIA pursuant to Section 206 or any other applicable provision of the Federal Power Act and FERC's rules and regulations thereunder; provided that each Party shall have the right to protest any such filing and to participate fully in any proceeding before FERC in which such modifications may be considered. Nothing in this GIA shall limit the rights of the Parties or of FERC under Sections 205 or 206 of the Federal Power Act and FERC's rules and regulations thereunder, except to the extent that the Parties otherwise mutually agree as provided herein.

30.12 No Partnership. This GIA shall not be interpreted or construed to create an association, joint venture, agency relationship, or partnership among or between the Parties or to impose any partnership obligation or partnership liability upon any Party. No Party shall have any right, power or authority to enter into any agreement or undertaking for, or act on behalf of, or to act as or be an agent or representative of, or to otherwise bind, the other Parties.

IN WITNESS WHEREOF, the Parties have executed this GIA in multiple originals; each of which shall constitute and be an original GIA among the Parties.

Transmission Provider
Midcontinent Independent System Operator, Inc.

By: _____ May 29, 2025 kkh

Name: Aubrey Johnson

Title: VP, System Planning & Competitive Development

Transmission Owner
Entergy Louisiana, LLC

By: John T. Anderson Digitally signed by John T. Anderson DN: cn=John T. Anderson, o=Entergy, ou=Power Delivery, email=jande18@entergy.com Date: 2025.04.24 09:26:05 -05'00'

Name: John T. Anderson

Title: VP - Project & Portfolio Development

Interconnection Customer
Lacassine Solar II, LLC

By: _____

Name: David Groberg

Title: Manager

APPENDICES TO GIA

Appendix A Interconnection Facilities, System Protection Facilities, Distribution Upgrades, Generator Upgrades and Network Upgrade

Appendix B Milestones

Appendix B-1 Pre-Certification Generation Test Notification Form

Appendix C Interconnection Details

Appendix D Security Arrangements Details

Appendix E Commercial Operation Date

Appendix F Addresses for Delivery of Notices and Billings

Appendix G Interconnection Requirements for a Non-synchronous Generating Facility

Appendix H Interconnection Requirements for Provisional GIA

Appendix I Requirements Applicable to Surplus Interconnection Service

Appendix A To GIA
In
Interconnection Facilities, System Protection Facilities, Distribution Upgrades, Generator Upgrades and Network Upgrade

1. Description of Generating Facility

Interconnection Customer shall install a 112.53 MVA facility, rated at 101.277 MW gross and 100 MW net, with all studies performed at or below these outputs. The Generating Facility is composed of thirty-one (31) Power Electronics FS3510 inverter units in a solar farm rated at 3.63 MVA each. Interconnection Service provided under this agreement is 100 MW of conditional ERIS that will become 100 MW of ERIS upon completion of all Network Upgrades under this GIA and transmission assumptions listed in Table A10-1 of Exhibit A10.

Interconnection Customer shall install a switchyard with the appropriate protection equipment coordinated per Appendix C to this GIA. The Switchyard shall contain one (1) generator step-up transformer 138/34.5 kV 66/88/110 MVA, one (1) 138kV, 1200 A circuit breaker connected in series fashion as described in Exhibit A1.

The generating facility shall provide reactive compensation as necessary to meet the FERC Order 827.

2. Interconnection Facilities

(a) Point of Interconnection.

i. The Point of Interconnection (POI) shall be at the point where the J1795 lead line connects to the 138 kV bus located in the Transmission Owner's new J1795 interconnection substation.

ii. The Point of Change of Ownership (PCO) shall be at the dead-end structure outside Transmission Owner's new J1795 interconnection substation.

iii. The Metering Point will be located on the Transmission Owner Interconnection Facilities and compensated to the Point of Interconnection. Refer to Exhibit A2 one-line diagram for the demarcation points of these facilities.

(b) Interconnection Facilities (including metering equipment) to be constructed by Interconnection Customer. Interconnection Customer shall construct Interconnection Customer Interconnection Facilities and are detailed in Exhibit A1. These facilities shall include:

- Approximately 1 mile of 138kV generator lead line 795 kcmil.
- A servitude agreement will need to be executed between Interconnection Customer and Entergy for the switching station area, transmission line, PCO to POI, and access road(s). Actual size and final location will be determined after detail design is complete. Below is a list of the servitudes and their preliminary dimensions:
    a. The transmission servitude will be approximately 120 feet wide 2640

feet long, approximately 7.3 acres.
b. The substation pad servitude will be approximately 320 feet wide by 482 feet long, approximately 3.54 acres.
c. The PCO to POI servitude will approximately 120 feet wide 300 feet long, approximately 0.8 acres.

- The Interconnection Customer will provide to Entergy a copy of wetland delineation for the new switching station site, access road and transmission line ROW, ESA Phase I report, and cultural resources constraint analysis/archeological survey, if required.
- Interconnection customer shall use SEL-411L relays at the interconnection end of the customer line and provide a communications link between the Transmission Owner and the IC's collector station for relaying purposes. Entergy require that the customer use fiber for this communication.

(c) Transmission Owner Interconnection Facilities to be constructed by Transmission Owner.

i. J1795 POI Switching Station:

The Transmission Owner's Interconnection Facilities (TOIFs) are the facilities and equipment at the interconnection substation that are owned by the Transmission Owner from the Point of Interconnection (POI) to the Point of Change of Ownership (PCO). Refer to **Error! Reference source not found.** for a p reliminary one-line diagram with demarcation of the facilities within the interconnection substation.

The general design assumption is that the new J1795 POI station will be constructed adjacent to L-258 138 kV transmission line (Bayou Cove – Richard) and interconnected via a 138 kV tie-line to the PCO. A single dead-end structure owned by the Interconnection Customer is needed in order to adequately accommodate the connection from the Interconnection Customer. This transmission line dead-end structure will be installed off of Entergy's switchyard property and will be the PCO.

The installation of the dead-end structure at the PCO, including grounding and the fiber splice box, shall be the sole responsibility of the Interconnection Customer. Only the conductor, insulators, shield wire, and attachment hardware required for the span between the PCO and the dead-end inside the POI switching station are included in the TOIF transmission line scope and estimate. A servitude between the PCO structure and the edge of Entergy's switchyard property will be provided by the Interconnection Customer for the installation and maintenance of the span between the PCO structure and the POI switching station dead-end structure.

The Transmission Owner's Interconnection Facilities shall consist of a 138 kV high bus dead-end structure, a section of 138 kV tubular bus tapping the 138 kV ring bus, a motor operated 138 kV vertical break disconnect switch with ground switch, metering accuracy current transformers, and a coupling capacitor voltage

transformer. The facilities shall include associated support structures and associated foundations, as shown on **Error! Reference source not found.** which i s a preliminary electrical profile view of the TOIFs.

The TOIFs shall also include a suitable revenue accuracy meter and power quality meter for measuring power deliveries from the generator.

General

a) Customer shall transmit generator breaker status, telemetry data including MW, MWhr, and MVARs to Entergy's new station RTU via the fiber installed on the customer's line.
b) Customer shall design line protection compatible with the line relaying installed at the new station.

Foundations

a) The foundation work shall be in accordance with Entergy standards and design drawings.
b) Drilled pier and pad foundations will be used for all equipment support structures.
c) The foundation scope of work includes design and construction of the following:
    - One (1) set of two (2) 66-inch diameter 35-foot-long drilled piers for the three phase 138 kV high bus H-Frame dead-end structures.
    - Six (6) 36-inch diameter 15-foot-long drilled piers for the single phase 138 kV coupling capacitor voltage transformer, current transformer, and high bus support structures.
    - One (1) set of two (2) 42-inch diameter 20-foot-long drilled piers for the 138 kV high disconnect switch support structure.

Electrical

a) Steel Structures
    - One (1) three phase 138 kV H-frame dead-end structure with provisions for high bus supports.
    - One (1) three phase 138 kV high bus disconnect switch support structure.
    - Two (2) single phase 138 kV high bus support structures.
    - One (1) single phase 138 kV coupling capacitor voltage transformer support structure.
    - Three (3) single phase 138 kV current transformer support structures.
b) Station Post Insulators
    - Five (5) high strength, porcelain station post insulators rated for nominal system voltage of 138 kV, 650 kV BIL, minimum 116"

creepage distance for H-frame high bus support and single-phase high bus supports.
- Nine (9) standard strength, porcelain station post insulators rated for nominal system voltage of 138 kV, 650 kV BIL, minimum 116" creepage distance for 138 kV vertical break line disconnect switch.
- Three (3) station post insulators rated for nominal system voltage of 34.5 kV, 200 kV BIL, minimum 37" creepage distance for in-line bus isolation and CT metering.
- Installation includes fitting accessories.

c) Suspension Insulators
- Three (3) polymer suspension insulators rated for nominal system voltage of 161 kV, 71" long, 25,000 lb., ANSI 52-5.
- Installation includes fitting accessories.

d) 138 kV Disconnect Switch
- One (1) three phase motor operated vertical break disconnect switch, with ground switch attachment, mounted horizontally on the high bus switch structure.
- Switch to be rated 145 kV, 2000 A continuous, 100 kA momentary withstand capacity, 650 kV BIL.
- Includes installing control conduits to the motor operator cabinet, and control cable to the control building.

e) 138 kV Conductor
- The main bus will be 4" aluminum (6063-T6) schedule 40 rigid tubular bus.
- The H-frame vertical bus and current transformer equipment jumpers shall be 954 kcmil ACSR Rail, two conductors per phase.
- The equipment jumpers for the coupling capacitor voltage transformers shall be 954 kcmil ACSR Rail, one conductor per phase.
- Includes all necessary fittings.

f) 138 kV Instrument Transformers
- One (1) coupling capacitor voltage transformer to be rated 138 kV, 650 BIL, 700/1200:1:1:1 ratio, 0.3% accuracy, WXYZ,ZZ load burden, 1000 VA thermal burden with three (3) secondary windings.
- Three (3) current transformers to be rated 138 kV, 650 kV BIL, 0.15S% extended accuracy range, B1.8 burden. Actual CT ratio, thermal rating factor, and the extended range to be determined during detailed design.

g) Miscellaneous yard equipment shall include:
- One (1) single phase CCVT outdoor junction box.
- One (1) metering CT outdoor junction box.

h) Raceway & Conduit
- 2" PVC conduit with couplings, bends, adapters, liquid-tight fittings, and related hardware for motor operator, CTs and CCVT**.**

Relay

a) Two (2) relay panels are required for the protective relays at the interconnection switching station for customer interconnection.
    - One (1) protection panel shall include two (2) SEL-411L relays, Entergy Standard PM1803, using a line current differential scheme. Relay communication to IC collector station shall utilize direct fiber for all channels.
    - One (1) breaker control panel shall include a SEL-451, Entergy Standard PM0501, for breaker control.
b) Includes installing the panels in the control building, connection of control cable wiring, checkout and commissioning of the associated systems.
c) Entergy shall require the solar interconnection customer to use SEL-411L relays at the solar interconnection end of the customer line and provide a communications link between ELL and the IC's collector station for relaying purposes. Entergy requires that the customer use fiber for this communication.

Metering

a) One (1) metering panel will be required at the interconnection for revenue metering to the customer. The meters will be bidirectional.
    - Two (2) SEL-735 meters (or Entergy approved equivalent) shall be installed for the interconnection revenue metering.
b) One (1) power quality metering panel will be required at the interconnection for power quality metering of the customer. The meters will be bidirectional.
    - One (1) SEL-735 (or Entergy approved equivalent) shall be installed for interconnection power quality metering.
c) The current transformers listed under Section 2.1.1.4 shall be used for solar interconnection customer revenue metering purpose.
d) The third secondary winding of the capacitive coupled voltage transformers under Section 2.2.1.4 shall be used for solar interconnection customer revenue metering purpose.

ii. Transmission Owner's 138kV Transmission Line:

General

a) The transmission line scope involves connecting the new 138 kV Interconnection Customer transmission line to the new J1795 POI 138 kV switching station.
b) The interconnection will consist of a single span of conductor located between the PCO, which is a new dead-end structure on the new single-circuit 138 kV Interconnection Customer transmission line, and the J1795 POI 138 kV switching station dead-end structure.
c) New single 666.6 kcmil 24/7 Strand ACSR "Flamingo" conductors (rated for 882A at 212 degF and 211 MVA at 138kV) with a maximum tension of 3000 lbs and two (2) 7#7 Alumoweld shield wires with a maximum tension of 1500 lbs each will be installed on the interconnection span.

d) The standard 138kV H-Frame style substation dead-ends are being utilized, which have a max pull-off tension of 5217 lbs for the conductor and 2964 lbs for the shield. Therefore, this will be sufficient for our planned pull-off tensions.
e) Required phase to ground minimum clearance: 29ft.
f) Required phase to switching station bus minimum clearance: 13ft.
g) Phase verification will need to be performed to ensure the IC phase rotation matches Entergy's substation configuration. Interconnection Customer Dead-end phase separation is assumed to be sufficient to allow wires to be strung to the POI substation dead-end structure without violating required wire to wire clearances.
h) The PCO dead-end structure is assumed to be located within 300 ft of the north side of Entergy switching station fence and placed such that conductor pull-off angle at the switching station bay is minimized.
i) The PCO dead-end structure is assumed to be designed such that it is sufficient to support the loading of the flamingo conductor and alumoweld shield wires. Entergy can supply loads for the conductors between the PCO and the Entergy substation.
j) The PCO dead-end structure phase separation is assumed to be sufficient to allow wires to be strung to the POI switching station dead-end structure without violating required wire to wire clearances.
k) The transmission right-of-way (ROW) leading to POI Substation will be acquired by Interconnection Customer.
l) Entergy will have review and approval authority over design documentation prior to installation of any facilities crossing into Entergy's switching station and connecting to the switching station dead end.

Structures and Foundations

a) No intermediary structures or foundations between the PCO and POI are necessary.
b) The IC dead-end structure and foundation will be designed by Interconnection Customer.

Conductor and Insulators

a) New 666.6 kcmil 24/7 Strand ACSR "Flamingo" Conductor (1250 ft / 1070 lbs).
b) Three (3) 161 kV Polymer Dead-end Insulators with turnbuckles (DEP-TRB-161).
c) Three (3) 161 kV Polymer Dead-end Insulators (DEP-161).
d) Bolted dead ends will be used.

Shield Wire

a) New 7#7 Alumoweld (900 ft).

b) Four (4) DE insulator for 7#7 Alumoweld.

Entergy shall install insulator in the shield wire position to isolate the collector station ground grid from Entergy's ground grid.

These TOIFs are detailed in Exhibit A6.

3. Network Upgrades

(a) Stand-Alone Network Upgrades to be installed by Transmission Owner.
i. J1795 POI Switching Station:

The Stand-Alone Network Upgrades (SANU) shall include a new 138 kV POI switching station that will cut into the L-258 138 kV transmission line (Bayou Cove – Richard). A Remote Terminal Unit (RTU) will be provided to gather accumulated data from the Interconnection Customer's facility via a communication link.

General

a) Refer to Figure 1-1 above for the location of the SANU. Exhibit A2 and Exhibit A3 show the oneline and general arrangement drawings for the new J1795 switching station.
b) Construct a new 138 kV, three breaker, four position ring bus. Two transmission lines shall be installed in this build-out.
c) The L-258 138 kV transmission line (Bayou Cove – Richard) is an existing transmission line which will be cut into by the new J1795 138 kV switching station.
d) No planned deviations from the normal design package organizations are foreseen.

Site

a) General:
- Site development contractor shall furnish all labor, material, transportation, tools, utilities, equipment, appurtenances, and performance of all operations necessary for filling, excavating, and grading for the site development of the new station.
- Approximately 3.5 acres of property/easement will need to be provided to accommodate the proposed substation and access roads. The substation pad will be approximately 337' x 230'.
- The geotechnical report and topographical survey was not available when this facility study was developed. The site quantities in this estimate are based on a rough AutoCAD Civil3D model utilizing USGS topographic data obtained through Autodesk InfraWorks. Site soil conditions, if different from what was assumed, could cause an increase in the actual project cost. Site grading installation recommendations specified in the geotechnical report shall also form part of the installation requirements.

- The earthwork estimate is based on a combination of surface drainage design and subsurface drainage design. Subsurface drainage will be necessary between the cable trench runs. A sheet flow design will be utilized outside the cable trench runs.
- The proposed pad site is located within a FEMA zone X which is classified as an "area of minimal flood hazard" (Refer to FIRM map No. 22001C0175D).

b) The site scope of work includes:
   - The existing areas to be cleared and grubbed consists of light brush (3.5 acres). All waste involved in this activity will be disposed of off-site.
   - Strip all topsoil and unsuitable material within the grading limits of construction to a depth of 6 inches (2,396 cubic yard).
   - Rough and fine site grading (12,539 square yard) will be required to build the proposed substation pad and access roads to design elevations. The estimated volume of excavation is 505 cubic yard. The estimated volume of fill material is 14,172 cubic yard which will consider a compaction factor. Fill material shall be compacted in layers in accordance with ASTM D698.
   - Dispose all unused stripped material (2,396 cubic yard).
   - Supply and apply soil sterilization where specified by Entergy standards (1.8 acres).
   - Supply and install #610 gray crushed limestone surface material (2,659 cubic yard or 3,989 tons) for the substation yard area and new drive paths inside and outside the yard per Entergy Standard SL1201. A 12-inch layer shall be installed for drive path areas within the yard and access roads outside the yard. A 6-inch layer shall be installed for the remainder of the yard as well as to 5 feet outside substation fence.
   - Install 1444 linear feet of new 20 feet wide drive path/access road consisting of a 12-inch layer #610 rock as mentioned above and lined with Mirafi RS380i Fabric (30,322 square feet).
   - Install an asphalt access road apron (40' radii, 40 total length). The apron is an assumed jurisdictional requirement and shall be verified during detail design.
   - Supply and install a 12-inch layer of riprap to all pad side slopes (995 cubic yard or 1,989 tons). Riprap shall be lined with Contech C-80NW geotextile (28,200 square feet).
   - Supply and install topsoil, seed, and mulch on all disturbed areas not receiving other surfacing (1.0 acres)
   - Supply and install 3'x3' precast concrete drain inlets (2 each).
   - Supply and install 18" diameter reinforced concrete pipe for subsurface drainage and access road culverts (403 feet).
   - Supply and install 18" diameter reinforced concrete end sections for access road culverts and subsurface drainage outfall (7 each).

- Supply and install materials for a French drain beneath the cable trough (535 feet).
- Supply and install temporary sediment control fence (1,521 feet).
- Supply and install substation security fence in accordance with Entergy Standard SL0701 (1,054 feet).
- Supply and install 20’ double swing gates in accordance with Entergy Standard SL0701 (2 each).
- Supply and install standard substation signs including Entergy substation name sign and warning signs per the Entergy Substation & Switchyard Sign Standards.
- Spoils to be handled per direction of environmental sampling.
- Installation of the substation ground grid, which was assumed to be made up of 20’ x 20’ grid using #4/0 bare copper conductors. 5/8” x 10’ ground rods shall be installed at every other crossing along the fence and near major equipment such as surge arresters.

Foundations

a) The foundation work shall be in accordance with Entergy standards and design drawings.

b) Drilled pier and pad foundations will be used for all equipment support structures.

c) The foundation scope of work includes design and construction of the following:

- Two (2) set of two (2) 66-inch diameter 35-foot-long drilled piers for the three phase 138 kV high bus H-Frame dead-end structures.
- Fifty-seven (57) 36-inch diameter 15-foot-long drilled piers for the single phase 138 kV surge arrester, coupling capacitor voltage transformer, station service transformer, high bus, and low bus support structures.
- Ten (10) sets of two (2) 42-inch diameter 20-foot-long drilled piers for the 138 kV high and low disconnect switch support structures.
- Three (3) sets of one (1) 8’ x 16’ x 3’ slab for 138 kV gas circuit breakers and one (1) 3’ x 4’ x 1’ slab for the breaker operator platform stair landings.
- One (1) set of fifteen (15) 36-inch diameter 20-foot-long drilled piers for control building.
- One (1) 8’ x 4’ slab for the battery room entrance for the control building.
- One (1) 10’ x 7’ slab for the control building entrance door landing.
- Three (3) 60-inch diameter 25-foot-long drilled pier for the 75+5 ft. static mast.

- Eight (8) 3’ x 2’ x 2’ foundations for the 200W LED yard lights as depicted in drawing no. (J1795EA1).
- Twenty-two (22) traffic bollards to be installed as depicted in drawing no. (J1795EA1).

Electrical

a) Steel Structures
   - Two (2) three phase 138 kV H-frame dead-end structures with provisions for high bus supports.
   - Three (3) 138 kV breaker operator platform structures.
   - Seven (7) three phase 138 kV low bus disconnect switch support structures.
   - One (1) future low bus disconnect switch support structure with provisions for bus support insulators.
   - Two (2) three phase 138 kV high bus disconnect switch support structures.
   - Twenty-six (26) single phase 138 kV low bus support structures.
   - Nine (9) single phase 138 kV high bus support structures.
   - Nine (9) single phase 138 kV surge arrester support structures.
   - Eleven (11) single phase 138 kV coupling capacitor voltage transformer support structures.
   - Two (2) single phase 138 kV station service voltage transformer support structures.
   - Three (3) 75+5 ft. static mast structures for lightning protection.

b) Station Post Insulators
   - Forty-seven (47) high strength, porcelain station post insulators rated for nominal system voltage of 138 kV, 650 kV BIL, minimum 116” creepage distance for H-frame, future disconnect switch, single phase low bus and high bus supports.
   - Forty-two (42) standard strength, porcelain station post insulators rated for nominal system voltage of 138 kV, 650 kV BIL, minimum 116” creepage distance for 138 kV center break disconnect switches.
   - Eighteen (18) standard strength, porcelain station post insulators rated for nominal system voltage of 138 kV, 650 kV BIL, minimum 116” creepage distance for 138 kV Vertical break line disconnect switches.

c) Suspension Insulators
   - Six (6) polymer suspension insulators rated for nominal system voltage of 161 kV, 90” long, 25,000 lb., ANSI 52-5.
   - Installation includes fitting accessories.

d) Surge Arresters
   - Nine (9) substation class surge arresters rated for nominal system voltage of 138 kV, duty cycle of 132 kV, 106 kV MCOV, energy classification of “F”.

e) 138 kV Disconnect Switches
   - Seven (7) three phase center break disconnect switches.

- Two (2) three phase motor operated vertical break disconnect switches, with ground switch attachment.
- All switches to be rated 145 kV, 2000 A continuous, 100 kA momentary withstand capacity, 650 kV BIL.

f) 138 kV Gas Circuit Breakers
- Three (3) 145 kV, SF6 gas circuit breakers rated 3000 A, 63 kA interrupting capability.
- Includes installing foundations, control conduit to the circuit breakers, jumpers to the associated disconnect switches, and control cable to the control building.

g) 138 kV Conductor
- The main bus will be 4" aluminum (6063-T6) schedule 40 rigid tubular bus.
- The H-frame vertical bus and breaker equipment jumpers shall be 954 kcmil ACSR Rail, two conductors per phase.
- The equipment jumpers for the coupling capacitor voltage transformers, station service transformers and surge arresters shall be 954 kcmil ACSR Rail, one conductor per phase.
- Includes all necessary fittings.

h) 138 kV Instrument Transformers
- Eleven (11) coupling capacitor voltage transformers to be rated 138 kV, 650 kV BIL, 700/1200:1:1:1 ratio, 0.3% accuracy, WXYZ,ZZ load burden, 1000 VA thermal burden with three (3) secondary windings.

i) 138 kV Station Service Voltage Transformers
- Two (2) single phase station service voltage transformers rated 145 kV maximum system voltage, 80.5 kV line-to-ground, 50 kVA, 750 kV BIL, 125/250 Volts secondary, with a ratio correction factor between 0.9 and 1.1.

j) Control Enclosure
- Installation of a 20' X 48' prefabricated control enclosure is required to house relay panels, control panels, metering panels, remote terminal unit, miscellaneous equipment, and cable tray systems.
- Includes installing the panels in the control building, connection of control cable, wiring, checkout and commissioning of the associated systems.
- Miscellaneous equipment shall include, but not be limited to the following:
    i. One (1) stand-alone AC panel.
    ii. One (1) Automatic transfer switch (ATS).
    iii. One (1) stand-alone DC panel.
    iv. One (1) 125VDC 290 Ah battery set with rack.
    v. One (1) 50A battery charger.
    vi. One (1) DC switching panel.
    vii. Three (3) three phase, three winding CCVT indoor bus distribution boxes.

viii. Control Enclosure HVAC.

k) Miscellaneous yard equipment shall include:
- Eight (8) yard lights; (2) 400 W fixtures per light.
- Three (3) three phase, three winding CCVT outdoor junction boxes.
- Two (2) single phase CCVT outdoor junction boxes.
- Two (2) fused disconnect switches for SSVT feed to the ATS.

l) Raceway & Conduit
- One hundred and thirty-one (131) pre-cast concrete cable trench sections (4 feet each), traffic rated, with five (5) universal tee sections (4 feet each), as well as end walls and lids for cable routing between control building pull boxes.
- Four (4) 30" W x 48" L x 42" D cable pull boxes.
- 2" PVC conduit with couplings, bends, adapters, liquid-tight fittings, and related hardware everywhere throughout the station required.
- 3" PVC conduit with couplings, bends, adapters, liquid-tight fittings, and related hardware everywhere throughout the station required.
- 4" PVC conduit with couplings, bends, adapters, liquid-tight fittings, and related hardware to station service voltage transformer junction boxes.
- All conduits at road crossings shall be encased in concrete.

Relay

a) Two (2) relay panels are required for the protective relays for the Bayou Cove 138 kV transmission line and control of the 138 kV breakers.
- One (1) protective relay panel shall include two (2) SEL-411L relays, Entergy Standard PM1803, using a line current differential scheme. Relay communications to the Bayou Cove 138 kV Switching Station shall utilize direct fiber for all channels, through new OPGW.
- One (1) protective relay panel shall include one (1) SEL-451 relay, Entergy Standard PM0501 for breaker control.

b) Two (2) relay panels are required for the protective relays for the Richard 138 kV transmission line and control of the 138 kV breakers.
- One (1) protective relay panel shall include two (2) SEL-411L relays, Entergy Standard PM1803, using a line current differential scheme. Relay communications to the Richard 500/138 kV Substation shall utilize multiplexer communication for all channels, through existing backbone fiber on the line.
- One (1) protective relay panel shall include one (1) SEL-451 relay, Entergy Standard PM0501 for breaker control.

c) One (1) panel shall be installed for the digital fault recorder (DFR) to monitor equipment and relays within the J1795 station.

d) Includes installing the panels in the control building, connection of control cable wiring, checkout and commissioning of the associated systems.

Relay Settings

a) Study and apply NERC/SERC Regulatory compliance requirements (FAC-008-3, PRC-027, PRC-023, PRC-025). Perform analysis based on Entergy Standard compliance procedure for setting revisions and updates.
b) Develop new relay settings for one (1) Line Panel for the J1795 customer interconnect line for two (2) SEL-411L relays.
c) Perform NERC PRC-027 coordination for the J1795 customer interconnect line.
d) Develop new relay settings for one (1) Line Panel for 138 kV Bayou Cove Line for two (2) SEL-411L relays.
e) Develop new relay settings for one (1) Line Panel for 138 kV Richard Line for two (2) SEL-411L relays.
f) Develop new relay settings for three (3) Breaker Control Panels for the three (3) new breakers for three (3) SEL-451 relays.
g) Develop and support Test Plans for all newly developed settings.
h) Provide off site/on-site relay settings support during construction for setting database issues, relay software and firmware compatibility.
i) Upgrade existing relays to the latest Entergy standard firmware.
j) Perform setting logistics duties such as relay date base (.rdbs), relay viewable documents (.xls), project work folders for Web posting (IPS), and relay modeling in CAPE®.
k) Perform relay settings corrections and issue new “as built” settings based on returning “as left” settings.

Communications and SCADA

a) Telecom link via fiber optic cable will establish a new SCADA (Supervisory Control and Data Acquisition) communication at new J1795 POI station.
b) Entergy will complete ADSS fiber installation and splicing at the PCO structure. OPGW will be spliced inside the substation at the dead ends, transitioned to ADSS, and brought into the new control house.
c) Install one (1) new 19” integrated rack kit for telecom equipment.
d) Install one (1) new 19” rack for fiber panels and future equipment.
e) Install two (2) new fiber transport nodes.
f) Terminate one (1) new patch panel inside J1795 sub to new OPGW fiber facing Bayou Cove.
g) Terminate one (1) new patch panel inside J1795 sub to new OPGW fiber facing Richard.
h) Install two (2) new relay channel banks with circuits distributed as follows:
    - Channel bank 1: 3 circuits facing Bayou Cove. 3 circuits facing Richard.
    - Channel bank 2: 3 circuits facing Bayou Cove. 3 circuits facing Richard.
i) Test and label twelve (12) dark fibers from Bayou Cove to J1795 for dual 411L direct fiber relaying.
j) Install and turn up OTNet router and switch.
k) Terminate and test new Ethernet circuit for OTNet router.

l) Install two (2) 125V, 30A DC feeds at new telecom racks to power telecom equipment.
m) Install one (1) Entergy standard SEL-3530/SEL-2488 RTU panel which includes SEL-3530 RTAC, SEL-2488 Satellite-Synchronized Clock, and dual SEL-3610's.
n) Install one (1) Entergy standard I/O panel which includes eight (8) SEL-2440 DPAC's.
o) Develop new RTU configuration file.
p) Perform full station SCADA points commissioning and checkout.

ii. L-258 138kV Transmission Line Cut-in (Bayou Cove – Richard)

The Stand-Alone Network Upgrades (SANU) includes the following structures: two (2) custom steel self-supporting dead-end structures at the POI station, four (4) self-supporting running angle structures, and four (4) single pole steel vertical framed tangent structures.
The standard 138 kV H-Frame style substation dead-ends are being utilized, which have a max pull-off tension of 5217 lbs for the conductor and 2964 lbs for the shield. For this project, our max pull-off tensions on the substation dead-ends will be 3000 lbs and 1500 lbs for the conductor and shield wire respectively.
A Lidar Survey, Level C Survey, and soil borings will need to be obtained for the detailed design phase.
Additional ROW will need to be acquired between the transmission line dead-ends and the new J1795 substation. A 120ft wide right-of-way will be needed from the Bayou Cove substation to the J1795 sub. The approximate length of new right-of-way required is approximately 0.50 miles. In total, approximately 7.3 acres will be required.
This project falls within the NESC Light District Load Case which has 0.5” of extreme ice load, and an extreme wind speed of 140mph. The wind speed on this project is being increased to 150mph to account for pending standard changes.
Required phase to ground minimum clearance: 29ft.
Required phase to switching station bus minimum clearance: 13ft
No special environmental considerations have been identified at the point.

Structures and Foundations

- Project management team should include a 5% risk on structure cost due to quarterly steel pole cost fluctuations.
- Soil borings were not obtained for this project. Foundation designs were completed based on historical borings near this site, but not in the immediate project location.
- The direct embed structures will require concrete backfill foundations.

The following structures will be required in this scope:

a) Two (2) new, 80-foot tall, single circuit, self-supporting galvanized steel dead-end structures on drilled pier foundations, per Entergy Standard Drawing SDJ-V-DEP-S
    - Base plated drilled pier foundations with a diameter of 6', and a depth of 26' is assumed to be adequate for the 105' outer poles. Reveal is anticipated to be 2'. Final foundation type and dimensions may vary depending on actual site conditions.

b) Four (4) new, 105-foot, single circuit, self-supporting running angle structures, per Entergy Standard Drawing SC-OGA-V-BP2
    - Base plated drilled pier foundations with a diameter of 5.5', and a depth of 24' is assumed to be adequate for the 105' poles. Reveal is anticipated to be 2'. Final foundation type and dimensions may vary depending on actual site conditions.

c) Four (4) new, single circuit, self-supporting galvanized single pole steel vertical framed tangent structures direct embed, per Entergy Standard Drawing A-OGA-V-BP2
    - Base diameter of 4.0', is assumed to be adequate for the 105' poles which are on the span of the circuit running north from the cut-in structures towards the POI switching station
    - Embedment depth for all legs is assumed to be 20'. Final foundation type and dimensions may vary depending on actual site conditions.

(b) Network Upgrades to be installed by Transmission Owner.

i. Bayou Cove 138kV Substation

General

a) Perform relay settings work.
b) L-258 138 kV transmission line (Bayou Cove – Richard) is an existing transmission line which will be cut into by the new J1795 138 kV POI switching station.
c) No planned deviations from the normal design package organization are foreseen.

Site

a) Contractor shall restore disturbed site to pre-construction conditions or better.

Foundations

a) No change to existing foundations or installation of new foundations required.

Electrical

a) Modification of associated drawings to reflect new line destinations per L-258 138 kV transmission line (Bayou Cove – Richard) interconnect switching station.
b) No change to existing conduits or installation of new conduits required.

Relay

a) No Relay scope updates are required at this station.

Relay Settings

a) Study and apply NERC/SERC Regulatory compliance requirements (FAC-008-3, PRC-027, PRC-023, PRC-025). Perform analysis based on Entergy Standard compliance procedure for setting revisions and updates.
b) Review/Update relay settings for one (1) Line Panel for 138 kV J1795 POI substation Line for two (2) SEL-411L relays.
c) Review/Update relay settings for the SEL-411L relays at 138 kV Bayou Cove substation for Richard line L203.
d) Review/Update relay settings for the SEL-311C and Micom P544 relays at 138 kV Bayou Cove substation for NRG Line L1.
e) Review/Update relay settings for the SEL-311C and Micom P544 relays at 138 kV NRG substation for Bayou Cove Line L1.
f) Review/Update relay settings for the SEL-311C and Micom P544 relays at 138 kV Bayou Cove substation for NRG Line L2.
g) Review/Update relay settings for the SEL-311C and Micom P544 relays at 138 kV NRG substation for Bayou Cove Line L2.
h) Review/Update relay settings for the SEL-411L relays at 138 kV Bayou Cove substation for Jennings line.
i) Review/Update relay settings for the SEL-411L relays at 138 kV Jennings substation for Bayou Cove line.
j) Review/Update relay settings for the SEL-421 and SEL-411L relays at 138 kV Bayou Cove substation for Henning line.
k) Review/Update relay settings for the SEL-421 and SEL-411L relays at 138 kV Henning substation for Bayou Cove line.
l) Review/Update relay settings for the EM relays for BKR 27205 and BKR 27210.
m) Develop and support Test Plans for all revised settings.
n) Provide off site/on-site relay settings support during construction for setting database issues, relay software and firmware compatibility
o) Upgrade existing relays to the latest Entergy standard firmware.
p) Perform setting logistics duties such as relay date base (rdbs), relay viewable documents (xls), project work folders for Web posting (IPS), and relay modeling in CAPE®.
q) Perform relay settings corrections and issue new “as built” settings based on returning “as left” settings.

Communications and SCADA

a) Install two (2) new fiber transport nodes for OTNet connectivity.
b) Terminate fiber cable into fiber panel for OPGW facing J1795.
c) Install new router and switch for OTNet connectivity.
d) Terminate and test new Ethernet circuit for OTNet router.
e) Test and label twelve (12) dark fibers from J1795 to Bayou Cove for dual 411L direct fiber relaying.
f) Continue using GE D20 I/O boards with existing GE D20 RTU.
    - Two (2) GE D20 Analog boards.
    - Four (4) GE D20 Status boards.
    - One (1) GE D20 Control board with four (4) KI interposing relay modules.
g) Update RTU configuration file for GE D20 RTU.
h) Perform partial SCADA points commissioning and checkout for relays with updated relay settings.

Metering

a) No metering scope updates are required at this station.

ii. Richard 500/138 kV Substation

General

a) Perform relay settings work.
b) The L-258 138 kV transmission line (Bayou Cove – Richard) is an existing transmission line which will be cut into by the new J1795 138 kV switching station.
c) No planned deviations from the normal design package organization are foreseen.

Site

a) Contractor shall restore disturbed site to pre-construction conditions or better.

Foundations

a) No change to existing foundations or installation of new foundations required.

Electrical

a) Modification of associated drawings to reflect new line destinations L-258 138 kV transmission line (Bayou Cove – Richard) interconnect switching station.

Relay

a) No Relay scope updates are required at this station.

Relay Settings

a) Study and apply NERC/SERC Regulatory compliance requirements (FAC-008-3, PRC-027, PRC-023, PRC-025). Perform analysis based on Entergy Standard compliance procedure for setting revisions and updates.
b) Review/Update relay settings for one (1) Line Panel for 138 kV J1795 POI substation Line for two (2) SEL-411L relays.
c) Review/Update relay settings for the SEL-411L relays at 138 kV Richard substation for Bayou Cove line L203.
d) Review/Update relay settings for the SEL-411L relays at 138 kV Richard substation for Acadia line L303.
e) Review/Update relay settings for the SEL-411L relays at 138 kV Acadia substation for Richard line L303.
f) Review/Update relay settings for the SEL-411L relays at 138 kV Richard substation for Acadia line L304.
g) Review/Update relay settings for the SEL-411L relays at 138 kV Acadia substation for Richard line L304.
h) Review/Update relay settings for the SEL-411L relays at 138 kV Richard substation for Colonial Acadia line L679.
i) Review/Update relay settings for the SEL-411L relays at 138 kV Colonial Acadia substation for Richard line L679.
j) Review/Update relay settings for the SEL-411L relays at 138 kV Richard substation for Scott line L233.
k) Review/Update relay settings for the SEL-411L and KD, KRD relays at 138 kV Scott substation for Richard line L233.
l) Review/Update relay settings for the SEL-411L and SEL-421 relays at 138 kV Richard substation for Scott line L647.
m) Review/Update relay settings for the SEL-411L and SEL-421 relays at 138 kV Scott substation for Richard line L647.
n) Review/Update relay settings for the SEL-421 and SEL-311C relays at 138 kV Richard substation for SW Eunice line.
o) Review/Update relay settings for the SEL-421 and SEL-311C relays at 138 kV SW Eunice substation for Richard line.
p) Review/Update relay settings for the SEL-451 relay for BKR 27160.
q) Review/Update relay settings for the EM and SEL-487B relays for AT #1 and AT #1 Bus Node.
r) Review/Update relay settings for the EM and SEL-487B relays for AT #2 and AT #2 Bus Node.
s) Develop and support Test Plans for all revised settings.
t) Provide off site/on-site relay settings support during construction for setting database issues, relay software and firmware compatibility.
u) Upgrade existing relays to the latest Entergy standard firmware.
v) Perform setting logistics duties such as relay date base (.rdbs), relay viewable documents (.xls), project work folders for Web posting (IPS), and relay modeling in CAPE®.
w) Perform relay settings corrections and issue new “as built” settings based on returning “as left” settings.

Communications and SCADA

a) Install two (2) new fiber transport nodes.
b) Install two (2) new relay channel banks with circuits distributed as follows:
    - Channel bank 1: 3 circuits facing the J1795 POI station.
    - Channel bank 2: 3 circuits facing the J1795 POI station.
c) Continue using GE DNP I/O boards with existing dual SEL-3555 RTAC. RTU's.
    - Three (3) GE DNP I/O Analog boards.
    - Twelve (12) GE DNP I/O Status boards.
    - Three (3) GE DNP I/O Control boards with eleven (11) KI interposing relay modules.
d) Update RTU configuration files for dual SEL-3555 RTAC RTU's.
e) Perform partial SCADA points commissioning and checkout for relays with updated relay settings.

Metering

a) No metering scope updates are required at this station.

iii. L-258 138 kV Transmission Line Cut-in (Bayou Cove – Richard)

a) The Network Upgrades includes installing a .50 mi transmission line tap into and out of the J1795 POI substation from the existing L-258 138 kV transmission line (Bayou Cove- Richard).
b) The cut-in will be located between structures #410 and #411 of existing L-258. This scope includes replacing cut-in structures #410 and #411 and associated hardware. The new structures to be installed will be two custom steel self-supporting dead-end structures.
c) L-258 is configured as a single circuit comprised of mostly wooden H-frame structures, but the section of line closest to the cut-in location is all single pole steel construction.
d) The majority of existing L-258 consists of a double bundled 336.4 ACSR (26/7) "Linnet" conductor with a rating of 940A and 225 MVA at 138 kV Per the Facility Rating Workbook. This is indicated to be the limiting element on this line. The section of line serving Bayou Cove where the cut-in for J1795 will occur, is a single conductor which will need field verification to determine the size. This line is shielded with an OPGW cable.
e) Single 954 MCM ACSR Cardinal, which is rated at 1088A and 260MVA at 138 kV and 212deg F, will be used for the cut-in on L-258. The rating of this conductor exceeds that of the existing conductor. If additional capacity is required in the future 954ACSS or 1272MCM ACSS conductor can be utilized.
f) OPGW (DNO-12160) and 7 no.7 Alumoweld shield wire will be installed from cut-in dead end structures to J1795 POI Sub.
g) The standard 138 kV H-Frame style substation dead-ends are being utilized, which have a max pull-off tension of 5217 lbs for the conductor and 2964 lbs

for the shield. For this project, our max pull-off tensions on the substation dead-ends will be 3000 lbs and 1500 lbs for the conductor and shield wire respectively.

h) A Lidar Survey, Level C Survey, and soil borings will need to be obtained for the detailed design phase.
i) A temporary tie may be needed between the two new dead-end structures to keep L-258 energized while the J1795 substation construction is underway.
j) This project falls within the NESC Light District Load Case which has 0.5" of extreme ice load, and an extreme wind speed of 140mph. The wind speed on this project is being increased to 150mph to account for pending standard changes. There are no special environmental conditions affecting this site. The J1795 substation site and cut-in line are on farmland. The standard SWPPP plan will be adequate for this site.
k) Phase verification will need to be completed on L-258 prior to the detailed design phase.
l) One distribution line, which runs parallel to Bergeaux Rd, will be crossed by the cut-in line. Bergeaux Rd will also be crossing by the cut-in line. This road is located just south of the J1975 substation. No other crossings will be encountered on this project.
m) Required phase to ground minimum clearance: 29ft.
n) Required phase to switching station bus minimum clearance: 13ft.

Structures and Foundations

a) Two (2) new, 80-foot tall, single circuit, self-supporting galvanized steel dead-end structures on drilled pier foundations, per Entergy Standard Drawing SDJ-V-DEP-S
    - Base plated drilled pier foundation with a diameter of 5.5', and a depth of 36' is assumed to be adequate for the 80' poles. Reveal is anticipated to be 2'. Final foundation type and dimensions may vary depending on actual site conditions.

Conductor and Insulators

- The existing double bundled 336.4 ACSR conductor will be dead ended at new dead-end structures #410 and #411 and new conductor will be installed from these points into the new J1795 substation. Single 954 MCM ACSR 54/7 Cardinal will be the new conductor utilized for this project. This conductor is rated at 1088A at 212 degrees F, which supplies 260MVA at 138 kV. This line will require 22614 lbs (18400ft) of the Bittern conductor.
- Bolted dead end assemblies will be used for the new 954 ACSR "Cardinal" conductor.
- Twenty-four (24) 161 kV Polymer Dead-end Insulators with turnbuckles (DEP-TRB-161).
- Six (6) 161 kV Polymer Dead-end Insulators (DEP-161).
- Twenty-four (24) 161 kV Polymer Braced Post Insulators (BP2-161).

- Twelve (12) 161 kV Polymer Jumper Line Post Insulators (JLP-161).

Shield Wire

a) Shield wire:
   - 220ft of 7#7 Alumoweld shield-wire.
   - 6750ft of OPGW (DNO-12160) Fiber.

b) OPTGW Assembly Hardware required:
   - Eight (8) – OPTGW Dead-end Assemblies.
   - Two (2) – OPTGW Suspension Assemblies.
   - Four (4) – OPTGW Splice Boxes.

Splice boxes will be located at structures #410, #411, and at both substation dead-ends. The existing fiber will be cut between structures #410 and #411 then routed down new structure 411 into the splice box. A new section of fiber will be run from structure #410 into the Bayou Cove substation.

Removals

a) Existing structure #410 and #411 will be removed along with the conductor and shield wire between the two new dead-ends. The following items are contained in this removal:
   - Two (2) Steel Pole RU5309.
   - Twelve (12) 138 kV Insulators RU5325.
   - 450ft of Conductor RU5323.
   - 150ft of OPGW Shield Wire RU7530.
   - ~1200ft of Guy Wire R75301.
   - 10 Anchors RU5302.

These facilities are detailed in Exhibit A6.

(c) Shared Network Upgrade(s) to be funded by Interconnection Customer. None.

4. System Protection Facilities

(a) System Protection Facilities not listed in Section 2 or 3 to be constructed by Interconnection Customer. None.

(b) System Protection Facilities not listed in 2 or 3 to be constructed by Transmission Owner. None.

5. Distribution Upgrades

(a) Distribution Upgrades to be constructed by Transmission Owner. None.

6. **Contingency List** See Exhibit A10.

7. **Affected System Upgrades List** None.

8. **Common Use Upgrades List** None.

9. **Exhibits** – The following exhibits are included:

A1. Interconnection Customer One-Line and Site Map
- A1-1: Interconnection Customer One-Line Diagram
- A1-2: Interconnection Project Site Map

A2. Transmission Owner One-Line

A3. Transmission Owner Substation General Arrangement Drawing

A4. Transmission Owner Plan and Profile
- A4-1: Transmission Line drawing
- A4-2: Structure Types
- A4-3: Structure Types
- A4-4: 3-D Interconnection Layout

A5. Facilities to be Constructed by Transmission Owner

A6. Detailed Costs of Facilities to be Constructed by Transmission Owner

A7. Facilities to be Constructed by Interconnection Customer

A8. Detailed Cost of Facilities to be Constructed by Interconnection Customer

A9. Facilities Subject to Transmission Owner Reimbursement

A10. Contingent Facilities

A11. Interconnection Customer Milestones

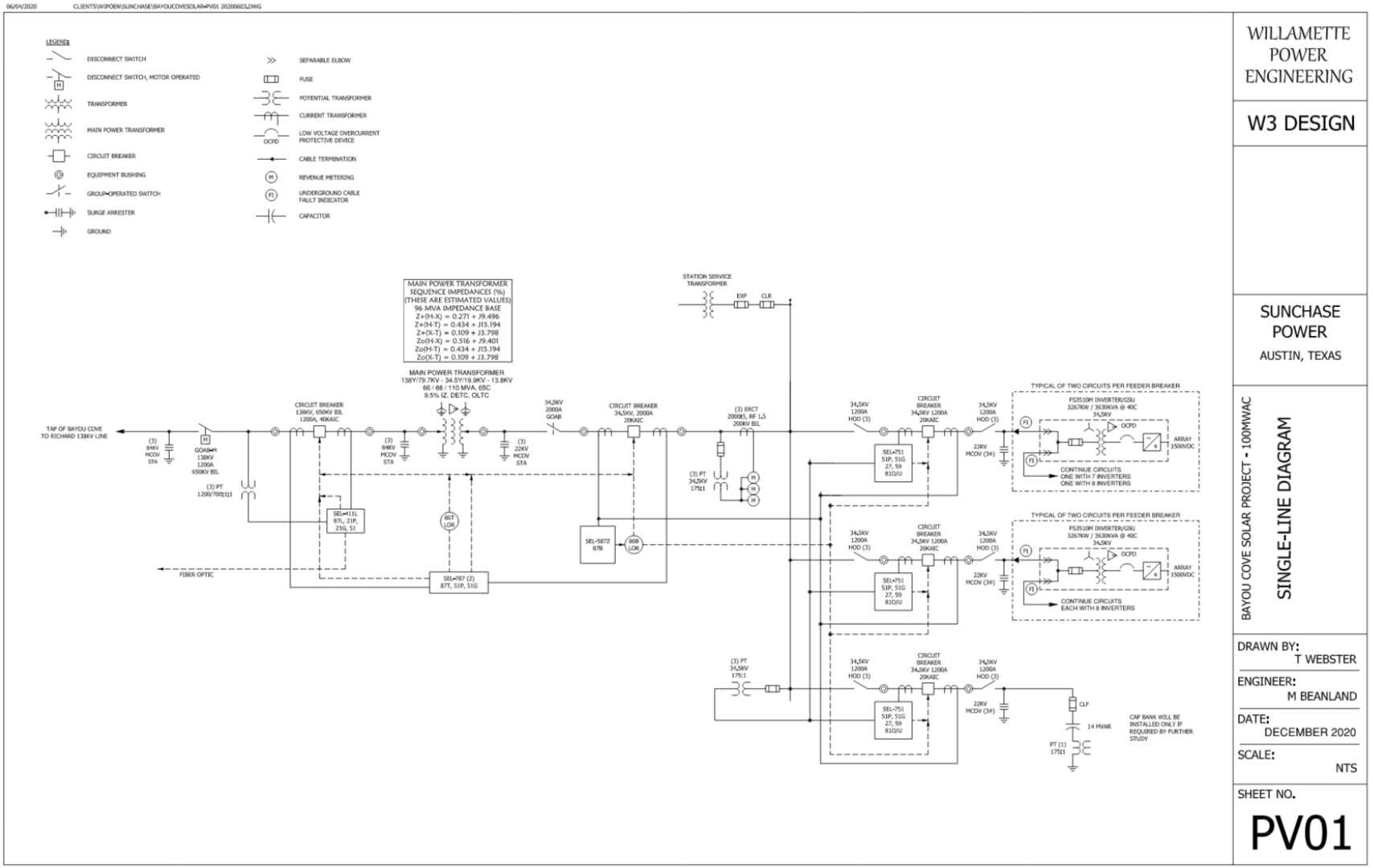
A12. Construction and Coordination Schedules

A13. Permits, Licenses, Regulatory Approvals and Authorization

A14. Interconnection and Operating Guidelines

Exhibit A1. Interconnection Customer One-Line and Site Map

A1-1: Interconnection Customer One-Line Diagram



A1-2: Interconnection Project Site Map

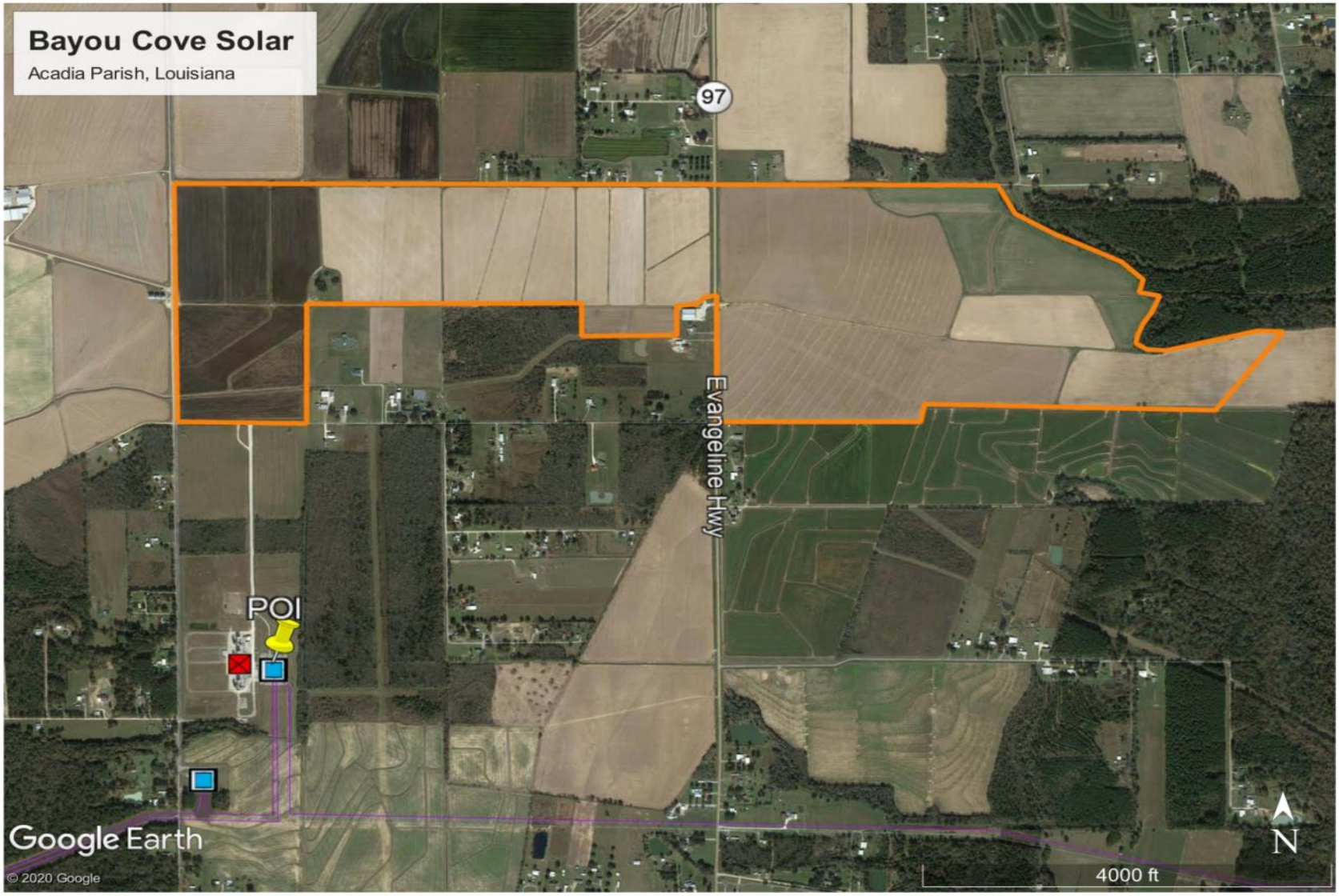


Exhibit A2. Transmission Owner One-Line

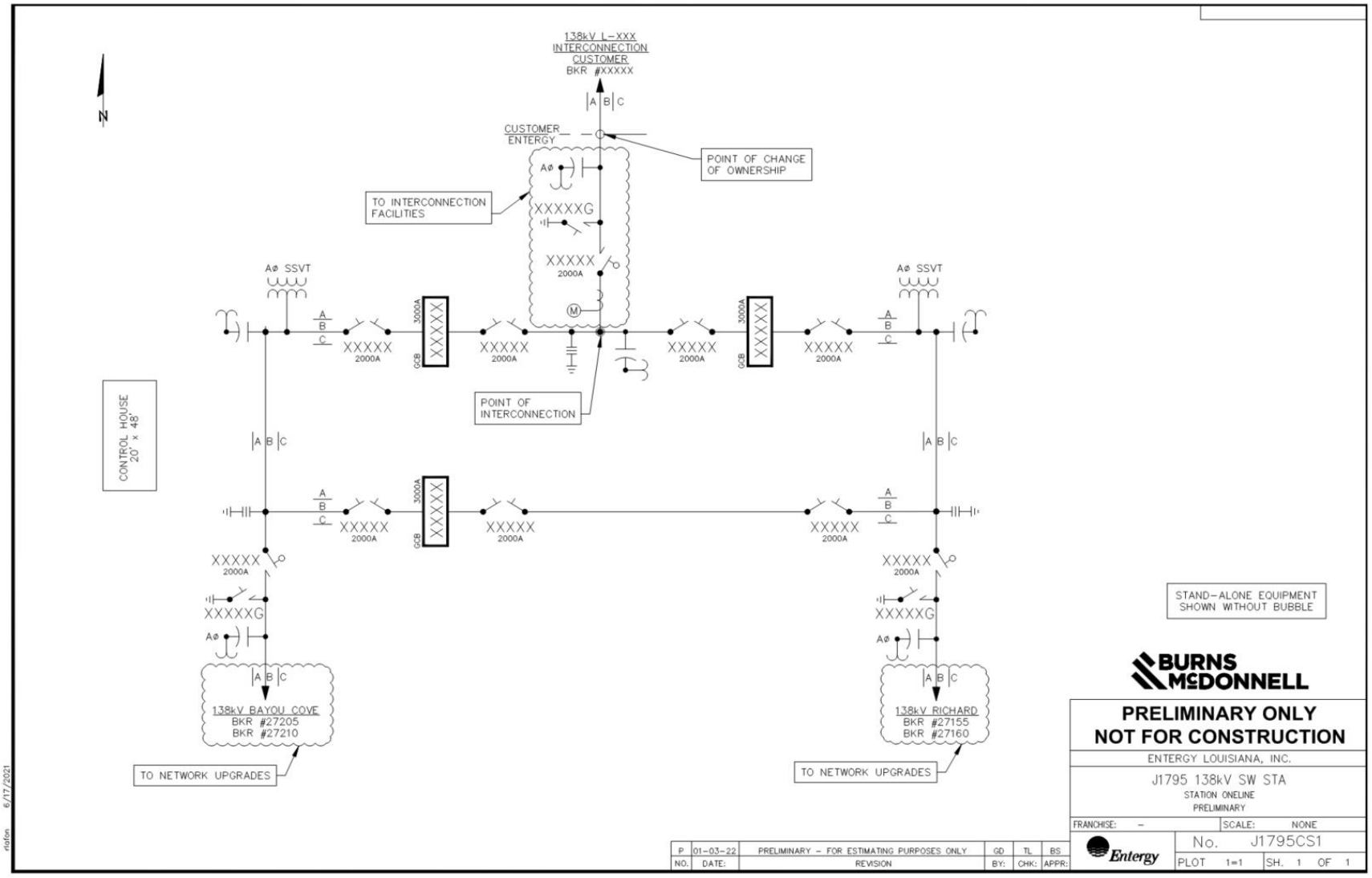


Exhibit A3. Transmission Owner Substation General Arrangement Drawing

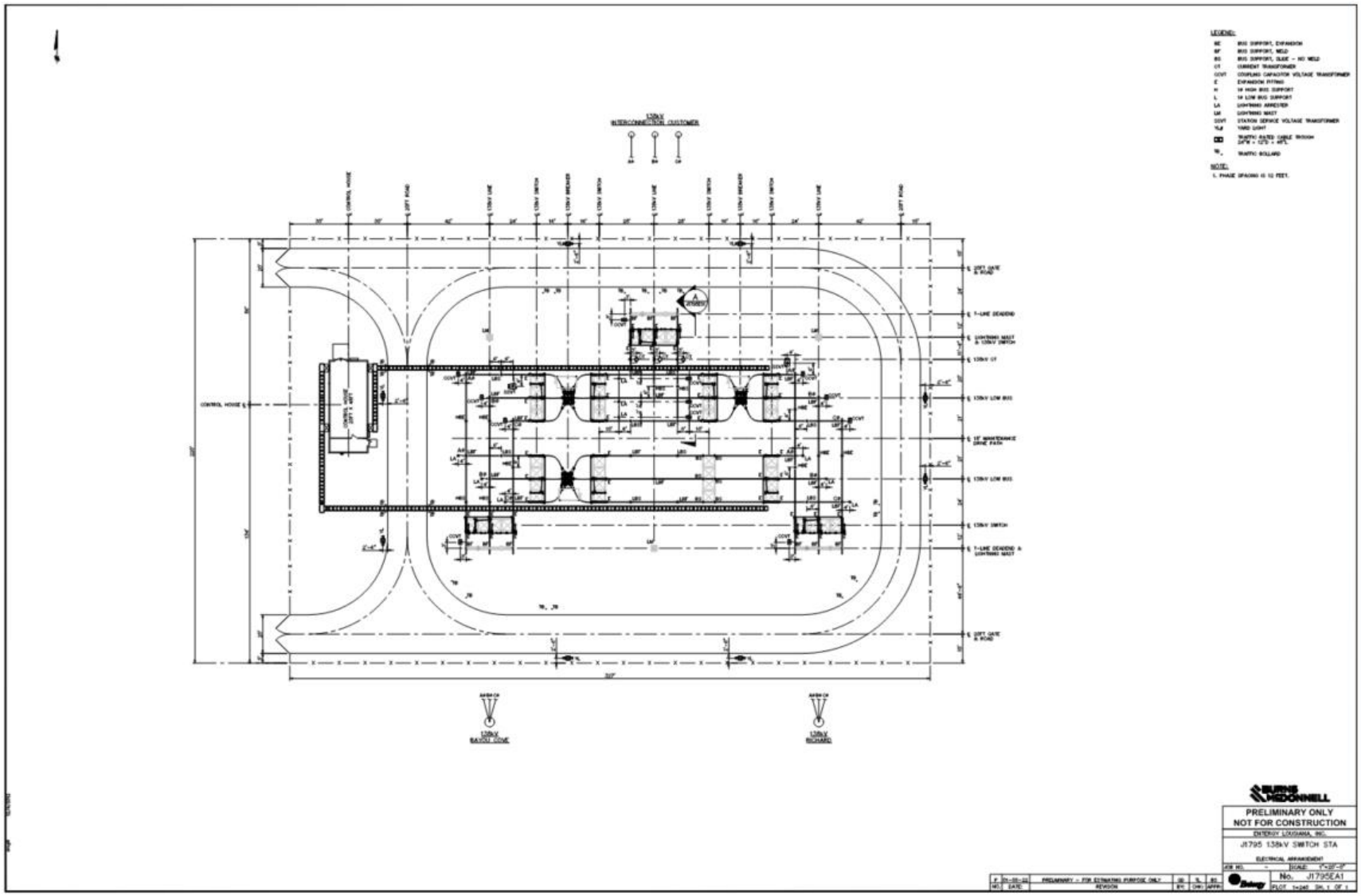
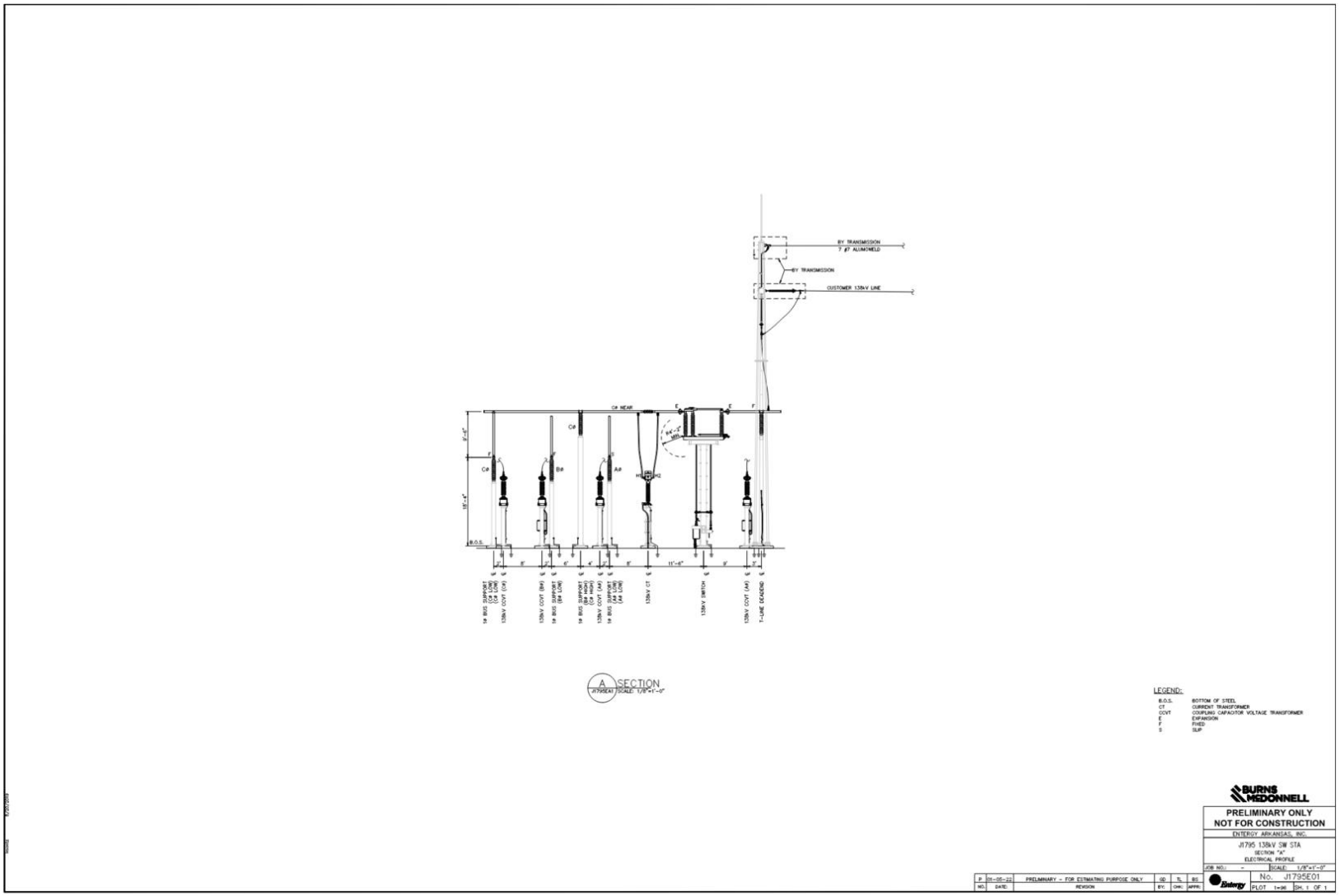
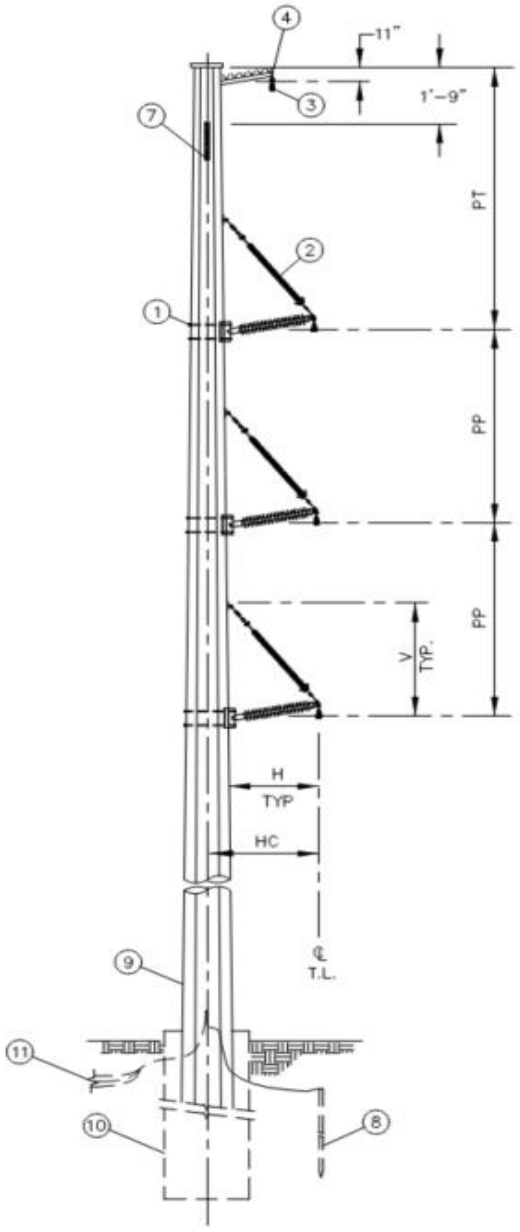


Exhibit A4. Transmission Owner Plan and Profile
A4-1: Transmission Line Drawing



A4-2: Structure Types

0°–1.5°, VERTICAL, SINGLE CIRCUIT, BRACED POST 2–1/2", WITH 3' OHGW ARM, STEEL



ASSEMBLY LIST		
ITEM	QTY.	ASSEMBLY/DRAWING
1	3	BLT–P–S
2	3	INSUL (SEE TABLE 1)
3	1	OHG–SUS–XX
4	1	ARM–OHG–SUS–S
5		
6		
REFERENCE DRAWINGS		
7	1	SGN–S
8	–	GND–S–XX
9	–	SEE MFR. DWG.
10	1	FOUNDATION TYPE T.B.D.
11	–	ANODE–XX (IF REQ.)

NOTES:
1. For General Notes, See Dwg. TMS220.

TABLE 1

VARIABLE DIMENSIONS & REF. DWGS.			
	69 kV	161 kV	230 kV
PT	8'–9"	10'–7"	11'–11"
PP	8'–0"	13'–0"	14'–0"
V	6'–8"	8'–6"	9'–10"
H	5'–0"	6'–4"	7'–10"
HC	6'–0"	7'–2"	9'–0"
INSUL	BP2–69–XX	BP2–161–XX	BP2–230–XX

A/OGA–V–BP2–S

ENTERGY SERVICES, INC.

Transmission LineDesign Standard
0–1.5° SC COMPACT VERT, BP2, STL
STRUCTURE DRAWING & DETAIL

STD NO. | SCALE: NONE

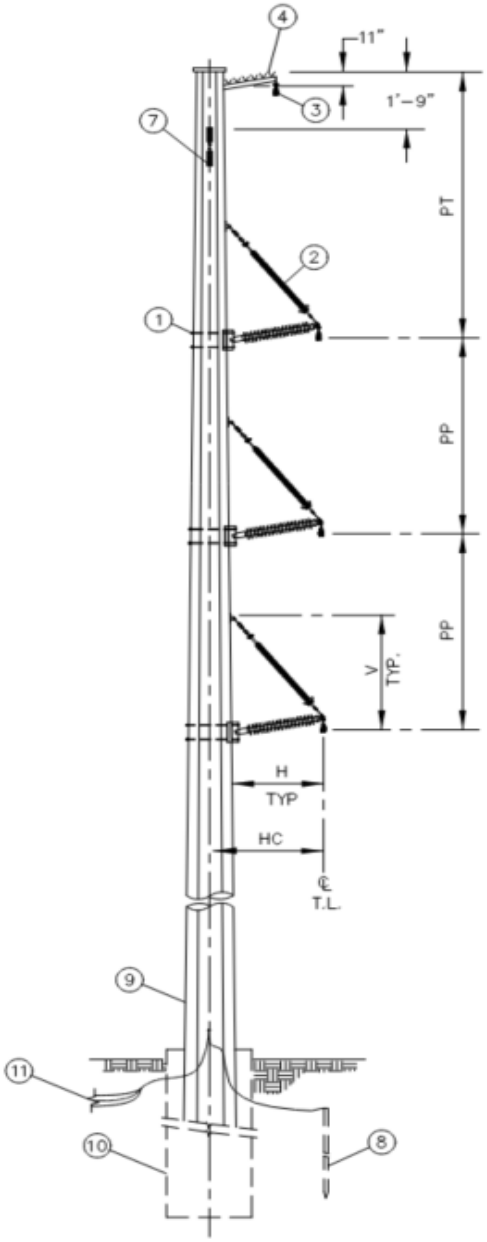
Entergy | No. TMS389A0 | PLOT 1=1 | SH. 1 OF 1

NO.	DATE:	REVISION	BY:	CHK:	APPR:
0	01–02–16	REPLACES DWS TCS723A1, & TDS664A1	WRK	KC	ECW

rkugl90 12/3/2015 A/OGA–V–BP2–S

A4-3: Structure Types

6'–20', VERTICAL, SINGLE CIRCUIT, BRACED POST 2 1/2", WITH 3' OHGW ARM, STEEL



ASSEMBLY LIST		
ITEM	QTY.	ASSEMBLY/DRAWING
1	3	BLT–P–S
2	3	INSUL (SEE TABLE 1)
3	1	OHG–SUS–XX
4	1	ARM–OHG–SUS–S
5		
6		
REFERENCE DRAWINGS		
7	1	SGN–S
8	–	GND–S–XX
9	–	SEE MFR. DWG.
10	1	FOUNDATION TYPE T.B.D.
11	–	ANODE–XX (IF REQ.)

TABLE 1

VARIABLE DIMENSIONS & REF. DWGS.			
	69 kV	161 kV	230 kV
PT	8'–9"	10'–7"	11'–11"
PP	8'–0"	13'–0"	14'–0"
V	6'–8"	8'–6"	9'–10"
H	5'–0"	6'–4"	7'–10"
HC	6'–0"	7'–2"	9'–0"
INSUL	BP2–69–XX	BP2–161–XX	BP2–230–XX

NOTES:
1. For General Notes, See Dwg. TMS220.
2. Line post adapters are required where line angles greater than 3°, or phase transitions pull suspension hardward toward the pole. See Dwg. TMD415 for additional materials.

SC/OGA–V–BP2–S

ENTERGY SERVICES, INC.

Transmission LineDesign Standard
6–20' SC VERT, BP2, STL
STRUCTURE DRAWING & DETAIL

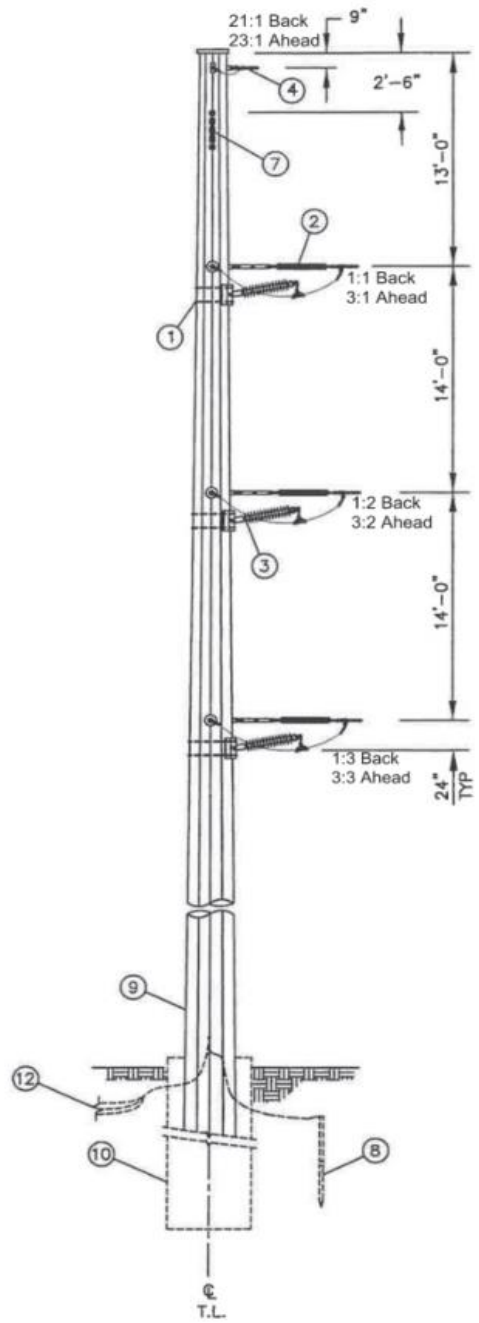
STD NO. | SCALE: NONE

Entergy | No. TMS619A0 | PLOT 1=1 | SH. 1 OF 1

NO.	DATE:	REVISION	BY:	CHK:	APPR:
0	01–02–16	REPLACES DWS TCS721A1, & TDS662A1	WRK	KC	ECW

rkugl90 12/2/2015 SC/OGA–V–BP2–S

5°–70°, SELF SUPPORTING, DE, JUMPER, VERTICAL,
SINGLE CIRCUIT, DE POLYMER, STEEL 161kV



ASSEMBLY LIST		
ITEM	QTY.	ASSEMBLY/DRAWING
1	3	BLT–P–S
2	6	DEP/TRB–161–XX
3	3	JLP–161–XX
4	2	OHG–DE–XX
REFERENCE DRAWINGS		
7	1	SGN–S
8	–	GND–S–XX
9	–	POLE INDEX
10	1	FOOTING
11	–	PVO, SHT 1, VW PV–19
12	–	ANODE–XX (IF REQ.)

NOTES:

1) All dimensions are to Centerline of top attachment or BLT–XX.
2) See Pole Fabricator's drawings for attachment details.
3) See Staking Sheet and CU Drawings for line angles, pole, footing, grounding, and sign requirements.
4) Refer to Assembly drawings for part details.
5) Conductor attachment vangs shall be placed within 5 degrees of Ahead and Back locations.
6) This structure shall not be used for single insulator bundled conductor applications unless the combined phase tension is less than 19.4 kips. For two insulator attachment bundled conductor applications, use the appropriate bundled framing.

Set:Phase

NO.	DATE:	REVISION	BY:	APPR:
3	8–18–04	REVISED DIMENSION FOR SIGN	ITRON	
2	4–3–03	GENERAL DRAWING REVISIONS	ITRON	
1	9–14–01	CHANGED ANGLE RANGE	LSFT	

SDJ–V–DEP–S 161
ENTERGY STANDARD DWG.

DESIGN APPROVAL	STANDARDS APPROVAL
SIGNED / DATE	SIGNED / DATE

DEADEND STEEL 161kV
5°–70° SINGLE CIRCUIT

APPROVED BY:	DATE: 10–17–96
CHECKED BY:	SCALE: NONE
DRAWN BY: LINESOFT	ESI NO. TCS444A3

Entergy

No.

PLOT 1=130 | SH. 1 OF 1

Figure A4-4: 3-D Interconnection Layout

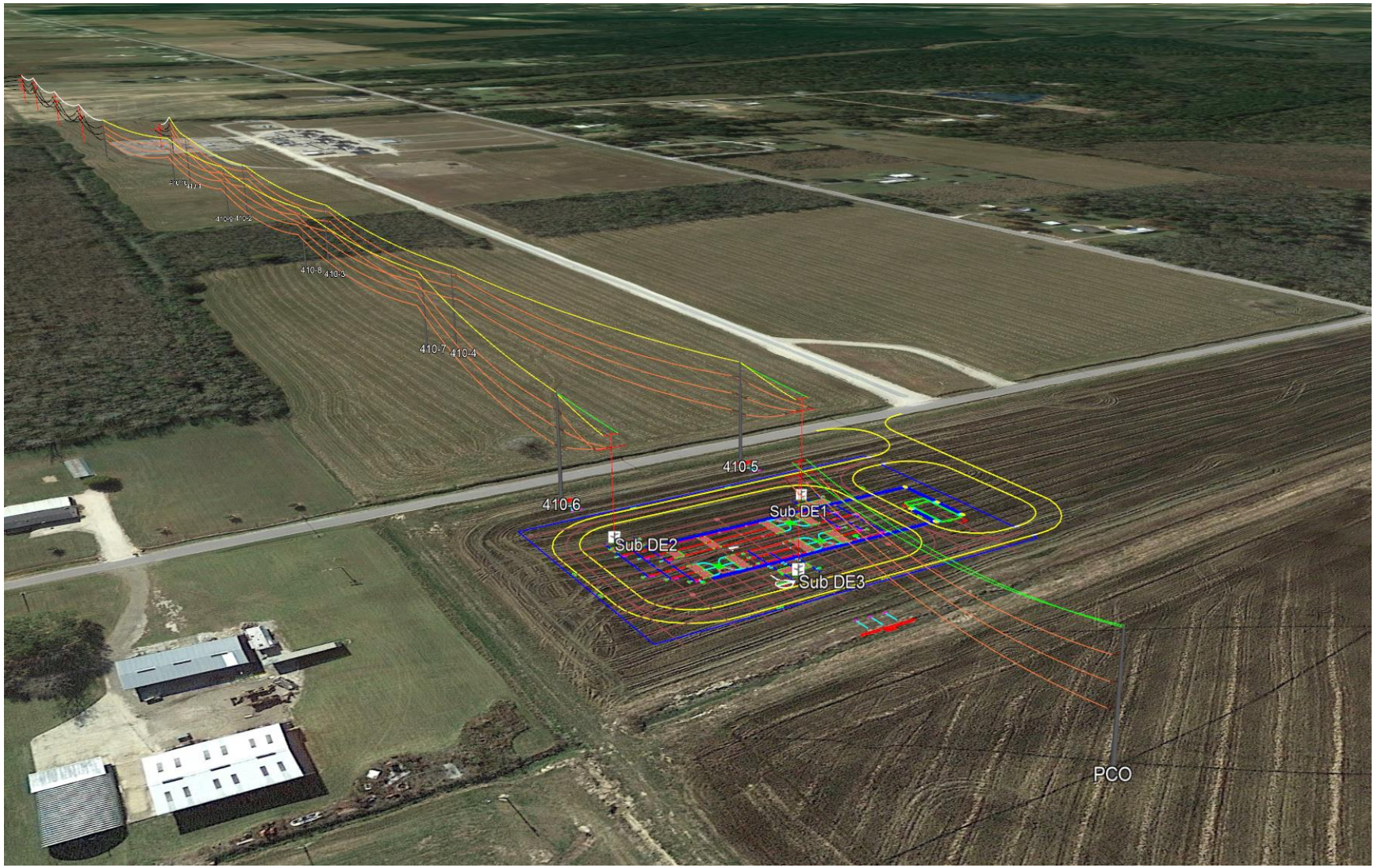


Exhibit A5. Facilities to be Constructed by Transmission Owner

Description of Upgrade	Upgrade Classification	Estimated Cost*
J1795 138kV Switching Station: Transmission Owner Interconnection Facilities shall consist of a high bus dead-end structure, a section of 3Ø tubular bus tapping off the ring bus, a motor operated disconnect switch with a ground switch attachment, three metering accuracy current transformer, and one capacitor voltage transformer. The facilities shall include associated support structures and associated foundations, as shown on Exhibit A4 which is a preliminary electrical profile view of the TOIFs. The TOIFs shall also include a suitable revenue accuracy meter panel and power quality meter panel for measuring power deliveries from the solar photovoltaic generator. One line protection panel and one breaker control panel shall also be installed.	TOIF	$ 1,474,000
J1795 138 kV Transmission Line: ELL's Interconnection Facilities shall consist of 666.6 kcmil 24/7 Strand ACSR "Flamingo", and 7#7 alumoweld shieldwire, along with all attachment hardware.	TOIF	$ 281,000
J1795 138kV Switching Station: The Stand-Alone Network Upgrades shall include a new interconnection three-position ring-bus switching station that will cut into the L-258 138 kV transmission line (Bayou Cove-Richard). A Remote Terminal Unit (RTU) shall be provided to gather accumulated data from the Interconnection Customer's facility via a fiber communication link. Two line protection panels and two breaker control panels shall be installed. One Digital Fault Recorder (DFR) with Digital Disturbance Recorder (DDR) shall be installed for independent fault condition and disturbance monitoring.	SANU	$ 11,561,000
L-258 138 kV Transmission Line: The Stand-Alone Network Upgrades for ELL's cut-in will comprise of two single circuit, self-supporting galvanized steel deadend structures on drilled pier foundations, four self supporting running angle structures on drilled pier foundations, and four single circuit self supporting single pole steel vertical framed tangent structures direct embedded to connect the existing Bayou Cove – Richard Sub line to the POI switching station.	SANU	$ 2,027,000

L-258 138 kV Transmission Line: Network Upgrades for ELL's cut-in will comprise of two steel self-supporting single pole dead-end structures on drilled pier foundations located on existing ROW and connect the existing Bayou Cove-Richmond Sub line to the POI switching station. Two existing deadend structures will be removed along with associated conductor and shield wire New 954 kcmil ACSR 54/7 "Cardinal" conductors and 7#7 Alumoweld shield wires and DNO 12160 OPTGW will be installed on the cut-in spans.	NU	$ 1,917,000
Bayou Cove 138 kV Substation: No relay or station upgrades have been identified for this project. Relay settings and Telecom updates shall be performed.	NU	$ 268,000
Richard 138 kV Substation: No relay or station upgrades have been identified for this project. Relay settings and Telecom updates shall be performed.	NU	$ 168,000
	Total Cost:	**$17,696,000**

Exhibit A6. Detailed Costs of Facilities to be Constructed by Transmission Owner

Item Description	Estimated Total	TOIF	SANU	NU
Contract Construction	$7,539,000	$726,000	$5,815,000	$998,000
Contract Project Management	$1,399,000	$171,000	$960,000	$268,000
ROW, Permitting Support & Environmental	$233,000	$15,000	$188,000	$30,000
Materials	$3,215,000	$305,000	$2,665,000	$245,000
Contract Engineering	$893,000	$85,000	$696,000	$112,000
Telecom	$232,000	$0	$155,000	$77,000
Payroll	$435,000	$72,000	$234,000	$129,000
Distribution	$0	$0	$0	$0
Expense	$82,000	$15,000	$51,000	$16,000
Total Direct Cost:	**$14,028,000**	**$1,389,000**	**$10,764,000**	**$1,875,000**
Capital Suspense:	$3,608,000	$360,000	$2,778,000	$470,000
AFUDC:	$0	$0	$0	$0
Other Overheads:	$60,000	$6,000	$46,000	$8,000
Total In-direct Cost:	**$3,668,000**	**$366,000**	**$2,824,000**	**$478,000**
Total Cost:	**$17,696,000**	**$1,755,000**	**$13,588,000**	**$2,353,000**

*Estimated Costs are in 2022 dollars

Exhibit A7. Facilities to be Constructed by Interconnection Customer

None.

Exhibit A8. Detailed Cost of Facilities to be Constructed by Interconnection Customer

None.

Exhibit A9. Facilities Subject to Transmission Owner Reimbursement

No facilities are subject to Transmission Owner reimbursement pursuant to Attachment FF.

Exhibit A10. Contingent Facilities

Higher queue and/or same DPP group study Interconnection Requests that may create contingencies pursuant to Article 11.3.1 are listed in tables below. Table A10-1 describes transmission assumptions modeled in the studies that were deemed necessary to allow for the Interconnection Service as described in Appendix A of this GIA and is not related to Article 11.3.1, i.e., does not describe projects associated with a higher queued and/or same DPP group study Interconnection Request. Nevertheless, if the transmission assumptions are not completed or significantly modified, the Interconnection Service granted under this GIA may be restricted until such time as the Interconnection Customer funds a study to determine the applicable ERIS and NRIS level that results due to the changes in Table A10-1.

The list of higher-queued and/or same DPP group study projects in Tables A10-2 and A10-3, not yet in service, were included in the interconnection study for this project. However, a project's inclusion in the System Impact Study does not necessarily mean that these facilities would be contingencies for the Interconnection Customer's Generating Facility. In the event that any of the higher queued and/or same DPP group study generators were to drop out, then the Interconnection Customer may be subject to restudy pursuant to Article 11.3.2.

Table A10-1: Transmission Assumptions

MTEP ID	Project Name	Facility ID	Facility Description	ISD	Status
None					

Table A10-2: Higher Queued Projects

Project #	Service Type	TO	County, State	Point Of Interconnection	Pmax (MW)	Fuel Type	Status
J328	NRIS Only	Entergy	Jefferson County, AR	5301 Fairfield Rd, Pine Bluff, AR	160	Gas	Done
J348	NRIS	Entergy	Arkansas County, AR	Entergy Stuttgart Rcuskey-Almyra115 kv line	81	Solar	Done
J373	NRIS Only	CLECO Corporation	St. Mary,St. Mary Parish, LA	Near Morgan City Substation on 138 kV side	71.5	Gas	Done
J376	NRIS Only	Entergy	Union County, AR	El Dorado substation	1270	Gas	Done
J396	NRIS	Entergy	St. Charles Parish, LA	Entergy Little Gypsy Power Station	1020	Gas	Done
J472	NRIS	Entergy	Montgomery County, TX	Entergy 138 kV and 230kV Lewis Creek Substations	1044.8	Gas	Done
J477	NRIS	Entergy	Hinds County, MS	Close to Entergy Hinds Power Station	36.4	Gas	Done
J481	NRIS	Entergy	Orleans Parish, LA	Existing Michoud Power Station	134.82	Gas	Done
J483	NRIS	Entergy	Liberty County, TX	Line tap at 138 kV line approximately 6,000 ft SW of Gordon Sub	99.96	Solar	Done
J484	NRIS	Entergy	Calcasieu, Calcasieu Parish, LA	Existing Entergy Nelson Power Station	1056.19	Gas	Done

J544	NRIS	Entergy	Morehouse Parish, LA	Galion 115kV Substation	98.1	Solar	Done
J552	NRIS	Entergy	Lee County, AR	Brinkley East - Moro 115kV Line	100	Solar	Done
J586	NRIS	Entergy	Mississippi County, AR	EAI Crooked Lake 161kV Substation	175	Solar	Done
J603	NRIS	Entergy	, AR	Tap on Montrose Mist 115 kV line	99.96	Solar	Done
J604	NRIS	Entergy	Sunflower County, MS	Ruleville 115kV Sub	99.96	Solar	Done
J620	NRIS	Entergy	Chicot County, AR	Tap 1396 - Tap 1398 115kV	100	Solar	Done
J473	ERIS	South Mississippi Electric Power Association	Lamar County, MS	Sumrall II substation	52	Solar	Done
J482	NRIS	Entergy	St. Charles Parish, LA	Little Gypsy Power Station	26.2	Gas	Done
J486	NRIS Only	Entergy Texas, Inc.	Grimes County, TX	Grimes 345kV Substation	51	Combined Cycle	Done
J663	NRIS	Entergy	Phillips County, AR	Robert E Ritchie 230kV Substation	100	Solar	Done
J697	NRIS	Entergy	Morehouse Parish, LA	Oak Ridge 115 kV Substation	150	Solar	Done
J830	NRIS	Entergy	Madison County, MS	Canton South to Hoy Road 230 kV Line	100	Solar	Done
J834	NRIS	Entergy	Phillips County, AR	Robert E Ritchie 230kV Substation	100	Solar	Done
J866	NRIS	Entergy	Tunica County, MS	Batesville - Robert E. Ritchie 230 kV line	200	Wind	Done

J868	NRIS	Entergy	Concordia Parish, LA	Plantation 115 kV Substation	37.2		Done
J888	NRIS	South Mississippi Electric Power Association	George County, MS	Benndale 69kV Substation	25	Gas	Done
J893	NRIS	Entergy	White County, AR	Searcy Price Sub 161kV	100	Solar	Done
J907	NRIS	Entergy	Monroe County, AR	Brinkley East - Aubrey 230 kV Line	200	Solar	Done
J908	NRIS	Entergy	Tallahatchie County, MS	Webb 115kV Substation	100	Solar	Done
J909	NRIS	Entergy	Tangipahoa Parish, LA	Amite 115 kV Substation	100	Solar	Done
J919	NRIS	Entergy	Jackson County, AR	Cash - Newport AB 161 kV Line Tap	180	Solar	Done
J934	NRIS	Entergy	Crittenden County, AR	Kuhn 161kV Substation	200	Solar	Done
J944	NRIS	Entergy	New Madrid County, MO	Portageville - New Madrid 161kV Line	200	Solar	Done
J1425	NRIS Only	Entergy Arkansas, LLC	Jefferson County, AR	Pine Bluff International Paper 115 kV substation	114	Gas	Done
J679	NRIS	Entergy	Carroll County, MS	Greenwood - Winona 115kV	100	Solar	Done
J680	NRIS	Entergy	Ashley County, AR	Crosset West 115kV Substation	100	Solar	Done
J683	NRIS	Entergy	West Baton Rouge Parish, LA	Wilbert - Livonia 138kV Substation (~1.5 mi from Wilbert)	50	Solar	Done

J1007	NRIS	Entergy Arkansas, LLC	Crittenden County, AR	Marion - Marked Tree 161kV Line Tap	200	Solar	Done
J1030	ERIS	Entergy Texas, Inc.	Polk County, TX	Poco - Menard 138kV Line Tao	150	Solar	Done
J1060	NRIS	Entergy Arkansas, LLC	St. Francis County, AR	West Memphis - Keo 500 kV Line Tap	500	Solar	Done
J1076	NRIS	Entergy Louisiana, LLC	St. James Parish, LA	Vacherie Sub 230 kV Substation	200	Solar	Done
J1125	NRIS	Entergy Arkansas, LLC	Cross County, AR	Wynne South - Wynne East AECC 161 kV Line Tap	135	Solar	Done
J1142	NRIS	Entergy Louisiana, LLC	St. James Parish, LA	Vacherie 230kV Substation	20	Solar	Done
J1144	NRIS	Cooperative Energy	Covington County, Lamar County, MS	Moselle - Columbia 161kV Line Tap	96	Solar	Done
J1146	NRIS	Entergy Arkansas, LLC	Mississippi County, AR	San Souci 500 kV Substation	10	Coal	Done
J1155	NRIS	Entergy Arkansas, LLC	Mississippi County, AR	Driver 230 kV Substation	150	Solar	Done
J1158	NRIS	Entergy Louisiana, LLC	St. James Parish, LA	Vacherie 230 kV Substation	70	Solar	Done
J1184	NRIS	Entergy Louisiana, LLC	Washington Parish, LA	Lagen Holton-Franklinton 115kV Line Tap	50	Solar	Done
J962	ERIS	Entergy Mississippi, LLC	DeSoto County, MS	Twinkletown 230 kV Substation	100	Solar	Done

J1205	ERIS	CLECO Corporation	Allen Parish, LA	Centennial - West Bay 138kV Line	125	Solar	Done
J1215	NRIS	Entergy Arkansas, LLC	Arkansas County, AR	Stuttgart Ricuskey - Woodward 230 kV	200	Solar	Done
J1219	NRIS	Entergy Louisiana, LLC	Tangipahoa Parish, LA	Chicken Farm - Amite 115 kV Line	100	Solar	Done
J1239	ERIS	Entergy Louisiana, LLC	Ouachita Parish, LA	Rilla 115 kV Substation	100	Solar	Done
J1246	NRIS	Entergy Louisiana, LLC	Iberville Parish, LA	Bayou Labutte 500kV Substation	725.2	Gas	Done
J1247	ERIS	CLECO Corporation	St. Mary,St. Mary Parish, LA	Ivanhoe 138kV Substation	16.98	Waste Heat Recovery	Done
J1249	NRIS	Cooperative Energy	Lamar County, MS	Morrow 161 kV Substation	600	Gas	Done
J1257	NRIS	Entergy Louisiana, LLC	Tangipahoa Parish, LA	Chicken Farm - Amite 115 kV Line	20	Battery Storage	Done
J1258	NRIS	Entergy Arkansas, LLC	Lonoke County, AR	KEO EHV 500 kV Substation	50	Solar	Done
J1281	NRIS	Entergy Arkansas, LLC	Ashley County, AR	Crossett - Monticello East 115kV Line	75	Solar	Done
J1286	ERIS	CLECO Corporation	Evangeline Parish, LA	Manuel 138 kV Substation	150	Solar	Done
J1294	NRIS	Entergy Mississippi, LLC	Sunflower County, MS	Indianola - Cleveland 115kV Line	99	Solar	Done
J1347	ERIS	Entergy Mississippi, LLC	Scott County, MS	Pelahatchie - Scott 115 kV Line Tap	175	Solar	Active

J1361	NRIS	Entergy Louisiana, LLC	Washington Parish, LA	Lagen Holton - Franklinton 115kV Line (J1184)	50	Solar	Done
J1367	NRIS	CLECO Corporation	St. Mary, St. Mary Parish, LA	Caneland 230kV Substation	95	Solar	Done
J1368	NRIS	Entergy Louisiana, LLC	Pointe Coupee, Pointe Coupee Parish, LA	Fancy Point - Waterloo 230 kV Line	200	Solar	Active
J1372	NRIS	Entergy Louisiana, LLC	Pointe Coupee, Pointe Coupee Parish, LA	Fancy Point - Waterloo 230 kV Line (J1368 POI)	100	Solar	Active
J1373	NRIS	Entergy Arkansas, LLC	White County, AR	Griffithville - Searcy Price 115 kV Line Tap	95	Solar	Done
J1402	NRIS	Entergy Arkansas, LLC	Jackson County, AR	Cash - Newport AB161kV Line	0	Battery Storage	Done
J1415	NRIS	Entergy Arkansas, LLC	Lonoke County, AR	KEO EHV 500 kV Substation	250	Solar	Done
J1421	NRIS	Entergy Louisiana, LLC	East Feliciana Parish, LA	Moler 230 kV Substation	300	Solar	Active
J1424	ERIS	CLECO Corporation	Beauregard Parish, LA	Cooper - Penton Road 230kV Line	200	Solar	Done
J1431	NRIS	Entergy Louisiana, LLC	St. James Parish, LA	Vacherie 230kV Substation	200	Solar	Done
J1434	NRIS	Entergy Arkansas, LLC	Monroe County, AR	J907 230kV Sub, Brinkley East - Aubrey 230 kV Line	100	Solar	Done
J1436	ERIS	Entergy Louisiana, LLC	Morehouse Parish, LA	Oak Ridge 115 kV Substation	50	Solar	Done

J1437	NRIS	Arkansas Electric Coop Corp	Carroll County, AR	AECC Dry Creek 161kV Switchyard Station	180	Wind	Done
J1440	NRIS	Entergy Mississippi, LLC	Tunica County, MS	Lula - Tunica 115 kV Line	160	Solar	Done
J1442	ERIS	Entergy Texas, Inc.	Polk County, TX	Rich (SHECO) - Livingston 138 kV Line Tap	165	Solar	Done
J1458	NRIS	Entergy Mississippi, LLC	Washington County, MS	Ray Braswell - Tinnin - Gerald Andrus 230kV Line	200	Solar	Done
J1463	ERIS	Entergy Louisiana, LLC	West Baton Rouge Parish, LA	Addis - Big Cajun 230 kV Line Tap (J639)	150	Solar	Active
J1465	NRIS	Entergy Louisiana, LLC	Pointe Coupee, Pointe Coupee Parish, LA	Livonia - Colonial Springs 138kV line	150	Solar	Done

Table A10-3: Similar Queued Projects

Project #	Service Type	TO	County, State	Point Of Interconnection	Pmax (MW)	Fuel Type	Status
J1509	NRIS	Entergy Mississippi, LLC	Leflore, MS	Greenwood - Pickens 115 kV	100	Solar	Active
J1514		Entergy Arkansas, LLC	Mississippi, AR	Driver 230kV	100	Solar	Active
J1542	NRIS	Entergy Louisiana, LLC	St. James, LA	Thibodaux - Vacherie 230kV	135	Solar	Active
J1558	NRIS	Entergy Arkansas, LLC	Pope, AR	Moreland - St. Vincent 161kV line	200	Solar	Active

J1559	NRIS	Entergy Arkansas, LLC	Conway, AR	Morrilton East Substation (161kV bus)	200	Solar	Active
J1561	NRIS	Entergy Louisiana, LLC	Acadia, LA	Richard 138kV	75	Solar	Active
J1562	NRIS	Entergy Arkansas, LLC	Mississippi, AR	Blytheville I-55 to AECC Blytheville North 161kV	200	Solar	Active
J1564	NRIS	Entergy Louisiana, LLC	Washington, LA	Talisheek - Bogalusa 230 kV Line	100	Solar	Active
J1576	NRIS	Entergy Louisiana, LLC	Washington, LA	Angie to Adams Creek 230 kV	200	Solar	Active
J1577	NRIS	Entergy Arkansas, LLC	Grant, AR	Hardin West SS (Endpoint A) to Sheridan (Endpoint B) 115 kV	100	Solar	Active
J1583	NRIS	Entergy Louisiana, LLC	Iberville, LA	Donaldsonville - Evergreen 230kV	175	Solar	Active
J1598	NRIS	Entergy Louisiana, LLC	Calcasieu, LA	Solac - Chalkley 230kV	325	Solar	Active
J1599	NRIS	CLECO Corporation	Iberia, LA	Morbihan - Hopkins 138kV	100	Hybrid	Active
J1602	NRIS	Entergy Louisiana, LLC	Ouachita, LA	Rilla 115kV substation (Same as J1239)	25	Battery Storage	Active
J1607	NRIS	Entergy Arkansas, LLC	Grant, AR	Hardin West SS to Sheridan 115 kV	0	Battery Storage	Active
J1612	NRIS	Entergy Arkansas, LLC	Union, AR	El Dorado East - Sterlington 115 kV	100	Solar	Active
J1644	NRIS	CLECO Corporation	St. Landry, LA	Champange - Plaisance	150	Solar	Active

J1669	ERIS	Entergy Louisiana, LLC	Jefferson Davis, LA	LC Bulk to Henning 138kV Line (Circuit 28)	165	Solar	Active
J1670		Entergy Arkansas, LLC	Mississippi, AR	Crooked Lake 161kV at J586 POI	50	Solar	Active
J1671	ERIS	Entergy Texas, Inc.	Walker, TX	Rocky Creek to Lewis Creek 230kV	150	Solar	Active
J1672	NRIS	Entergy Mississippi, LLC	Hinds, MS	Tinnin Road 230kV	150	Solar	Active
J1673	NRIS	CLECO Corporation	Sabine, LA	Fisher to Mansfield Compressor 230 kV	200	Solar	Active
J1686	NRIS	CLECO Corporation	Evangeline, LA	Coughlin - Plaisance 138 kV	100	Hybrid	Active
J1709	ERIS	Entergy Texas, Inc.	Grimes, TX	138 kV Grimes	100	Battery Storage	Active
J1710	NRIS	Entergy Arkansas, LLC	Lee, AR	Entergy Keo-Memphis 500kV Line Tap	300	Solar	Active
J1747	NRIS	Entergy Mississippi, LLC	Madison, MS	J830 POI	50	Solar	Active
J1748	NRIS	Entergy Mississippi, LLC	DeSoto, MS	Twinkletown 230 kV	100	Solar	Active
J1759		CLECO Corporation	Sabine, LA	Fisher to Mansfield Compressor 230 kV	0	Battery Storage	Active
J1760	NRIS	Entergy Texas, Inc.	Liberty, TX	Timberland 230kV	205	Solar	Active
J1790		Entergy Arkansas, LLC	Ashley, AR	Crossett - Monticello East 115 kV Line	0	Battery Storage	Active
J1794	ERIS	Entergy Louisiana, LLC	Acadia, LA	Richard to Scott 138kV line	150	Solar	Active

J1795	ERIS	Entergy Louisiana, LLC	Acadia, LA	Bayou Cove to Richard 138kV (Circuit 258)	100	Solar	Active
J1813		Entergy Mississippi, LLC	DeSoto, MS	Entergy Freeport 230 kV Substation	205	Gas	Active
J1816	NRIS	Entergy Arkansas, LLC	Cross, AR	Fisher (Endpoint A) to Cherry Valley (Endpoint B) 161 kV line	135	Solar	Active
J1818	NRIS	Lafayette Utilities System	Lafayette, LA	Northeast (Moss) 69kV	80	Solar	Active
J1819	NRIS	Entergy Arkansas, LLC	Lee, AR	Entergy Keo - West Memphis 500kV Line Tap	0	Battery Storage	Active
J1820	NRIS	Entergy Arkansas, LLC	Lee, AR	Entergy Keo-West Memphis 500kV Line Tap	100	Solar	Active
J1821	NRIS	Entergy Arkansas, LLC	Lee, AR	Entergy Keo -West Memphis 500kV Line tap	0	Battery Storage	Active
J1827	NRIS	Entergy Louisiana, LLC	Morehouse, LA	Bastrop tap point	90	Solar	Active
J1832	NRIS	CLECO Corporation	St. Landry, LA	Plaisance 138kV	145	Solar	Active
J1833	NRIS	Entergy Louisiana, LLC	Lafourche, LA	Leeville 115kV	115	Gas	Active
J1834	NRIS	Entergy Texas, Inc.	Orange, TX	Sabine 138/230 kV Substation	126	Gas	Active
J1842	ERIS	Entergy Arkansas, LLC	Cross, AR	Forrest City to Wynne S 161 KV	135	Wind	Active

Exhibit A11. Interconnection Customer Milestones

This Exhibit A11 is provided for reference only and is not binding.

Interconnection Customer has requested a desired in-service date of 8/31/2023 for Transmission Owner's interconnection substation. The customer milestones include, but are not limited to those shown in the table below:

Table A11-1: Milestone Schedule

IC Milestone Schedule	
Task Name	**Proposed Finish**
GIA Execution:	02/17/2023
GIA Payment:	04/03/2023
Landowner Access Approval (soil borings):	05/08/2023
Site Acquisition (servitude agreements):	06/20/2023
Receipt of Environmental Permitting:	06/06/2023

Exhibit A12. Construction and Coordination Schedules

This Exhibit A12 is provided for reference only and is not binding. Binding milestones can be found in Appendix B of this GIA.

The proposed schedule is dependent on obtaining a suitable substation site, on timely receipt of all approvals associated with constructing the Transmission Owner interconnection substation and lines, outage availability, equipment lead-time, and resource availability. This schedule does not include upgrades that may be required because of the system impact studies. An Engineering and Procurement Agreement between the parties could potentially be utilized to begin work before the generation interconnection agreement is executed.

A detailed schedule will be prepared subsequent to customer approval to proceed with the project. Based on the Task duration schedules listed below, the overall project in-service date is projected to be 12/04/2025 The following milestone dates are based on an assumed Customer Notice to Proceed date of 02/17/2023.

Table A12-1: Milestone Schedule (Reference only)

Task Name	Proposed Start	Proposed Finish
GIA Execution:	02/17/2023	
Customer Approval & Funding:	04/03/2023	
Complete Project Execution Plan:	11/02/2023	
Update/ Baseline Schedule:	11/02/2023	
Detail Design:	11/03/2023	01/31/2025
Order Material:	01/23/2024	02/19/2025
Receive Material:	02/03/2025	08/08/2025
Issue for Construction Design Packages:	02/05/2024	01/31/2025
Environmental Permits:	11/23/2023	
Construction:	10/07/2024	12/04/2025
In-Service:	12/04/2025	

1. This schedule does not account for adverse weather conditions or major weather events.
2. All construction work requiring outages will be performed during acceptable periods of system load, which most often is the off-peak load season. Line outages will be discussed with the SOC and TOC and the assumption is made that line outages will be executed as planned. However, last minute denial of outages by the SOC/TOC along with resulting schedule delay is possible.
3. Substation construction will be coordinated with the transmission line outages when possible.
4. The above milestone schedule assumes construction resources will be available when required.

Exhibit A13. Permits, Licenses, Regulatory Approvals and Authorization

Permits, licenses, and approvals required to construct the Transmission Owner facilities may include, but are not limited to:

1. Revised station one line to be submitted to MISO, TCC and TOP for review and approval
2. Wetland delineation of servitudes and proposed construction access points
3. US Corps of Engineers permitting/approval for wetlands
4. US Corps of Engineers permitting/approval for river/levee permits
5. Environmental sampling and disposal of displaced soils
6. Develop and implement a Storm Water Pollution Prevention Plan (SWPPP)
7. Temporary landowner easements
8. Pipeline Letters of No Objection
9. Access Road Permits for Public Roads

Exhibit A14. Interconnection and Operating Guidelines

A14.1 Interconnection

The Interconnection Customer will comply with the Interconnection Guidelines for Transmission Interconnection Producer-Owned Generation Greater Than 20 MW dated March 2007, as amended from time to time and available at: http://entergy.com/energydelivery/facility_requirements.aspx ("ENTERGY FACILITY CONNECTION REQUIREMENTS").

A14.2 Power Factor

In accordance with FERC Order No. 827, all newly interconnected generators (synchronous or non-synchronous) are required to provide dynamic reactive power support within the range of +/- 0.95 power factor at the transmission voltage side of the generator step-up transformer at all dispatch levels. Static reactive power devices can only be used to make up for reactive losses that occur between the inverters and the transmission voltage side of the generator step-up transformer. Any additional reactive power must be supplied by a dynamic reactive power device.

The +/-0.95 power factor requirement corresponds to a triangle-shaped reactive power capability with proportional reduction in reactive power capability at lower active power (i.e. real power) output, as shown in Figure 14.1. However, reactive power capability outside the triangular-shaped requirement yet within the reactive power capability of the plant should be utilized to the greatest extent possible. Inverters shall not have artificial settings imposed to limit reactive power output to the triangular boundary.

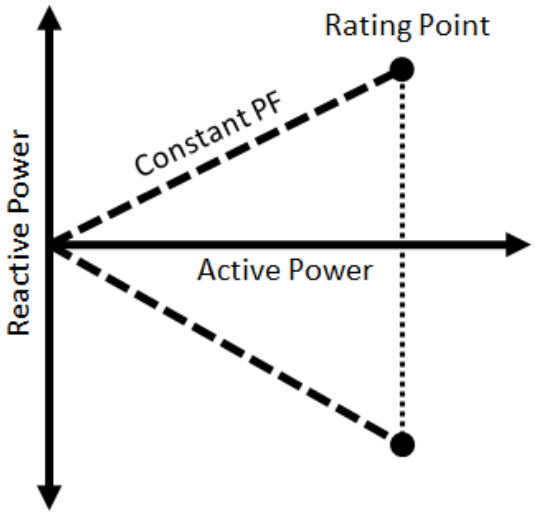


Figure A14-1: Inverter-based Resource Plant Reactive Power Capability

A14.3 Primary Frequency Response

In accordance with FERC Order No. 842, the Interconnection Customer shall ensure the primary frequency response capability of its Generating Facility by installing, maintaining, and operating a functioning governor or equivalent controls. The hardware and/or software that provides frequency responsive active power control shall be able to sense changes in system frequency and autonomously adjust the Generating Facility's real power output in accordance with the droop and deadband parameters provided by Entergy Transmission and to the extent the Generating Facility has operating capability in the direction needed to correct the frequency deviation. The Interconnection Customer is required to install a governor or equivalent controls with the capability of operating with a maximum 5% droop and a maximum ± 0.036 Hz deadband. This response must be timely and sustained rather than injected for a short period and then withdrawn. The primary frequency response requirements shall apply to all inverter-based generating resources including energy storage during both charging and discharging states.

The Generator Owner shall provide the status and settings of the governor or equivalent controls to Entergy Transmission upon request. If the Generator Owner needs to operate the Generating Facility with its governor or equivalent controls not in service, the Generator Owner shall immediately notify Entergy Transmission and make reasonable efforts to return its governor or equivalent controls to service as soon as practical.

A14.3.1 Dynamic Frequency Response Requirements for Inverter-Based Resources

For a step change in frequency at the Point of Interconnection (POI), inverter-based resources (IBR) shall have the capability to meet the dynamic characteristics shown in Table A14.1. The active power-frequency response shall be sustained by the resource until such time that the control signals (e.g., Balancing Authority automatic generation control (AGC)) return the generator to a new set point value.

Table A14.1: Dynamic Active Power-Frequency Performance

Parameter	Description	Performance Requirement
Reaction Time	Time between the step change in frequency and the time when the IBR's active power output begins responding to the change.	< 500 ms
Rise Time	Time in which the IBR has reached 90% of the new steady-state (target) active power output command.	< 4 s
Settling Time	Time in which the IBR has entered into, and remains within, the settling band of the new steady-state active power output command.	< 10 s
Overshoot	Percentage of rated active power output that the IBR can exceed while reaching the settling band.	< 5%
Settling Band	Percentage of rated active power output that the resource should settle to within the settling time.	< 2.5%

A14.4 Frequency and Voltage Ride Through

The frequency and voltage ride-through curves described in NERC Reliability Standard PRC-024-2 apply to the POI and not the inverter terminals themselves. The Interconnection Customer, in

coordination with their inverter manufacturer, shall reflect the ride-through requirements at the POI to the inverter terminals to ensure that the inverters can ride through and continue current injection during disturbances.

The PRC-024-2 voltage and frequency ride-through curves specify a “No Trip Zone”. Operations outside of the “No Trip Zone” shall not be interpreted as a “Must Trip Zone” but rather a “May Trip Zone”.

A14.4.1 Frequency Measurements and Protection

Frequency shall be calculated over a period of time (no less than 6 cycles) and filtered to take control action on the fundamental frequency component of the calculated signal. Calculated frequency shall not be susceptible to spikes caused by phase jumps on the Bulk Electric System (BES).

Inverter-based resources shall also inhibit frequency tripping during and immediately after low voltage conditions. Inverter-based resources shall ensure that the frequency measurement and protection settings are set such that these resources are able to ride through and not trip for phase jumps or other grid disturbances where calculated frequency is affected but grid frequency is within the ride-through curves of PRC-024-2.

A14.4.2 Momentary Cessation

Inverter-based resources interconnected to the transmission system are expected to continue current injection inside the “No Trip” zone of the PRC-024-2 frequency and voltage ride through curves. Newly interconnecting inverter-based resources should eliminate the use of momentary cessation unless necessary in areas of the BES with low short circuit strength (SCR < 3).

A14.5 Voltage Control

BES-connected resources are required to operate in automatic voltage control mode as per NERC Reliability Standard VAR-002-4. All resources, including inverter-based resources, shall operate in a closed-loop, automatic voltage control mode to maintain voltage at the POI to within the specified voltage schedule tolerance band provided by the Transmission Operator as per NERC Reliability Standard VAR-002-4.

A14.5.1 Small Disturbance Reactive Power-Voltage Control Performance

Individual inverters shall operate in automatic voltage control at all times to support BES voltage schedules, post-contingency voltage recovery, and voltage stability. Each inverter shall continually respond to all changes in terminal voltage in a closed loop fashion to maintain the set point voltage level.

Inverter-based resources shall have the capability to meet the performance characteristics shown in Table A14.2. These characteristics are specified for the response of reactive power of the overall closed-loop response of the inverter-based resource plant.

Table A14.2: Dynamic Reactive Power-Voltage Performance for Small Disturbance

Parameter	Description	Performance Requirement
Reaction Time	Time between the step change in voltage and when the IBR's reactive power output begins responding to the change. Shall not include any intentional time delay.	< 500 ms
Rise Time	Time between a step change in control signal input (reference voltage) and when the reactive power output changes by 90% of its final value.	< 2-30 s
Overshoot	Percentage of rated reactive power output that the IBR can exceed while reaching the settling band.	< 5%

A14.5.2 Large Disturbance Reactive Power-Voltage Performance

The reactive power-voltage control shall be stable over all expected operating and grid conditions and be tuned to provide this stable response. The dynamic response of inverter-based resources shall be programmable by the Interconnection Customer (in coordination with the inverter manufacturer) to enable changes based on changing grid conditions once installed in the field.

Inverter response to fault events on the BES shall abide by the following principles:

- **Fault Inception**: During the inception of a fault, priority shall be given to delivering as much current to the system as quickly as possible to support protective relay systems to detect and clear the fault.
- **On-Fault Current Injection**: For the remaining on-fault period after the first couple cycles up to fault clearing, priority shall be given to accurately detecting and controlling the type of current needed based on terminal conditions, and providing a combination of active and reactive current as necessary.
- **Post Fault Current Injection**: Priority shall be given to ensuring sufficient local voltage support before attempting to maintain or return to pre-disturbance active current injection.
- **Post-Fault Voltage Overshoot Mitigation**: Inverter reactive current response shall not exacerbate transient overvoltage conditions on the BES.

Inverters shall be designed to have the capability to meet the performance specifications shown in Table A14.3.

Table A14.3: Dynamic Reactive Power-Voltage Performance for Large Disturbance

Parameter	Description	Performance Requirement
Reaction Time	Time between the step change in voltage and when the IBR's reactive power output begins responding to the change.	< 16 ms
Rise Time	Time between a step change in control signal input (reference voltage) and when the reactive power output changes by 90% of its final value.	< 100 ms

A14.7 Transmission System Studies

NERC Reliability Standard FAC-002-2 requires each Transmission Planner to assess the reliability impact of interconnecting new generation as well as any material modifications to existing interconnections of generation. Steady-state, dynamic, and short circuit studies are performed as part of the MISO Definitive Planning Phase (DPP) process. However, there are other considerations that may need to be assessed based on the interconnection location, IBR control functionality, and nearby resources (SVCs, series capacitors, synchronous generation, IBR generation, etc.).

A14.7.1 Controls Interactions and Controls Instability

Inter-plant coordination is increasingly important as the penetration of inverter-based resources continues to grow. When multiple resources attempt to control the same location on the BES (e.g., controlling voltage at the same POI or close proximity), their control systems need to be coordinated with one another in order to avoid reactive power output imbalance, circular reactive power flows, controls overshoot, voltage control hunting, and unstable oscillations. Coordination issues shall be evaluated prior to the resource connecting to the BES. Prior to performing this analysis, the consultant selected by the Interconnection Customer shall be approved by Entergy Transmission. This analysis shall be performed during the plant design phase prior to plant backfeed. The results of the analysis shall be provided to Entergy Transmission with the cost of the study borne by the Interconnection Customer.

A14.7.2 Weak Grid Analysis (SCR < 3)

The changing nature of the electric grid and the growing complexity of IBR control systems can impact the performance of inverter-based resources and shall be assessed to ensure reliable operation of the BES. One of these changes is the continued reduction of short circuit levels in areas of high penetration of inverter-based resources. For interconnections with a short circuit ratio < 3, an EMTP study is required to assess potential SSR/SSCI and the proposed control design. Prior to performing this analysis, the consultant selected by the Interconnection Customer shall be approved by Entergy Transmission. This analysis shall be performed during the plant design phase prior to plant backfeed. The results of the analysis shall be provided to Entergy Transmission with the cost of the study borne by the Interconnection Customer.

A14.7.3 Power Quality

Due to converter characteristics associated with IBR generation, there is increased possibility of generating harmonic currents; therefore, the Interconnection Customer shall make provisions to install power quality monitoring devices to ensure that the IEEE 519 and IEEE 1453 requirements are met at the POI with the IBR. The data retention period for the power quality meter shall be at least 30 days. If the power quality meter at the POI identifies harmonic currents (total demand distortion), harmonic voltages (total harmonic distortion), or other power quality issues in excess of the power quality criteria specified in IEEE 519 and IEEE 1453, and either complaints related to power quality are received from other customers on the system in the vicinity of the IBR plant, or Entergy experiences equipment performance issues that can be attributed to power quality then operating

restrictions will be placed on the IBR plant by Entergy Transmission until it has been adequately demonstrated that the power quality issues have been resolved.

The Interconnection Customer shall provide harmonics modeling data for the inverters to be installed and modeling data for the IBR collector system in sufficient detail to perform a harmonics study of the IBR plant. In the event that a harmonics study is required due to power quality meter measurements at the POI in excess of the power quality criteria, the costs to conduct such studies will be the responsibility of the Interconnection Customer. Background measurements for harmonics, flicker, and voltage unbalance shall be supplied to Entergy Transmission prior to plant backfeed.

The IBR shall adhere to Entergy Standard AM3902 *Transmission Power Quality Guide.*

A14.7.4 Subsynchronous Resonance and Subsynchronous Torsional Interaction

Additional studies, such as subsynchronous resonance and subsynchronous torsional interaction, may be performed depending on the interconnection location. No poorly damped subsynchronous modes or interactions shall be permitted on the utility system.

Despite the results of the above-mentioned studies, if the IBR performance is found unsatisfactory after the commencement of commercial operation, such as problems associated with overvoltages, resonance, power quality, damping, or any adverse interactions impacting the reliability of the BES, the Interconnection Customer shall be responsible for mitigating the unsatisfactory performance.

A14.8 Modeling

The steady-state and dynamic models used to represent inverter-based resources shall accurately capture the small and large disturbance aspects of the resource. The following modeling information shall be provided to Entergy Transmission within 60 calendar days prior to plant backfeed.

- Steady state model including collector system equivalent as recommended by NREL, collector station transformer, padmount transformer, and transmission gen-tie line.
- Dynamic models using the 2nd Generation Renewable Energy Models [PSS/E Models REPC, REGC, and REEC].
- Frequency and voltage ride through characteristics and corresponding dynamic models [PSS/E VTGTPA, FRQTPA]. The low-voltage ride-through capability of the PV plant shall be demonstrated either through detailed modeling of the PV controls or by field/ factory tests.
- Inverter PQ curve.
- CAPE short circuit model using a Voltage Controlled Current Source Model.
- EMTP-RV or PSCAD model of the IBR plant.
- Harmonic current spectrum information for modeling of the PV plant in harmonic scan studies.

Model verification shall be performed as per NERC Reliability Standards MOD-026-1 and MOD-027-1 and transmitted to the Transmission Planner within 365 calendar days after the commissioning date.

A14.9 Disturbance Monitoring

Various measurement and monitoring technologies are necessary to capture the performance of IBRs in response to small and large disturbances on the system. Sequence of events recording (SER) data provides information related to what occurred and when each action was taken. Digital fault recorder (DFR) and dynamic disturbance recorder (DDR) data captures the dynamic response of the resource at the POI and within the plant.

Measurements are taken throughout the inverter-based resource, from the individual inverters to the POI, using the technologies described above. The requirements relating to the types of data, resolution of that data, and retention of that data required of the Interconnection Customer is provided in Table A14.4. This table is not intended to be all-inclusive; rather, it provides an overview of the types of measurements and associated resolution and retention requirements.

All data within an inverter-based resource plant shall be time synchronized to a common reference time.

The data from the below-listed measuring devices shall be provided to Entergy Transmission within 30 calendar days of a request.

Table A14.4: Measurement Data and Retention Requirements

Data Type	Measurement/Data Points	Resolution	Retention
Sequence of Events Recording (SER) Data	SER devices shall be sized to capture and store thousands of event records and logs. SER events records can be triggered for many different reasons but include, at a high level, the following: • Event date/time stamp (synchronized to common reference) • Event type (status changes, synchronization status, configuration change, etc.) • Description of action • Sequence number (for potential overwriting)	≤ 1 millisecond	90 days
Digital Fault Recording (DFR) Data	This data shall be captured for at least the plant-level (e.g., at the POI) response to BES events. It is typically high resolution (kHz) point-on-wave data and triggered based on configured settings. Data points shall include: • Bus voltage phase quantities • Bus frequency (as measured/calculated by the recording device) • Current phase quantities • Calculated active and reactive power output • Dynamic reactive element voltage, frequency, current, and power output	> 960 samples per second, triggered	90 days
Dynamic Disturbance	A DDR (e.g., a PMU or digital relay with this capability) shall capture the plant-level response during normal and disturbance events. This data	> 30 samples	1 year

Recorder (DDR) Data	shall be captured continuously at the POI and can be used for multiple purposes including event analysis and disturbance-based model verification. Data points shall include: • Bus voltage phasor (phase quantities and positive sequence) • Bus frequency • Current phasor (phase quantities and positive sequence) • Calculated active and reactive power output	per second, continuous	

A14.10 Communication Paths

The Interconnection Customer shall coordinate with the Transmission Owner to ensure that Transmission Owner's microwave paths are not obstructed by wind turbines, as applicable, associated with the Generating Facility. Obstruction shall be defined as any encroachment on the 3D cylindrical ellipse, as determined by the Worst Case Fresnel Zone calculation, that would interfere with communication signals propagating between the transmitter and receiver. At a minimum, this coordination shall consist of one of the following:

The Interconnection Customer shall provide the Transmission Owner with geographical coordinates, along with tower heights and blade diameters, of each proposed wind turbine associated with the Generating Facility for review against the Transmission Owner's Licensed Microwave Paths. The Transmission Owner may elect to hire an RF Consultant to determine whether the proposed turbine locations will obstruct its Licensed Microwave Paths Costs, such a study shall be allocated to the Interconnection Customer and shall not be subject to reimbursement.

The Interconnection Customer shall provide the Transmission Owner with a microwave path study, including evaluation of the Transmission Owner's Licensed Microwave Paths. The Transmission Owner will review and approve or deny the completed study within 10 business days after the study is submitted, where the approval of the Transmission Owner is not to be unreasonably withheld or denied. The Interconnection Customer shall gain approval of the submitted study prior to constructing the Generating Facility.

A14.11 Generic Requirements for Fiber Optic Construction in Support of MISO Generation Projects

- All fiber shall be singlemode, and its performance shall meet or exceed ITU-TG.657A1 specifications for Single-mode Optical Fiber, Dispersion Unshifted, with Low Water Peak.
- Minimum per-cable fiber count shall be forty-eight (48) strands.
    - To facilitate inter-company splicing, this count shall be incorporated into the fiber optic cable in four (4) bundles/tubes of twelve (12) fibers each, for a total of 48 fibers.
    - Higher count cables may be coordinated with Entergy.

- Fiber cable design shall be appropriate for the intended purpose (OPGW versus ADSS versus all-dielectric duct).
- All fiber optic joinery shall be accomplished via fusion splicing. Minimum standards for finished fusion splice performance are found in Table 1, below.
    - Mechanical fiber splicing is not permitted.
    - All fibers in the cable shall be spliced through at all intermediate splice points.
- Fiber termination connector type shall be SC ultra-polished physical contact or LC ultra-polished physical contact.
    - Mechanical fiber terminations are not permitted.
    - Connector Pigtails shall be attached to entrance cable via fusion splicing.
    - All fibers in the cable shall be terminated.
- At the designated meet point(s), Entergy will provide an all-dielectric fiber optic cable stub of sufficient length, to which the customer shall splice their fiber optic cable in a customer-provided splice enclosure appropriate to the cable and construction type.
    - This interface will be the point of demarcation for ownership and operational responsibility of the new fiber optic cable construction.
    - All fibers in the cable shall be spliced through at the point of demarcation.
- Entergy reserves the right to inspect customer-owned fiber optic facilities for compliance with these minimum requirements to ensure reliable operation of inter-company relay and communication systems.

Table A14.5: Summarized Fiber Optic Splice Performance Grid

Summarized Fiber Optic Splice Performance Grid (all measurements to be recorded at 1550 nm)		
Parameter	**Measured Value**	**Action Required**
Splice Loss as estimated by Fusion Splicer:	$loss \leq 0.05$ dB	TARGET
	$0.05 \text{ dB} < loss < 0.10$ dB	acceptable
	$loss \geq 0.10$ dB	Break and re-splice up to three times; if no improvement, note the splice location and the fiber number for communication to Entergy.
Splice Loss as measured by OTDR:	$loss \leq 0.05$ dB	TARGET
	$0.05 \text{ dB} < loss < 0.10$ dB	acceptable
	$loss \geq 0.10$ dB	Note the splice location and fiber number for communication to Entergy.

Appendix B To GIA
Milestones

1. Selected Option pursuant to Article 5.1: Interconnection Customer selects the Standard Option as described in Article 5.1.1. Articles 5.1.2, 5.1.3 and 5.1.4 shall not apply to this GIA.

2. Milestones: The description and date entries listed in the following tables are provided solely for the convenience of the Parties in establishing their applicable Milestones consistent with the provisions of this GIA and the GIP.

A. Interconnection Customer Milestones

No.	Description	Date
1a.	Provide initial payment to Transmission Owner (GIA 11.5) in form of security for $ 17,696,000 (100% of total Network Upgrade and Transmission Owner Interconnection Facilities cost). Coordinate Initial Payment confirmation with Transmission Owner and notify Transmission Provider.	Within the later of a) 45 Calendar Days of the execution of the GIA by all Parties, or b) 45 Calendar Days of acceptance by FERC if the GIA is filed unexecuted and the payment is being protested by Interconnection Customer, or c) 45 Calendar Days of the filing if the GIA is filed unexecuted and the initial payment is not being protested by Interconnection Customer. – Complete.
1b.	Provide security, *i.e.*, a guarantee, surety bond, letter of credit or other reasonably acceptable form of security to Transmission Owner (GIA 11.6).	Not Applicable.
1c.	Enter into a Facilities Service Agreement ("FSA") with Transmission Owner for the Network Upgrades associated with this GIA.	Not Applicable.
1d.	Provide to Transmission Owner W9 document from entity responsible for payments to Transmission Owner.	Within 15 Calendar Days of the execution of the GIA by all Parties.
1e.	Enter into amendment of this GIA, if required, for Interconnection Requests that did not elect optional negotiation period adjustment for Interconnection Facilities Study pursuant to GIA Section 11.3.1.	Within 90 Business Days of the posting of the final Interconnection Facilities report.
2.	Provide Certificate of Insurance (GIA 18.4.9).	The earlier of the construction work commencement date or the milestone date; thereafter,

		within 90 Calendar Days of end of fiscal year or insurance renewal date.
3.	i) Provide to Transmission Provider evidence of continued Site Control after execution of this GIA (GIP 7.2.2). ii) Provide evidence of one or more of the following milestones being achieved: (1) execution of contract for (a) fuel supply or transport; (b) cooling water supply; (c) engineering procurement of major equipment or construction; (d) execution of a contract for the sale of electric energy or capacity from the Generating Facility, or a statement signed by an officer or authorized agent of Interconnection Customer attesting that the Generating Facility is included in an applicable state resource adequacy plan; or other information that Transmission Provider deems to be reasonable evidence that the Generating Facility will qualify as a designated network resource; or (2) documentation of application for state or local air, water, land, or federal nuclear or hydroelectric permits and that the application is proceeding per regulations (GIP 11.3).	As may be agreed to by the Parties. Within 180 Calendar Days of Effective Date.
4.	Cash Payments in Milestones 4a-4d below are required on dates shown for the Transmission Owner to meet their schedule for Transmission Owner Interconnection Facilities and Network Upgrades. Cash Payment and security obligations are subject to quarterly review and update by Transmission Owner; who will take into account actual costs, updated cost projections and schedule revisions and make changes to the Interconnection Customer's Cash Payments and security obligations as required.	See Below.
4a.	$877,000 - Engineering & Procurement for Long Lead Equipment.	07/26/2024 COMPLETE
4b.	$802,000 – Procurement.	02/14/2026
4c.	$9,025,000 – Preconstruction.	09/15/2026
4d.	$6,992,000 – Construction.	06/15/2027
5.	Invoice Transmission Owner for the estimated amount to be expended by the Interconnection Customer to construct any Stand Alone Network Upgrades for which the Interconnection Customer has exercised its Option to Build if the Transmission Owner has	Not Applicable.

	elected to Self Fund (GIA 5.2.13).	
6.	Pre-construction meeting.	As may be agreed to by the Parties.
7.	Provide initial design and specifications for Interconnection Customer's Interconnection Facilities to Transmission Owner and Transmission Provider for comment (GIA 5.10.1).	180 Calendar Days prior to initial synchronization date.
8.	Provide final design and specifications for Interconnection Customer's Interconnection Facilities to Transmission Owner and Transmission Provider for comment (GIA 5.10.1).	90 Calendar Days prior to initial synchronization date.
9.	Deliver to Transmission Owner and Transmission Provider "as-built" drawings, information and documents regarding Interconnection Customer's Interconnection Facilities (GIA 5.10.3).	Within 120 Calendar Days of Commercial Operation Date.
10.	Provide Transmission Owner final cost invoices to construct Stand Alone Network Upgrades for which the Interconnection Customer has exercised its Option to Build if the Transmission Owner has elected to Self Fund (GIA 5.2.13).	Not Applicable.
11.	Notify Transmission Provider and Transmission Owner in writing of Local Balancing Authority where Generating Facility is located (GIA 9.2).	Three months prior to Initial Synchronization Date.
12.	Pre-energization meeting.	As may be agreed to by the Parties.
13.	Initial Synchronization Date.	5/1/2028
14.	Commercial Operation Date.	4/1/2026*
15.	Interconnection Customer shall provide the Parties with notice on the status of the Generating Facility, including COD, under Article 15 of this GIA and shall also send such notice by email to ResourceIntegration@misoenergy.org. Notification shall include Interconnection Customer's name, and as applicable Market Participant(s) name(s), and project number.	6 months prior to Initial Synchronization Date.
16.	Interconnection Customer shall provide notice to the Parties of a test plan in advance of conducting tests for the Generating Facility. The notice shall be in the form below and should be provided under Article 15 of this GIA, and a copy of such notice should be emailed to ResourceIntegration@misoenergy.org**.**	5 Business Days prior to testing.

17.	In the event the Interconnection Customer makes any modifications to the design of the site layouts or interconnection facility routes after execution of this GIA, Interconnection Customer shall notify the Parties of such changes immediately upon identifying the need for such changes. After providing such notification, the Interconnection Customer shall provide to Transmission Provider evidence of continued Site Control for land sufficient to accommodate the changes in site layouts and/or interconnection facility routes (GIP 7.2.2).	90 Calendar Days after Interconnection Customer provides notice to Parties.

* Modified from the original Commercial Operation Date of December 1, 2023 contained in the Interconnection Request. Interconnection Customer anticipates utilizing a portion of the three-year extension provided under GIA Section 2.3.1 and will reach the Commercial Operation Date on or before August 1, 2028.

B. Transmission Owner Milestones

No.	**Description**	**Date**
0.	Transmission Owner to enter Network Upgrade information into Transmission Provider's MTEP database and model on demand.	10 Business Days after Effective Date.
1.	Provide Certificate of Insurance (GIA 18.4.9).	The earlier of the construction work commencement date or the milestone date; within 90 Calendar Days of end of fiscal year or insurance renewal date.
2.	Commence design of Interconnection Facilities (GIA 5.5 et seq.). Commence equipment procurement.	As agreed to by the Parties (after completion of Interconnection Facilities Study, receipt of written authorization from Interconnection Customer and Interconnection Customer's deposit of security.
3.	Commence construction of Interconnection Facilities (GIA 5.6 et seq.).	As agreed to by the Parties (after receiving approval of Governmental Authority, receipt of written authorization and security from Interconnection Customer).

4.	Comment on Interconnection Customer's final design and specifications.	Within 30 Calendar Days of Interconnection Customer's submission of final design and specifications.
5.	Deliver to Interconnection Customer and Transmission Provider "as-built" drawings, information and documents regarding Transmission Owner's Interconnection Facilities (GIA 5.11).	Within 120 Calendar Days of Commercial Operation Date.
6.	Provide payment for invoiced estimated costs to construct for which the Interconnection Customer has exercised its Option to Build if the Transmission Owner has elected to Self Fund (GIA 5.2.13).	Not Applicable.
7.	Provide Interconnection Customer final cost invoices (GIA 12.2 *et seq.*).	Within (6) six months of completion.
8.	Refund overpayment of estimated costs (GIA 12.2).	Refunds within 30 Calendar Days of final invoice.
9.	In-Service Date/Backfeed Date.	05/01/2028

C. Affected System Owner Milestones

Task	**Date Due**
None.	

D. Transmission Provider Milestones

No.	**Description**	**Date**
1.	Transmission Provider to determine conditional limit for interconnection service.	Prior to Commercial Operation.

Appendix B-1 To GIA
Pre-Certification Generation Test Notification Form

The following form would need to be submitted to MISO Real Time Operations at least five (5) Business Days prior to the first date of testing.

Project Number:

Project Name:

Point of Interconnection:

Dispatcher Contact Information:

Date	Start Time (in EST)	End Time (in EST)	Expected MW Output	Expected MVAR Output (Only needed if beyond normal power factor)

Appendix C To GIA
Interconnection Details

This Appendix C is a part of this GIA among Interconnection Customer, Transmission Owner and Transmission Provider.

1. The unique requirements of each generation interconnection will dictate the establishment of mutually agreeable Interconnection and/or Operating Guidelines that further define the requirements of this GIA. The Interconnection and/or Operating Guidelines applicable to this GIA consist of the following information. Additional detail may be provided through attachment to this Appendix C or through electronic means via the web address specified.
    - (a) System Protection Facilities;
    - (b) Communication requirements;
    - (c) Metering requirements;
    - (d) Grounding requirements;
    - (e) Transmission Line and Substation Connection configurations;
    - (f) Unit Stability requirements;
    - (g) Equipment ratings;
    - (h) Short Circuit requirements;
    - (i) Synchronizing requirements;
    - (j) Generation and Operation Control requirements;
    - (k) Data provisions;
    - (l) Energization inspection and testing requirements;
    - (m) Control equipment, if any;

    Entergy Facility Connection Requirements website:
    http://www.entergy.com/energydelivery/facility_requirements.aspx.

    Reference material regarding the Entergy Facility Connection Requirements can be found on the two (2) links below located on the above mentioned web address.

    (1) Entergy Facilities Connection Requirements Document, March 2007; and
    (2) Entergy Facilities Connection Spreadsheet, October 2008.

    In the event that the Interconnection and/or Operating Guidelines conflict with this GIA, this GIA will govern.

2. Information required pursuant to Section 8.4 for Variable Energy Resource Generating Facilities: Not Applicable.

Appendix D To GIA
Security Arrangements Details

Infrastructure security of Transmission or Distribution System equipment and operations, as applicable, and control hardware and software is essential to ensure day-to-day Transmission and Distribution System reliability and operational security. The Commission will expect all Transmission Providers, market participants, and Interconnection Customers interconnected to the Transmission or Distribution System, as applicable, to comply with the recommendations provided by Governmental Authorities regarding Critical Energy Infrastructure Information ("CEII") as that term is defined in 18 C.F.R. Section 388.113(c) and best practice recommendations from the electric reliability authority. All public utilities will be expected to meet basic standards for system infrastructure and operational security, including physical, operational, and cyber-security practices.

Appendix E To GIA
Commercial Operation Date

This Appendix E is a part of this GIA between Transmission Provider, Transmission Owner and Interconnection Customer.

[Date]

Midcontinent Independent System Operator, Inc.
Attn: Director, Transmission Access Planning
720 City Center Drive
Carmel, IN 46032

Re: _____ Generating Facility

Dear _____:

On **[Date] [Interconnection Customer]** has completed Trial Operation of Unit No. ___. This letter confirms that **[Interconnection Customer]** commenced commercial operation of Unit No. ___ at the Generating Facility, effective as of **[Date plus one Calendar Day]**.

Thank you.

[Signature]

[Interconnection Customer Representative]

cc: Transmission Owner

Appendix F To GIA
Addresses for Delivery of Notices and Billings

Notices:

Transmission Provider:

MISO
Attn: Director, Transmission Access Planning
720 City Center Drive
Carmel, IN 46032

Transmission Owner:

Entergy Services, LLC
9585 Pecue Blvd
Baton Rouge, LA 70810
Attn: Manager, Sr. Mgr., Interconnection Planning and Customer Service

Interconnection Customer:

Lacassine Solar II, LLC
PO Box 303427
Austin, TX 78703
Email: mark@sunchasepower.com

Billings and Payments:

Transmission Provider:

MISO
Attn: Director, Transmission Access Planning
720 City Center Drive
Carmel, IN 46032

Transmission Owner:

Entergy Louisiana, LLC
P.O. Box 679505
Dallas, TX 75267-9505

Interconnection Customer:

Lacassine Solar II, LLC
PO Box 303427
Austin, TX 78703
invoices@sunchasepower.com

Alternative Forms of Delivery of Notices (telephone, facsimile or email):

Transmission Provider:

MISO
Phone: (317) 249-5700
Email: misotap@misoenergy.org or
MISOTransmissionAccessPlanning@misoenergy.org

Transmission Owner:

Entergy Louisiana, LLC
Voice telephone: (504) 576-7940
Email address: GIA@entergy.com

Interconnection Customer:

Lacassine Solar II, LLC
Telephone: (512) 466-4554

Appendix G To GIA
Interconnection Requirements for a Non-synchronous Generating Facility

Appendix G sets forth requirements and provisions specific to a non-synchronous generating facility. All other requirements of this GIA continue to apply to non-synchronous generating facility interconnections.

A. Technical Standards Applicable to a Non-synchronous Generating Facility

i. Abnormal Voltage Ride-Through Capability

A non-synchronous generating facility shall be able to remain online during voltage disturbances up to the time periods and associated voltage levels set forth in the standard below.

1. Non-synchronous generating facilities are required to remain in-service during three-phase faults with normal clearing (which is a time period of approximately 4-9 cycles) and single line to ground faults with delayed clearing, and subsequent post-fault voltage recovery to prefault voltage unless clearing the fault effectively disconnects the generator from the system. The clearing time requirement for a three-phase fault will be specific to the non-synchronous generating facility substation location, as determined by and documented by the transmission provider. The maximum clearing time the non-synchronous generating facility shall be required to withstand for a three-phase fault shall be 9 cycles after which, if the fault remains following the location-specific normal clearing time for three-phase faults, the non-synchronous generating facility may disconnect from the transmission system. A non-synchronous generating facility shall remain interconnected during such a fault on the transmission system for a voltage level as low as zero volts, as measured at the high voltage side of the GSU.

2. This requirement does not apply to faults that would occur between the non-synchronous Generator terminals and the high side of the GSU.

3. Momentary cessation (ceasing to inject current into the transmission grid during a fault without mechanical isolation) is prohibited in North American Electric Reliability Corporation (NERC) reliability standard PRC-024 no trip zone.

4. Non-synchronous generating facility inverters may not trip or cease to inject current for momentary loss of synchronism. Any inverter may trip if the phase lock loop is unable to regain synchronism 150 milliseconds after loss of synchronism.

5. Non-synchronous generating facilities may be tripped after the fault period if this action is intended as part of a Special Protection System.

6. Non-synchronous generating facilities may meet the abnormal voltage ride-through requirements of this standard by the dynamic performance capability of the generators, by installing additional equipment (*e.g.* Static VAr Compensator) within the non-synchronous generating facility or by a combination of generator performance and additional equipment.

7. Existing individual generator units that are, or have been, interconnected to the network at the same location at the effective date of the Appendix G abnormal voltage ride-through Standard are exempt from meeting the Appendix G abnormal voltage ride-through Standard for the remaining life of the existing generation equipment. Existing individual generator units that are replaced are required to meet the Appendix G abnormal voltage ride-through Standard.

ii. Power Factor Design Criteria (Reactive Power)

The following reactive power requirements apply only to a newly interconnecting non-synchronous generating facility that has completed a System Impact Study as of the effective date of the Final Rule establishing the reactive power requirements for non-synchronous generators in section 9.6.1 of this GIA (Order No. 827). A non-synchronous generating facility to which this provision applies shall maintain a factor within the range of 0.95 leading to 0.95 lagging, unless Transmission Provider has established different requirements that apply to all Generating Facilities in the Local Balancing Authority on a comparable basis, measured at the high side of the generator substation. The power factor range standard can be met by using, for example, power electronics designed to supply this level of reactive capability (taking into account any limitations due to voltage level, real power output, etc.) or fixed and switched capacitors if agreed to by Transmission Provider, or a combination of the two. Interconnection Customer shall not disable power factor equipment while the non-synchronous generating facility is in operation. Non-synchronous generating facilities shall also be able to provide sufficient dynamic voltage support in lieu of the power system stabilizer and automatic voltage regulation at the generator excitation system.

iii. Supervisory Control and Data Acquisition (SCADA) Capability

The non-synchronous generating facility shall provide SCADA capability to transmit data and receive instructions from Transmission Provider to protect system reliability. Transmission Provider and Interconnection Customer shall determine what SCADA information is essential for the proposed non-synchronous generating facility, taking into account the size of the plant and its characteristics, location, and importance in maintaining generation resource adequacy and transmission system reliability in its area.

iv. Transient Data Recording Equipment for Facilities above 20 MW

Non-synchronous generating facilities with generating capacities of more than 20 MW must monitor and record data for all frequency ride-through events, transient low-voltage disturbances that initiated reactive current injection, reactive current injection or momentary cessation for transient high-voltage disturbances, and inverter trips. The data may be recorded and stored in a central plant control system. The following data must be recorded:

Plant Level

(1) Plant three-phase voltage, current and power factor
(2) Status of ancillary reactive devices
(3) Status of all plant circuit breakers
(4) Status of plant controller

(5) Plant control set points
(6) Status of main plant transformer no load taps
(7) Status of main plant transformer tap changer (if applicable)
(8) Protective relay trips (relay target data)

Inverter Level Data

(1) Frequency, current, and voltage during frequency ride-through events
(2) Voltage and current during momentary cessation for transient high-voltage events (when used)
(3) Voltage and current during reactive current injection for transient low or high-voltage events
(4) Inverter alarm and fault codes
(5) DC current
(6) DC voltage

The data must be time synchronized to a one millisecond level of resolution. All data except phase angle measuring unit data must be sampled at least every 10 milliseconds.

The non-synchronous generating facility shall store this data for a minimum of 60 calendar days. The non-synchronous generating facility, upon request from MISO or the transmission owner, shall make this data available within 10 calendar days of the request.

v. Protection Settings

A non-synchronous generating facility, while complying with the under-frequency and over-frequency ride through requirements, shall use accurately calculated and filtered frequency measurement over a time window to set frequency protection and should not use an instantaneously calculated value to set frequency protection.

A non-synchronous generating facility, while complying with the under-voltage and over-voltage ride through requirements should use filtered Root-Mean-Squared (RMS) voltage measurement to set voltage protection.

Inverter rate-of-change-of-frequency (ROCOF) protection shall be disabled unless an equipment limitation exists that requires the inverter to trip on high ROCOF.

Appendix H To GIA – Not Applicable
Interconnection Requirements for Provisional GIA

Provisional Agreement

This GIA is being provided in accordance with Section 11.5 of the Transmission Provider's GIP, which provides among other things, that an Interconnection Customer may request that Transmission Provider provide Interconnection Customer with a Provisional Generator Interconnection Agreement that limits the transfer of energy by Interconnection Customer commensurate with that allowed for Energy Resource Interconnection Service. Interconnection Customer requested Transmission Provider to provide a Provisional Generator Interconnection Agreement for limited operation at the discretion of Transmission Provider based upon the results of available studies (by Interconnection Customer and by Transmission Provider).

A Provisional Interconnection Study, the results of which are posted on the confidential portion of the Transmission Provider's internet website, was performed by Transmission Provider in order to confirm the facilities that are required for provisional Interconnection Service and to require them to be in place prior to commencement of service under the GIA.

Interconnection Customer represents that the Interconnection Customer facilities (including Network Upgrades, Interconnection Facilities, Distribution Upgrades, System Protection Upgrades and/or Generator Upgrades) that are necessary to commence provisional Interconnection Service and meet the requirements of NERC, or any applicable regional entity for the interconnection of a new generator are in place prior to the commencement of generation from the Generating Facility and will remain in place during the term of the service. The requisite Interconnection Studies were performed for the Generating Facility. Interconnection Customer shall meet any additional requirements (including reactive power requirements) pursuant to the results of applicable future Interconnection System Impact Studies. Until such time as the applicable Interconnection Studies and any identified facilities are completed, the output of the Generating Facility will operate within the output limit prescribed in a future, if applicable, operating guide.

The maximum permissible output of the Generating Facility under Appendix A will be updated by Transmission Provider on a quarterly basis, determined in accordance with Section 11.5 of the GIP, by finding the transfer limit of energy commensurate with the analysis for Energy Resource Interconnection Service ("ERIS"). This study shall be performed assuming the system topology represented by the base cases used to calculate Available Flowgate Capability, as described in Attachment C of the Tariff, with dispatch and optimization algorithms posted on the MISO internet site and operation above those limits will be deemed as unauthorized use of the Transmission System and subject to provisions in the Tariff surrounding that use.

Use of interim operating guide

Implementation of interim operating guide, if applicable, will constitute an interim solution that will permit Interconnection Customer to operate the Generating Facility under conditional Interconnection Service until planned Network Upgrades are constructed. Any interim operating

guide will be subject to the approval of Transmission Owner and Transmission Provider. Minimum requirements for an interim operating guide are as indicated below.

* Transmission Operator will have control of breaker(s) dedicated to the Generating Facility and will be able to trip the Interconnection Customer's Generating Facility.
* Protection schemes must be tested and operative.
* Interconnection Customer will provide continuous communication capability with the Generator Operator.
* Interconnection Customer and the owner of the Existing Generating Facility will enter into an operating agreement or similar agreement which designates, among other things, the responsibilities and authorities of each of the parties and shall be subject to the acceptance of Transmission Provider and Transmission Owner.
* A termination date consistent with completion of construction of Network Upgrades will be included as part of all operating guides accepted by Transmission Owner and Transmission Provider.

Interconnection Customer assumes all risks and liabilities with respect to changes, which may impact the Generator Interconnection Agreement including, but not limited to, change in output limits and responsibilities for future Network Upgrade and cost responsibilities that have not yet been identified on the direct connect Transmission System as well as all affected Transmission, Distribution or Generation System(s) including non-Transmission Provider Systems. Such upgrades will be determined pursuant to the Tariff and Policies in effect at the time of the Interconnection Studies.

Appendix I To GIA – Not Applicable
Requirements Applicable to Surplus Interconnection Service

Where this GIA provides for Surplus Interconnection Service, Interconnection Customer acknowledges, agrees to, and will be required to operate under the following conditions:

1) The combined Real-Time Offers, including Energy and Operating Reserves, of the Generating Facility and the Existing Generating Facility with which Interconnection Customer has an executed Energy Displacement Agreement must be less than or equal to Interconnection Service limit (MW, MVAR, MVA output) provided in Exhibit I-1 (Monitoring and Consent Agreement) (hereinafter, "Interconnection Service limit"). In the event that the sum of the simultaneous energy output of the Surplus Interconnection Service Generating Facility and the Existing Generating Facility exceeds such Interconnection Service limit, MISO reserves the right to curtail and/or disconnect the Generating Facility immediately.

   In the event that the sum of the emergency and/or economic maximum offer limits of the Generating Facility and the Existing Generating Facility exceeds the Interconnection Service limit, MISO reserves the right to curtail and/or disconnect the Generating Facility immediately.

2) The total MW, MVAR, MVA output at the Point of Interconnection resulting from the combined output of the Generating Facility with Surplus Interconnection Service and the Existing Generating Facility with which Interconnection Customer has an executed Energy Displacement Agreement shall not at any time exceed the Interconnection Service limit.

3) The Existing Generating Facility with which Interconnection Customer has an executed Energy Displacement Agreement is not relieved of any applicable requirements under the RAR of the Tariff.

4) The Interconnection Customer shall submit to the Transmission Provider a report by the seventh Calendar Day of each month showing the prior month's output, by 15 minute increment, the combined real-time offers and cleared energy injection. The Existing Generating Facility and the Interconnection Customer shall cooperate consistent with other provisions in the Tariff to the extent necessary to ensure accuracy of the report. Transmission Provider shall provide a template for this report.

Exhibit I-1 (Completed Monitoring and Consent Agreement - Appendix 11 of the GIP)

Exhibit I-2 (Completed Energy Displacement Agreement - Appendix 12 of the GIP)